

SURG Project Record: Data-Center Load, Volatility, and the Grid

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Repository: github.com/tonyyu2170/surg

This is an accumulation of all data, sources, and papers read and analyzed throughout this summer undergraduate research project. Every number here is copied from the project's own log of record (decisions.md), the research notes (research-notes/), or the pre-registration documents (plans/); each section specifies where information in greater depth lives. Figures are in assets/final_report/ and are generated from the project's own scripts from the data on disk. The raw data is not in the repository because of its size (about 28 GB), but every dataset can be re-pulled: the fetch commands are in the root README.md, and [§2](#) says, per source, what each pull needs and where its constraints are documented.

0. Orientation

Before discussing the datasets and analysis of the actual project, it is worth going over the original proposal in greater depth. The full text of the original proposal is reproduced in [Appendix A](#).

The original research proposal. Hyperscale AI data centers in Northern Virginia were assumed to have more volatile load profiles than other large customers, since AI training and inference draw power differently from traditional data-center uses like storage and cloud services. The original proposal asked two questions. At what level of load *volatility* (in MW per minute) do congestion prices in the Dominion zone undergo a non-linear phase transition (a sudden change of regime rather than a gradual one)? And in what year would the growth forecast from JLARC (Virginia's Joint Legislative Audit and Review Commission) make that the chronic state of the grid?

Proposal Creation: The Reading Behind the Question

The proposal was built from a dozen sources, and it is worth recording what each contributed. The full citations are in [§6](#).

- JLARC's *Data Centers in Virginia* (Report 598, December 2024) provided a baseline understanding of the data center buildout in Virginia, as well as growth estimates for the second half of the original research question. It forecasts Virginia's electricity consumption doubling by 2040 under scenarios of 1.95, 2.23, and 2.90 times current load, and it does so on annual and monthly consumption. However, the report treats data center load as a flat baseline. The proposal's premise was that this flat baseline misses what happens at short timescales, and that these volatile events can have negative impacts on the DOM region's transmission grid.
- Li and Li (arXiv 2502.01647) supplied the volatility argument for the original research question. They argue from a power-electronics view that AI training makes load "spiky": GPUs run flat out during a compute phase, but then drop to near idle during a communication phase while they exchange data. A synchronized job swings between the two phases several times a second, causing the volatile swings. Unfortunately, the measurements behind this argument rest on single desktop

GPUs, which is unideal because the megawatt-scale swings the proposal was about are an extrapolation from a single workstation to large, hyperscale data centers ([§3.12](#)).

- Chen et al. (arXiv 2509.07218) supplied a broader perspective: a survey of the electricity demand of AI data centers and of the grid problems they raise, including load volatility. Unfortunately, as a survey it carries the volatility claim without measuring it, and the primary sources behind that claim turned out to be single-GPU measurements and operator statements with no published data ([§3.12](#)).
- Hogan and Harvey (2022) and PJM's Manual 11 explained how the price variable, locational marginal pricing (LMP), worked. It is a sum of a system energy price, a congestion price, and a marginal loss price, where a high LMP generally means the grid is unable to move power to where it is needed. LMP will be explained in greater detail shortly.
- Monitoring Analytics' 2024 State of the Market report described the location. Monitoring Analytics is PJM's independent market monitor, and its report names the Pleasant View to Ashburn line and the Goose Creek transformer in the DOM zone as constraints whose congestion costs had already risen. This is why most of the project's pricing nodes (the specific points on the grid at which PJM computes a price) sit in Loudoun County, with a few outliers to compare against. Of course, this was only in the original proposal's plan, before expanding to the rest of the US and even Europe due to data constraints.
- Hines et al. (2009), *Cascading failures in power grids*, supplied the "phase transition" idea in the original proposal. It reviews how large blackouts propagate: a grid carries load in a roughly linear regime until one component trips, the redistributed flow overloads its neighbors, and the failure cascades. This results in the size distribution of blackouts being heavy-tailed (the self-organized criticality view): many small blackouts, a few enormous ones, and far more of the enormous ones than a bell curve would predict. The proposal borrowed this to argue that congestion prices would likewise move from a linear to a heavy-tailed regime past some stress threshold, which is the "critical volatility threshold" it set out to find.
- Yang et al. (2023) model a data-center park under distribution-level locational marginal prices, with the park scheduling its own flexible resources (storage, shiftable compute, cooling) against the price signal until it reaches an equilibrium with the distribution utility. It shows LMPs reaching down to the distribution level, which matters here because the project's two Ashburn nodes sit on the distribution side (35 kV), where such prices would apply.
- Lindberg et al. (2021) use locational marginal CO₂ emissions to show that a hyperscale operator can shift load between regions to cut emissions. Its point for the proposal was that hyperscale load is something its operator actively moves, not a fixed local demand, so the load seen at one location is a management decision as much as a physical one.
- PJM's 2023 paper on price formation during reserve shortages supplied the method as first planned: a generalized Pareto tail model (a statistical model of the largest values in a distribution) tied to the reserve-shortage price steps. However, upon a more in-depth reading, this method was later changed to threshold regression (which searches for the point at which a relationship changes) plus quantile regression (which measures how the top of the price distribution, rather than its average, responds), with Hansen's (1996, 2000) threshold-regression papers added at set-up.

An initial read of these sources missed several details that made the original proposal difficult to sustain. The "spiky load" premise had never been measured at a facility; the phase-transition idea came from cascading-failure physics, not from anything observed in PJM prices; the reserve-shortage mechanism lands in a price component that is uniform across PJM, so it cannot carry a Loudoun-specific signal; and the proposal's own isolation filter (shoulder months, meaning spring and fall when heating and cooling loads are lowest, and the 2 to 5 AM hours) removes the very price events it was meant to explain. By the first few

weeks of the project, the possibility that the transition simply does not exist in this market was on the table, and a very appealing conclusion to make.

How PJM Prices Form

Given that pricing is one of the main variables involved in the original proposal, it makes sense to do a deeper reading into how PJM computes prices at specific nodes. The primary sources were PJM Manuals M11 (revision 137), M12 (revision 57), and M03 (revision 71), plus PJM's 2023 paper on reserve-shortage pricing, all vendored under `reference/`. The write-up is `reference/pjm-manuals/pjm-lmp-formation.md`.

Prices are formulated as locational marginal prices (LMP), calculated by the formula below.

LMP = system energy price + congestion price + marginal loss price

Given the nature of this project, it makes sense to focus primarily on system energy price and congestion price.

System energy price. When PJM runs short of reserves (generation held in standby to cover a sudden loss of supply), a penalty from its operating reserve demand curve (ORDC) is added to the price. The curve has the same two steps for every reserve product: every megawatt short of the reliability requirement is valued at \$850/MWh, the next 190 MW at \$300/MWh, and nothing beyond that (Figure 1). That penalty lands in the system energy component, which is the same everywhere in PJM, regardless of zone or node. This was verified independently in the data: across 350,174 rows, the largest difference in system energy between any two DOM nodes was 9.09×10^{-13} .

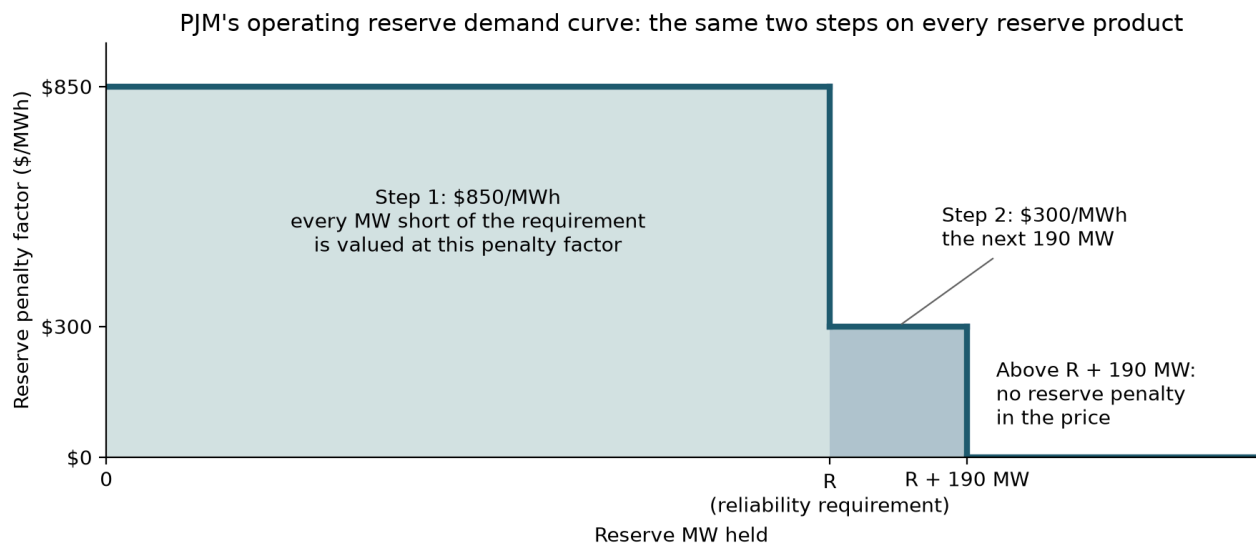


Figure 1. PJM's two-step operating reserve demand curve, drawn from Manual 11 section 4.3.3 [4]. The horizontal axis is how much reserve PJM is holding relative to its requirement R ; the vertical axis is the penalty added to every PJM price when reserves fall short. The first step (\$850/MWh) applies to every megawatt short of the requirement and the second (\$300/MWh) to the next 190 MW; beyond that there is no penalty. Drawn by `scripts/plot_ordc.py`.

Congestion price. PJM always dispatches the cheapest generation that the wires can deliver. When a transmission line or transformer reaches its limit, the cheap generator on the far side can no longer serve the load behind the constraint, so a more expensive generator on the near side must run instead. The

difference is charged to the nodes behind the constraint as their congestion price. Manual 11 §2.2 defines it as the effect on transmission congestion costs of "consumption by the resource on transmission line loadings," and it is computed for each node from the shadow price of every binding constraint (the cost of moving one more MW across a line or transformer that is already at its limit, capped at PJM's \$2,000/MWh transmission penalty factor) weighted by how much that node's consumption loads the constraint. That is why the congestion price differs from node to node while the system energy price does not, why it sits near zero most of the time and then jumps when a limit binds, and why the Loudoun nodes pay it when the Pleasant View to Ashburn line or the Goose Creek transformer is at its limit.

Working Conclusion as of the End of This Project

At hourly timescales, data-center load barely fluctuates, and the originally hypothesized "volatility" can't be seen in the data explored throughout this project (although none of the datasets were ideal). Modern mitigation techniques such as voltage ride-through (staying connected through a voltage dip instead of tripping to backup power), fast frequency response (batteries or loads reacting within a second to a frequency deviation), soft-start (ramping a facility's load up gradually) and on-site batteries appear to be absorbing what happens below one second, and other factors are flattening the slower parts of the curve: behind-the-meter generation (a facility's own power plant, whose output never passes through the utility meter) takes load off the meter altogether, and policy such as the Texas grid operator ERCOT's proposed 10 MW per 5-second limit, the North American Electric Reliability Corporation's (NERC) ride-through guideline, and Texas Senate Bill 6 demand response is aimed at the sub-second and emergency ends.

Repo Structure

The repository is a research pipeline. Data is pulled from an API or archive into `data/raw/`, cleaned and joined into analysis tables in `data/interim/`, and every analysis reads those tables and writes its results to `outputs/`. Nothing is passed between stages in memory.

Note: because this project's direction changed several times, the repository is not tidy. It holds dead ends, findings that carry little weight, and elaborate work that did not pan out. For anything in the repo to make sense, make sure to read the high-level overview below before looking at any specific code.

```
surg/
├── README.md           setup, API keys, the fetch and analyze commands
├── pyproject.toml      package definition; entry points surg-pull, surg-prep,
                        surg-analyze (hourly suite), surg-run-5min (5-minute suite)
├── src/surg/          the PJM pipeline as a Python package
│   ├── acquisition/   API clients and pull orchestration
│   │   ├── client.py   PJM Data Miner 2 client, throttled to ~6 requests/min
│   │   ├── gridstatus_client.py  gridstatus.io client; rotates the six free keys
│   │   ├── gridstatus_pull.py    the 5-minute backfill and (gridstatus_validate.py) its checks
│   │   ├── entsoe_parse.py       ENTSO-E XML to rows, A03 curves left unexpanded
│   │   ├── chunking.py, storage.py  date chunks; skip what is already on disk
│   │   ├── pull.py              the surg-pull entry point
│   │   └── targets.py           the 11 pricing nodes
│   ├── preprocessing/  raw pulls to analysis tables
│   │   ├── loaders.py, schema.py, features.py, build.py           the hourly DOM panel
│   │   ├── loaders_5min.py, schema_5min.py, build_5min.py        the 5-minute DOM panels
│   │   ├── <market>_features.py  one per ISO: caiso, ercot, ieso, isone, miso, nyiso, spp
│   │   └── entsoe_zones.py, entsoe_expand.py, entsoe_panel.py     the European panel
│   └── analysis/       the statistics
│       ├── tar.py       threshold regression (Hansen)
│       ├── qr.py, qr_full.py  quantile regression
│       ├── gpd.py, gpd_components.py, gpd_continuous.py  generalized Pareto tail models
│       └── mechanism.py, robustness.py, year_fe_diagnostic.py  the conditional-Z test and its checks
```


- └─ tail_risk_curves.py, ashburn_diagnostic.py, panel.py, bootstrap_strategies.py
 - └─ entsoe_seasonal.py the summer-minus-winter solar identification
 - └─ run.py, run_5min.py the hourly and 5-minute suites
- └─ diagnostics/
 - └─ stage1.py the level-versus-volatility check every market driver calls
- └─ scripts/ 39 stand-alone drivers, run by path
 - └─ <market>_fetch.py, <market>_diagnostic.py caiso, ercot, ieso, isone, miso, nyiso, spp
 - └─ entsoe_fetch.py, entsoe_ireland.py, entsoe_solar.py, entsoe_italy_stage1.py, cso_fetch.py
 - └─ ukpn_fetch.py, ukpn_flatness.py
 - └─ pecanstreet_fetch.py, pecanstreet_headroom.py, pecanstreet_1sec.py, pecanstreet_lib.py
 - └─ aidc_psd.py (GPU spectrum), aidc_edges.py (GPU edge speed), plot_aidc_timeseries.py
 - └─ necec_price_test.py, run_5min_nofilter.py
 - └─ compute_figure_inputs.py, plot_subq1_results.py, figures/ the figure pipeline
 - └─ plot_ordc.py, plot_hourly_series.py the two figures drawn only for this record
 - └─ poll_gridstatus_usage.py, verify_retention.py, build_report_docx.py, entsoe_probe.py
- └─ tests/ 585 test functions
 - └─ acquisition/, analysis/, preprocessing/, figures/ unit tests by package
 - └─ test_<market>_features.py, test_<market>_fetch.py, test_stage1_core.py, test_pecanstreet_lib.py
 - └─ regression/ pins numbers from production runs; a failure there is a real behavior change
- └─ data/ gitignored, ~28 GB
 - └─ raw/ one folder per source, exactly as downloaded
 - └─ interim/ the assembled analysis tables (parquet)
- └─ outputs/ gitignored: JSON results and figures, one folder per analysis
- └─ notebooks/ 01_data_miner_spike.ipynb, the day-one Data Miner API spike
- └─ logs/ run logs, gitignored
- └─ docs/
 - └─ decisions.md the log of record: 60-plus entries, ~6,100 lines, append-only
 - └─ Yu_Tony_SURG_Grant_Proposal.md original research proposal
 - └─ final_report.md this record
 - └─ final_reflection.md the Office of Undergraduate Research reflection
 - └─ research-notes/ the findings, one note each; INDEX.md is the entry point
 - └─ A-primary-verify.md ... H-event-catalog.md the PJM and Virginia desk research (§3.13)
 - └─ I-advisor-links-2026-08.md Professor Wei's links (§3.13)
 - └─ J-ukpn-flatness.md §3.9
 - └─ K-ireland-dc-shape.md, L-solar-metering-artifact.md §3.10
 - └─ M-pecanstreet-xfra-headroom.md §3.11
 - └─ N-subsecond-provenance-and-filters.md §3.12
 - └─ EU-0-synthesis.md ... EU-5-policy-renewables.md the Europe scoping (§3.13)
 - └─ external-context-research-2026-08.md the external-context digest (§3.13)
 - └─ sources/ what each API allows and where it breaks
 - └─ pjm-api-constraints.md, gridstatus-api-constraints.md, ukpn-api-constraints.md
 - └─ entsoe-api-constraints.md, entsoe-endpoint-reference.md
 - └─ pecanstreet-access-constraints.md, data-catalog.md
 - └─ availability/ one depth memo per market (caiso, ercot, ieso, isone, miso, nyiso, spp)
 - and the cross-ISO summary
 - └─ plans/ 42 plans, pre-registrations, and errata
 - └─ advisor/ the weekly meeting agendas, with the notes taken in each meeting
 - └─ specs/ seven design documents
 - └─ reference/ vendored PJM manuals (M03, M11, M12) and the Data Miner API guide, PJM's reserve-shortage paper, the Pecan Street catalog and dictionary
 - └─ assets/final_report/ the figures in this record

1. Timeline

This research project's timeline at a high level, in order. Each entry points to the section that writes it up.

1. **Set-up and orientation.** Built the repository, validated the PJM Data Miner 2 API (PJM's public data portal), locked the 11 pricing nodes, and read PJM's reserve-shortage paper, which changed the planned method from a tail model to threshold regression plus quantile regression ([§3.1](#)).
 2. **Hourly analysis in PJM.** Ran the original proposal's tests on the hourly DOM panel: threshold regression, quantile regression, tail models, the conditional test of the mechanism, and the tail-risk curves ([§3.1](#)).
 3. **5-minute analysis on PJM's own data.** Re-ran the tests on the 27 days of 5-minute load that PJM keeps, and found that hourly publication hides most 5-minute price spikes ([§3.2](#)).
 4. **5-minute analysis on gridstatus.io data.** Pivoted to a new data source, gridstatus.io, and then pre-registered and ran the one-year two-sided test on three nodes; everything came back weaker than hourly ([§3.2](#)).
 5. **Dropping the filter.** Dropped the proposal's 2 to 5 AM shoulder-season filter and interpreted the extended 3.4-year panel: congestion is driven by load level, not volatility ([§3.2](#), [§3.3](#), [§3.4](#)).
 6. **Events.** Verified the 2024-07-10 ride-through event in the project's own data and built the catalog of thirty grid-stress events ([§3.5](#)).
 7. **ERCOT, then seven additional markets.** Ran the level-versus-volatility check in ERCOT, then in NYISO, CAISO, IESO, MISO, ISO-NE, and SPP; level beats volatility everywhere ([§3.6](#), [§3.7](#)).
 8. **Desk research and the macro pivot.** Swept the policy and market context (notes A to H) and shut down the original proposal's JLARC projection and event-correlation ideas. The project then pivoted to a more exploratory mode ([§3.13](#)).
 9. **ISO-NE follow-ups.** Tested the Canada-tie, solar, and NECEC explanations for New England's rising volatility ([§3.8](#)).
 10. **Facility-level data at last: UKPN.** Cut the flatness note on 96 UK data-center profiles ([§3.9](#)).
 11. **Europe.** Got the ENTSO-E token and ran Ireland against the Netherlands, the solar check across 12 zones and the Italian Stage 1 ([§3.7](#), [§3.10](#)).
 12. **Residential: Pecan Street and XFRA.** Pulled the 22 GiB corpus and cut the panel-headroom note ([§3.11](#)).
 13. **The sub-second claim.** Traced its provenance, mapped the filters between a GPU and a meter, and computed the project's own GPU spectrum ([§3.12](#)).
 14. **Close-out.** Sanitized the repository and wrote this record and the final reflection.
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2. Datasets

In total, around 28 GB was pulled and analyzed in this project. All data is gitignored and reproducible from the fetch commands in the root README .md. The API keys are not published, for obvious reasons, but most were easy to obtain: every source has a free tier, and while none of the free tiers was ideal, all of them were usable. Table 1 lists every dataset used, and Table 2 lists the two sources pulled but never used, left in just for record-keeping and consistency. Each source name links to its note below, and the notes say, per source, what the source is, how it was accessed, and the source's limitations, along with links to the data and its documentation. Two words recur in the tables: an ISO (independent system operator) is the organization that runs a regional grid and its wholesale electricity market, and a zone is the sub-region of an ISO for which load and prices are published.

Table 1. Every dataset used in this record.

#	Source	Dataset	What it is	Resolution	Window	Size (rows or files; on disk)	Used in
1	PJM Data Miner 2	rt_hrl_lmps	Real-time LMP with its three components, 11 nodes	hourly	2022-10-02 to 2026-05 (Ashburn from 2024-05)	~320k rows, 3.6 MB	§3.1 , §3.4
2	PJM Data Miner 2	rt_fivemin_hrl_lmps	Five-minute real-time LMP, 11 nodes	5-minute	2025-11-12 to 2026-05-10	570k rows	§3.2 (hidden-spike comparison)
3	PJM Data Miner 2	hrl_load_metered	DOM zonal metered load	hourly	2021-01-01 to 2026-05-07	47k rows, 650 KB	§3.1
4	PJM Data Miner 2	sync_reserve_events	PJM's calls on synchronized reserves in the Mid-Atlantic/Dominion (MAD) sub-zone	events	2023-01-26 to 2026-03-05	38 events, 71 KB	stress flag, §3.1 and §3.5
5	PJM Data Miner 2	reserve_market_results	Clearing prices for synchronized and primary reserves in MAD, averaged to hourly	5-minute to hourly	2022-10-02 to 2026-05-10	758k rows, 1.3 MB	the "stressed hour" definition, §3.1
6	PJM Data Miner 2	dom_pnodes_all	Registry of all 2,328 DOM pricing nodes	snapshot		2,328 rows	node name and ID resolution
7	gridstatus.io	pjm_load	DOM zonal aggregate 5-minute instantaneous load	5-minute	2023-02 to 2026-06	750 files, 47 MB (with 8)	the extended 5-minute panel, §3.2 to §3.4
8	gridstatus.io	pjm_lmp_real_time_5_min	5-minute LMP for LOUDOUN, PLEASANT VIEW, GOOSECRE, and SKFFSCRK	5-minute	2023-02 to 2026-06		the extended 5-minute panel, §3.2 to §3.4
9	gridstatus.io	isone_interchange_hourly	Flows on every ISO-NE external tie (a transmission line connecting ISO-NE to a neighboring grid)	hourly	2026-03-01 to 03-08	one week	NECEC dose check, §3.8
10	ERCOT	native_load_<YYYY>.zip	Load by weather zone (ERCOT's regional divisions, drawn around areas of similar climate), 9 zones	hourly	2017-01-01 to 2026-07-31	30 files, 179 MB (with 11)	§3.6
11	ERCOT	report 13061, RTMLZHBSP_<YYYY>.zip	Real-time settlement point prices (the prices at which energy is bought and sold at each hub or load zone), 15 hubs (trading locations whose price is the average	hourly	2022 to 2026-07-31		§3.6

#	Source	Dataset	What it is	Resolution	Window	Size (rows or files; on disk)	Used in
			over a group of nodes) and load zones				
12	MISO	df_al	Forecast-and-actual load by local resource zone (6 LRZs), one file per market day	hourly	2023-01 to 2026-08	1,317 files, 1.4 GB (with 13)	§3.7
13	MISO	da_expost_lmp	Day-ahead ex-post LMP at the hubs, with congestion and loss components	hourly	2023-01 to 2026-08		§3.7
14	SPP	hourly load by control zone	17 zones used	hourly	2016 to 2026-03-23	914 files, 3.7 GB (with 15)	§3.7
15	SPP	day-ahead LMP by settlement location	With LMP components	hourly	2017 to 2026-03-23		§3.7
16	NYISO	palIntegrated	Integrated actual load, 11 zones	hourly	2001-06 to 2026-08	928 files, 391 MB (16 to 18)	§3.7
17	NYISO	damlbmp_zone	Day-ahead zonal LBMP (NYISO's name for LMP) with congestion and loss columns	hourly	2001-06 to 2026-08 (available from 1999-11)		§3.7
18	NYISO	realtime_zone	Real-time zonal LBMP	hourly	2001-06 to 2026-08		§3.7
19	ISO-NE	SMD hourly workbooks	Load, day-ahead and real-time LMP with components, and weather, 8 zones	hourly	2016-01 to 2026-06	10 files, 83 MB	§3.7 , §3.8
20	IESO	PUB_DemandZonal_<YYYY>.csv	Demand for the 10 Ontario zones	hourly	2003 to 2026-08	73 files, 28 MB (20 to 22)	§3.7
21	IESO	PUB_Demand_<YYYY>.csv	Ontario total demand	hourly	2002 to 2026-08		§3.7
22	IESO	PUB_PriceHOEP PredispOR_<YYYY>.csv	The Hourly Ontario Energy Price, the single province-wide price until 2025-04-30	hourly	2003-05 to 2025-04		§3.7
23	CAISO	actual load by transmission access charge (TAC) area	PG&E, SCE, SDG&E and Valley Electric areas plus the ISO total	hourly	2009-04 to 2026-08	864 files, 22 MB (with 24)	§3.7
24	CAISO	day-ahead LMP	Four utility load aggregation points and three trading hubs (NP15, SP15, ZP26), with components	hourly	2023-04-12 to 2026-08		§3.7
25	UKPN	ukpn-data-centre-demand-profiles	Utilization ratio (observed apparent power over contracted import capacity) for 96 anonymized data centers	half-hourly	2023-01-01 to 2026-05-13	5,442,348 rows (47 MB for 25 to 27)	§3.9
26	UKPN	ukpn-data-centres-by-local-authority	Count of data centers per district	snapshot		45 rows	§3.9
27	UKPN	ukpn-large-demand-list	Large demand connection projects	snapshot		496 rows	§3.9
28	ENTSO-E	actual load, document 6.1.A	19 bidding zones: Ireland (IE_CTA), the Netherlands, the seven Italian zones, Germany-Luxembourg,	15, 30 or 60-minute, parsed to hourly	2015 to 2026 (2026 excluded from analysis)	376 parquet files, 34 MB (28 to 30)	§3.7 (Italy), §3.10

#	Source	Dataset	What it is	Resolution	Window	Size (rows or files; on disk)	Used in
			France, Spain, Finland, both Danish zones, the four Swedish zones				
29	ENTSO-E	day-ahead price, document 12.1.D	Ireland (the all-island Single Electricity Market, SEM) and Italy	hourly	2015 to 2025		§3.10
30	ENTSO-E	installed solar capacity, document 14.1.A (A68)	7 zones	annual	2015 to 2025		§3.10
31	CSO (Ireland)	table MEC02	Metered electricity consumption in GWh, split into data centers, all other customers, and the total	quarterly	2015Q1 to 2025Q4	44 quarters, 8.5 KB	§3.10 , the only measured data-center dose
32	CBS StatLine (Netherlands)	data-center share of electricity consumption	The Netherlands' data-center share of electricity, used as the "dose" for the control country in the Ireland comparison (§3.10)	annual	2017 to 2024	8 values	§3.10
33	Pecan Street	Austin bundle, 25 homes	Whole-home and circuit-level power	15-minute, 1-minute, 1-second	2018 to 2019	~13 GB	§3.11
34	Pecan Street	New York bundle, 25 homes	Whole-home and circuit-level power	15-minute, 1-minute, 1-second	six months, no winter	~6.3 GB	§3.11
35	Pecan Street	California bundle, 23 homes (in fact San Diego)	Whole-home and circuit-level power	15-minute, 1-minute	2018 to 2019		§3.11
36	Pecan Street	metadata and surveys	metadata.csv, the audits and surveys, EV and weather, indoor temperature			22 GB and 37 files for 33 to 36 and 39	§3.11
37	rs-7943457 (York University and IESO)	GPU telemetry	Per-GPU power, utilization, memory, and temperature for AI-training sessions on 8-GPU H100 and B200 nodes; 8 of 32 sessions downloaded, 15 minutes each	polled every 20 ms, refreshing every ~103 ms (about 9.7 Hz)	2025	143 MB on disk	§3.12

Table 2. Datasets pulled but not used in any analysis.

#	Source	Dataset	What it is	Resolution	Window	Size (rows or files; on disk)	Why it was not used
38	PJM Data Miner 2	da_hr1_1mps	Day-ahead LMP, 11 nodes	hourly	2024-05-26 to 2026-05-10	189k rows	pulled for a day-ahead versus real-time spread analysis that was never needed
39	Pecan Street	Puerto Rico, September 2023	Per-metric files: real power, apparent power, current, harmonic distortion (the phase-angle file is broken on the server)	15-minute, 1-minute, 1-second	2023-09	counted in dataset 36	one month of data from a grid the project never studied

Source 1: PJM Data Miner 2

Data: [Data Miner 2](#); the feeds used are [rt_hrl_lmps](#), [da_hrl_lmps](#), [rt_fivemin_hrl_lmps](#), [hrl_load_metered](#), [sync_reserve_events](#), [reserve_market_results](#) and [pnode](#).

Documentation: the [PJM API portal](#) and the vendored API guide ([reference/pjm-manuals/data-miner-2-api-guide.pdf](#)).

[Back to the dataset table](#).

PJM Interconnection is the grid operator for 13 states and the District of Columbia across the Mid-Atlantic and Midwest, from Virginia and North Carolina north to New Jersey and west to Illinois. Data Miner 2 is its public data portal: a web interface at [dataminer2.pjm.com](#) and a REST API at [api.pjm.com](#), documented in the API guide (vendored at [reference/pjm-manuals/data-miner-2-api-guide.pdf](#)) and the developer portal at [apiportal.pjm.com](#). An API key is required; the free tier allows about 6 requests a minute, and anything older than a specific number of days (differs between datasets) sits in a separate "Historic" tier with fewer filtering options available. Everything pulled from this source is for the Dominion (DOM) zone: Dominion Energy's service territory in Virginia, which contains the Loudoun County data-center corridor (data center alley). Documentation of data and API constraints can be found in [sources/pjm-api-constraints.md](#) in the repo.

Two of these datasets define what a "stressed" hour is for this project. Hours are "stressed" when the reserve clearing price ([dataset 5](#)) reached \$850/MWh, the first step of PJM's reserve-shortage penalty curve, so the mechanism tests could be run separately on stressed and normal hours. The event records in [dataset 4](#) were too sparse (38 events in 3.6 years) to define stress on their own, and "any nonzero reserve price" turned out to be too lax (44% of intervals).

What went wrong.

- The load is aggregated for the whole DOM zone, not for data centers and not even for individual pricing nodes, so what any one facility is doing is essentially invisible within the larger aggregate.
- The resolution is hourly for load and at best 5-minute for prices, which is too coarse of a time scale for a project questioning volatility at the second timescale.
- The API rate limit (about 6 requests a minute) made every data pull slow.
- The data runs out past a certain date: 5-minute instantaneous load is kept for only about 30 days (`inst_load`), operational reserves for 15 days, and the node-filterable 5-minute LMP for 186 days, which is one reason the project later pivoted to [gridstatus.io](#).
- The boundary between the Standard and Historic tiers slides forward one day per day, and the Historic tier cannot filter by node, only by node subtype, so recovering the two Ashburn 35 kV nodes' history would mean downloading all ~10,786 LOAD-subtype nodes (about 94.5M rows a year) and discarding 99.98% of it. This was deliberately not done, which is why Ashburn has two years of coverage instead of 3.6.
- The type column is the node *subtype*, not its type.

Source 2: [gridstatus.io](#)

Data: [pjm_load](#), [pjm_lmp_real_time_5_min](#) and [isone_interchange_hourly](#) on [gridstatus.io](#).

Documentation: [docs.gridstatus.io](#) and the [gridstatusio client source](#).

[Back to the dataset table.](#)

gridstatus is a hosted data warehouse that republishes ISO data with deeper history than the ISOs' own portals keep. In the context of this project, it was used for PJM 5-minute data because PJM itself keeps only 186 days for LMP and 30 days for load. Like Data Miner 2, an API key is required; since each key is limited to 500k rows and 250 requests a month, this project rotated between six free-tier keys. However, due to the volume of data pulled, the quota still ran out, and the ISO-NE pull ([dataset 9](#)) stopped after one week. Constraints are in `sources/gridstatus-api-constraints.md`.

What went wrong.

- The same two problems as PJM: nothing finer than 5 minutes and nothing at the data-center level for load.
- The rate limits were once again a pain: 500k rows and 250 requests a month per free key meant creating two more Google accounts and two Yahoo accounts to get to six keys, and the ISO-NE pull still ran out after a week.
- `pjm_load.dom` is empirically PJM's Southern-Region aggregate, not a strict DOM series.
- 0.1 to 0.4% of intervals are missing upstream and cannot be recovered, and a naive difference across a 5h50m hole reads as a -4,933 MW step, larger than anything real. Therefore, every gradient in the project is only computed between adjacent intervals and gap-masked.

Source 3: ERCOT

Data: [hourly load by weather zone](#) and [report 13061, historical real-time load zone and hub prices](#).

Documentation: the [ERCOT data product catalog](#) and [developer portal](#).

[Back to the dataset table.](#)

ERCOT (the Electric Reliability Council of Texas) is the independent grid operator for most of Texas. Its archives are public on `ercot.com` with no key and no quota: the hourly load archive by weather zone and report 13061, the real-time settlement point prices for load zones and hubs. Earlier load years were skipped because they use different zone names and file formats. More information is located at `sources/availability/ercot-data-availability-research.md`.

What went wrong.

- Hourly only, and load by weather zone, so once again the time granularity is too coarse, and there is no data-center-specific load.
- `Native_Load_2026.xlsx` republishes all of May 2026 as 744 duplicate rows, a publisher-side defect.
- The Panhandle hub price (`HB_PAN`) is negative in 20.4% of hours because of wind, so West-zone correlations are uninterpretable.

Source 4: MISO

Data: [MISO market reports](#); the files themselves are served from `docs.misoenergy.org/marketreports/`.

Documentation: the report descriptions on the same page, and `sources/availability/miso-data-availability-research.md`.

[Back to the dataset table.](#)

MISO is the Midcontinent ISO, covering 15 central US states and Manitoba. Its market reports are public at `docs.misoenergy.org/marketreports` with no key. Retention is the current year plus three calendar years, which is what set the 2023 start; each `df_a1` file carries actual load for only one day, so every market day needs its own file. Memo: `sources/availability/miso-data-availability-research.md`.

What went wrong.

- The same problems as before: hourly and by local resource zone (LRZ, MISO's six planning regions), with nothing at the data-center level.
- MISO publishes no zonal price, so each LRZ had to be paired with a hub, and LRZ3_5 and LRZ4 both map to ILLINOIS.HUB. MISO's 36 of 36 result in [§3.7](#) is therefore slightly inflated, since two of its cells are not independent.

Source 5: SPP

Data: [hourly load](#) and [day-ahead LMP by settlement location](#) on the [SPP file portal](#).

Documentation: the portal's page for each file set.

[Back to the dataset table.](#)

SPP stands for the Southwest Power Pool, 14 states in the central US. Its file portal (`portal.spp.org`) is public with no key: one annual zip per year through 2024 (about 2.5 GB), then daily CSVs from 2025 (1.6 GB more). Memo: `sources/availability/spp-data-availability-research.md`.

The panel stops on 2026-03-23 on purpose: the next day the load file's layout and the zone roster both change (SPP's new RTO West region joins).

What went wrong.

- Hourly and by control zone, like the rest, with no data-center load.
- SPP has no zonal price either. The zone price used is an unweighted mean of the nodal LMPs whose names match the zone prefix, which matches 64.5 to 75.1% of nodes.
- The 2016 price zip mixes two naming families.

Source 6: NYISO

Data: [palIntegrated \(report P-58B\)](#), [damlbmp_zone \(P-2A\)](#) and [realtime_zone \(P-24A\)](#).

Documentation: [NYISO energy market and operational data](#).

[Back to the dataset table.](#)

NYISO is the New York ISO. Its monthly CSV archives are public at `mis.nyiso.com` with no key. Memo: `sources/availability/nyiso-data-availability-research.md`.

What went wrong.

- Hourly and zonal, like the rest.
- The zone roster changed on 2005-01-31, when New York City and Long Island stopped being reported as one combined zone. Therefore, the market ships as two panels (a merged 10-zone roster that spans the whole archive, and a split 11-zone roster from 2005 on) that are not interchangeable.

Source 7: ISO-NE

Data: the [SMD hourly workbooks](#).

Documentation: the [pricing node tables](#) and [ISO-NE web services](#).

[Back to the dataset table](#).

ISO-NE stands for ISO New England, the project's control market due to its little data-center development: by the ISO's own statement, New England "has not experienced similar growth so far, and only a small amount is expected in the coming decade." Its "SMD hourly" annual workbooks (SMD is Standard Market Design, the 2003 market redesign whose name the public hourly data files still carry) are public static files on `iso-ne.com`, one per year, no key. Memo:

`sources/availability/isone-data-availability-research.md`.

What went wrong.

- Hourly and zonal, like the rest.
- The workbook switches its time convention in 2024 (checked: hour labels line up across the break).
- 2026 is a half-year due to when this project took place.
- Separately, EIA-930's ISNE → HQT interchange series omits the NECEC line entirely (EIA is the US Energy Information Administration; EIA-930 is its hourly grid monitor; NECEC is the New England Clean Energy Connect line, [§3.8](#)).

Source 8: IESO

Data: [DemandZonal](#), [Demand](#) and [PriceHOEPPredispOR](#).

Documentation: the [IESO data directory](#).

[Back to the dataset table](#).

IESO (the Independent Electricity System Operator), Ontario's grid operator, was the first look outside of the US. Its annual CSVs are in a public directory at `reports-public.ieso.ca`, no key. Also note that Ontario moved from a single province-wide price (HOEP, the Hourly Ontario Energy Price) to nodal prices on 2025-04-30. Memo: `sources/availability/ieso-data-availability-research.md`.

What went wrong.

- Hourly and zonal, like the rest.
- After the market change, prices are published in a roughly 90-day rolling window, and no collector was set up, so the price series ends in April 2025.

Source 9: CAISO

Data: [CAISO OASIS](#).

Documentation: the [OASIS interface specification](#), which defines the `SingleZip` query the `fetch` script uses.

[Back to the dataset table](#).

CAISO is the California ISO. Its OASIS (Open Access Same-time Information System) API (`oasis.caiso.com`) needs no key but has informal rate limits, so the `fetch` works in 28-day chunks with a 6-second pause (note that there may be a more efficient way to gather data; this is just what this project

found to work quickly). Although prices were requested as far back as 2009, OASIS only returns them from 2023-04-12, so the market ships as two panels (full depth and modern). Memo: [sources/availability/caiso-data-availability-research.md](#).

What went wrong.

- Hourly and by utility area, like the rest.
- An initial probe claimed 2010 price depth; the PRC_LMP archive actually starts on 2023-04-12.
- Silicon Valley's data-center cluster is invisible since Santa Clara is served by a municipal utility that has no load area in CAISO's data.

Source 10: UK Power Networks (UKPN)

Data: [ukpn-data-centre-demand-profiles](#), [ukpn-data-centres-by-local-authority](#) and [ukpn-large-demand-list](#).

Documentation: the portal's [API console](#).

[Back to the dataset table](#).

UKPN is the distribution network operator for London, the East, and the South East of England. It runs the local wires rather than the transmission grid, which is why it meters individual large customers. Its open-data portal ([ukpowernetworks.opendatasoft.com](#), built on Opendatasoft) publishes anonymized half-hourly profiles for the data centers connected to its network. A free API key is required; the quota is 100k calls a day, and an export counts as one call regardless of size, so the rate limit is effectively infinite, a breath of fresh air after PJM and gridstatus. Constraints: [sources/ukpn-api-constraints.md](#).

What went wrong.

- The large-demand list cannot be filtered to data centers, and there is no MW, no location, and no price: shapes only.
- Half-hourly resolution is far too coarse for the sub-second question, even though this is the one source with facility-level data.
- The utilization ratio is not bounded between 0 and 1 (it runs from 0 to 3.992).
- `local_timestamp` is UTC despite its name (verified across seven daylight-saving boundaries).
- 13.1% of rows are exact zeros across 69 of the 96 sites, four sites are dead for the whole period, and about 1,700 short dropouts manufacture fake full-scale ramps that would corrupt any volatility statistic.
- Site #67 has a 5.4× structural break: its mean utilization ratio is 1.396 before 2025-11-27 and 0.260 after, and it sits above 1.0 (that is, reporting more power than its contracted capacity) for 86% of its span. Running above contracted capacity continuously for nearly three years is not physically plausible, so the likely explanation is that the contracted capacity on file was stale and was revised on that date. Either way, any statistic pooled across the site's whole 3.4-year span is not interpretable.

Source 11: ENTSO-E Transparency Platform

Data: the [ENTSO-E Transparency Platform](#).

Documentation: the [RESTful API sitemap](#) and [how to get a security token](#).

[Back to the dataset table](#).

ENTSO-E is the association of Europe's transmission system operators, and its Transparency Platform (transparency.entsoe.eu) is their shared data platform. Its REST API needs a free token, which took three working days to obtain; the limit is 400 requests a minute per token, and a breach is allegedly a short-term ban. The vendor's documentation is a Zendesk help center plus a Postman collection; what matters from them is summarized in [sources/entsoe-api-constraints.md](#) and [sources/entsoe-endpoint-reference.md](#).

What went wrong.

- National and bidding-zone aggregates at 15 to 60 minutes, so the same aggregation and granularity problems as every ISO source. (A bidding zone is the area within which Europe's day-ahead market treats electricity as freely tradable at one price; most countries are one zone, Italy has seven.)
- Irish load lives on the CTA (control area) identifier, which covers the Republic only, not on SEM (all-island, about 1,015 MW higher), which voided an earlier "footprint" objection to using Ireland; price is the opposite (CTA returns "no data" on price).
- Resolution is per document (NL 15-minute, IE 30-minute since 2015).
- The platform's compressed "A03" curve format makes a naive one-row-per-point parse read spikier than reality.
- Price documents come back as whole days in the area's time zone, so year-boundary days arrive twice.
- The Irish feed omits the daylight-saving fall-back hour, and 3,774 of 203,604 slots are missing (1.85%, in 661 runs).
- Metered solar (document A75) is invisible to the transmission operator for rooftop PV (photovoltaic panels; the Netherlands: 204 MW metered against 27,980 MW installed), so the solar dose has to be installed capacity (A68), and even A68's coverage runs from about 100% of national PV in the Netherlands to about 3% in Finland.
- Annual capacity documents start at local midnight expressed in UTC, which is 31 December of the previous year, so every zone-year was initially joined to the following year's capacity.

Source 12: CSO (Ireland's Central Statistics Office)

Data: [table MEC02](#), or the [JSON-stat API call](#) that returns it.

Documentation: the [PxStat API wiki](#).

[Back to the dataset table.](#)

Ireland's Central Statistics Office (CSO) is the national statistics agency. Since 2015, it has published a quarterly table, MEC02, of metered electricity consumption split into data centers, all other customers, and the total, built from the meter readings the electricity networks report to it. The table is served through PxStat, the CSO's open statistics system, as a JSON-stat API at [ws.cso.ie](#) that needs no registration: one HTTP request returns the whole table.

What went wrong.

- It is a national quarterly total, so it says how much data centers consume, not when or how fast.
- The CSO has no data-center classification of its own. Sites are identified heuristically (name matching, business parks, meters above 1 GWh a year), and the CSO itself warns that new small sites may fall below its thresholds.

Source 13: CBS StatLine (Statistics Netherlands)

Data: the [CBS custom table](#).

Documentation: [StatLine open data](#).

[Back to the dataset table](#).

Statistics Netherlands (CBS) is the Dutch national statistics office, and StatLine is its open statistics database, with no key. The data-center series, electricity delivered to data centers from 2017 to 2024, is a custom table that CBS publishes as a downloadable file rather than a standing StatLine series: annual, national only, with no regional breakdown, no size classes and no facility list.

What went wrong.

- Annual national totals, so coarser still than the CSO's.
- The CSO and CBS definitions differ (CSO's heuristic against CBS's "main activity" classification), so the two countries' *levels* are only roughly comparable.

Source 14: Pecan Street Dataport

Data: [Pecan Street Dataport](#); the bulk route is the [JupyterHub](#) contents API.

Documentation: the vendored catalog and data dictionary ([reference/pecan-street/](#)) and [dataport.pecanstreet.org](#).

[Back to the dataset table](#).

Pecan Street is a research nonprofit that instruments volunteer homes with circuit-level eGauge monitors (meters that record each circuit in the home separately), mostly in Austin. The free University tier gives no database access; the bulk route is the JupyterHub contents API (`jupyterhub.pecanstreet.org`), which `scripts/pecanstreet_fetch.py` drives with a JupyterHub token. The vendor's catalog and data dictionary are vendored in `reference/pecan-street/`; access mechanics and traps are in `sources/pecanstreet-access-constraints.md` and the docstring of `scripts/pecanstreet_lib.py`.

What went wrong.

- Whole-home use is not a column and had to be reconstructed as `grid + solar + solar2`.
- The dictionary's `use` and `gen` columns do not exist in `metadata.csv`, and row 2 of that file is a dictionary row, not a home.
- The "California" bundle is San Diego homes stamped in Central time.
- 136 of the 1,198 homes with 1-minute data lack the `grid` column.
- Heavy intervention programs contaminate many homes. Pecan Street runs experiments on its volunteers, and the metadata flags the homes that took part: a CCET pricing trial (time-varying electricity prices), a Verizon low-income apartment program, an LG appliance swap, the SHINES solar-plus-storage demonstration, and Civita text-message nudges. A home in one of these programs is not a natural observation of household demand, because the program itself changed how and when the home used power, so analysis was rerun without those homes as a robustness check ([§3.11](#)).
- Austin home 7536 emits a 5,308.7 kW telemetry fault with ± 1.15 MV leg voltages (filtered out).

- The Austin 1-second files zero-fill gaps in 18 of 25 homes, so a resumption after a gap reads as a ~23 kW step that no plausibility filter can catch; it corrupts maximum-based statistics only.
- No electrical service size is recorded anywhere in the free tier (i.e. whether a home has a 200 A panel, a 150 A panel, or something else).

Source 15: rs-7943457 GPU Telemetry

Data: the [GitHub dataset](#).

Documentation: the [preprint rs-7943457](#).

[Back to the dataset table](#).

This source contains per-GPU power telemetry from real AI training runs, recorded at York University on two 8-GPU servers (one with NVIDIA H100 cards, one with B200 cards) by reading each card's built-in power sensor every 20 ms. The 32 sessions cover fine-tuning of 1B to 8B parameter language models and diffusion-model training, about 15 minutes each, with per-GPU power, utilization, memory, and temperature all recorded. It is the dataset behind the Research Square preprint rs-7943457 (Elsayed, Al-Obaidi and Farag, York University and Ontario's grid operator IESO) and was published alongside it on GitHub under CC BY 4.0 (a license that allows reuse with attribution). It is the only source in the project that is fast enough to see the second-scale cycle (other than the residential data from Pecan Street), at the cost of being only one server node rather than a full, hyperscale facility.

What went wrong.

- The file says 50 Hz, but 80.7% of consecutive samples are identical, so the sensor refreshes every ~103 ms. The instrument is mute above about 5 Hz by construction.
- The repository's "sequence length" sweep (seq1024, 2048, 4096) has identical memory use (within 0.01%) and power (within 0.1%) across a claimed 4× range, so those are not independent runs; because of this, two files were dropped.

Assembled Tables

Because most upstream data sources aren't clean, every analysis reads a single cleaned table rather than the raw files. These live in `data/interim/` and are built by the preprocessing code from the raw pulls: the raw files are joined, put on a common clock, gap-masked, and given the derived columns (load gradient, cluster means, event flags) the tests need. The main ones are the hourly DOM panel (31,536 rows by 23 columns), the 5-minute DOM panels (the extended one at 351k rows and the one-year two-sided one at 105k), one Stage-1 panel per market (from MISO's 31.6k rows to NYISO's 220k), the Irish, Dutch, and Italian hourly series, the UKPN site tables, and the Pecan Street reconstructed whole-home series.

3. Analyses and Findings

Each subsection describes what question was asked, what data it ran on, and what the results/findings were. Numbers in square brackets are 95% bootstrap confidence intervals unless stated otherwise: a bootstrap re-estimates a statistic on many random resamples of the data, and the interval brackets the middle 95% of those re-estimates, so it says how much the number could move by chance.

3.1 PJM/DOM Hourly: Answering the Original Proposal's Question

The question. The proposal's idea was that fast swings in data-center load push Loudoun's congestion prices into a different regime once the swings get big enough.

Figure 2 shows where the Dominion zone sits inside PJM. PJM is divided into transmission zones, one per transmission-owning utility; Dominion's zone covers most of Virginia and a corner of North Carolina, and every PJM pricing node in this record is inside it.

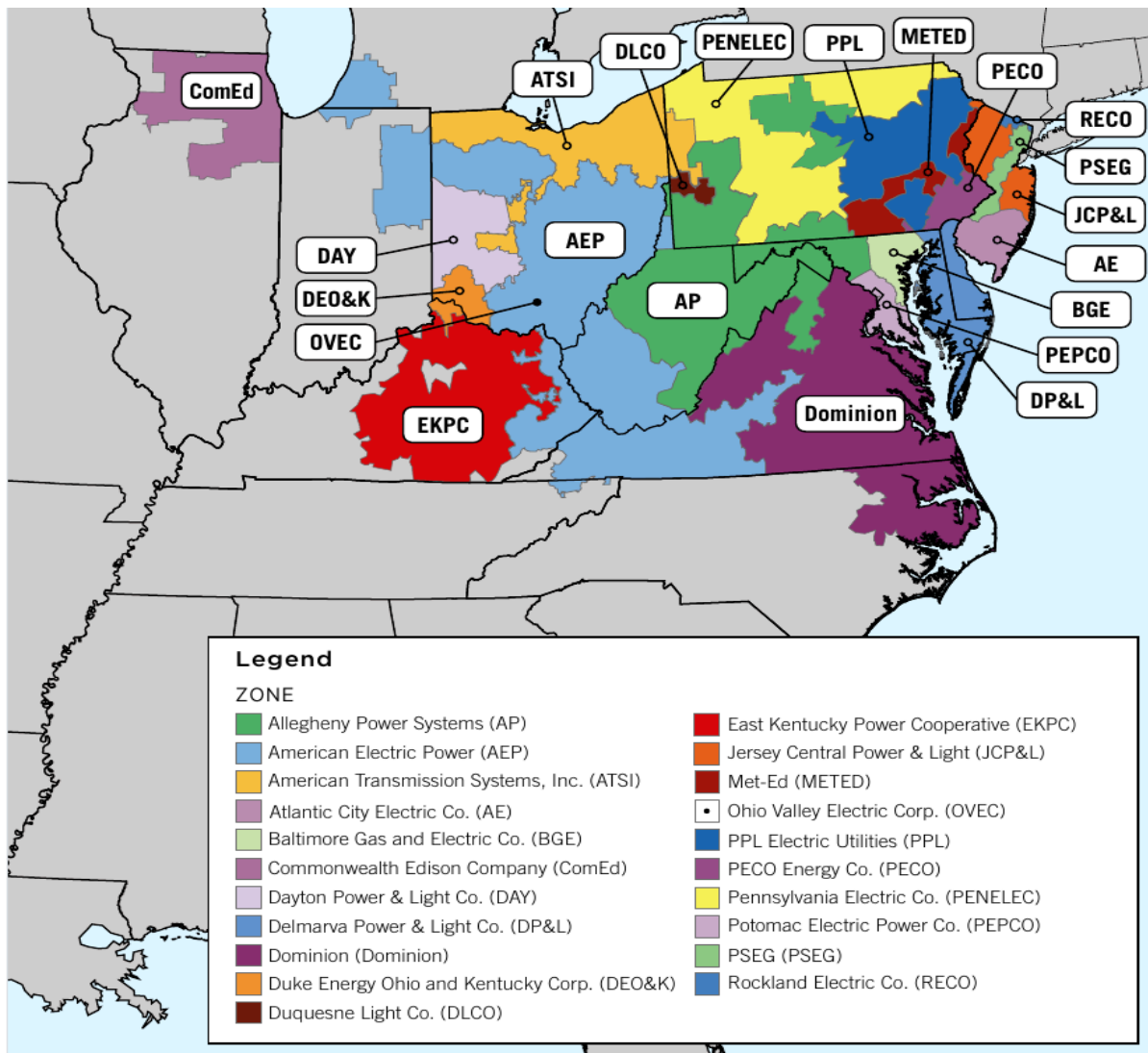


Figure 2. PJM's transmission zones, from PJM's own zone map (dated 2023-05-11) [106]. Dominion (purple, lower right) is the DOM zone of this record; Loudoun County sits at its northern tip, just west of Washington, DC.

The data. An hourly panel from 2022-10-02 to 2026-05, 31,536 hours. The price variable is the congestion component of the locational marginal price (LMP), averaged over the six 500 kV nodes in the Loudoun cluster (LOUDOUN, PLEASANT VIEW, GOOSECRE, BRAMBLET, MOSBY, and SKFFSCRK). After an initial round of analysis, several tests were repeated on total LMP as opposed to just the congestion price component. The volatility variable, called Z throughout, is how fast DOM load changes, measured as the absolute hour-over-hour change in MW per minute. Two 500 kV nodes outside the cluster (OX and BRISTERS) and the DOM zonal price served as controls, and the two 35 kV Ashburn nodes (data from May 2024) served as the distribution-side comparison. After filtering for the original proposal's isolation hours (shoulder months, 2 to 5 AM), only 2,027 hours remain. However, within those hours, very few high-MW events remained, so most tests were run on the full, unfiltered panel instead.

Figure 3 shows the two series over the whole panel, before any statistics are run on them. DOM load (top) has a strong seasonal shape, with summer and winter peaks near 24 GW and spring and fall troughs near 10 GW, and a daily cycle too fast to resolve at this scale. The cluster price (bottom) sits in a \$20 to \$60 band most of the time, with sharp spikes above \$1,000 that last a few hours. The congestion component is near zero most of the time, jumps when a transmission limit binds, and is occasionally negative, which [§3.5](#) explains.

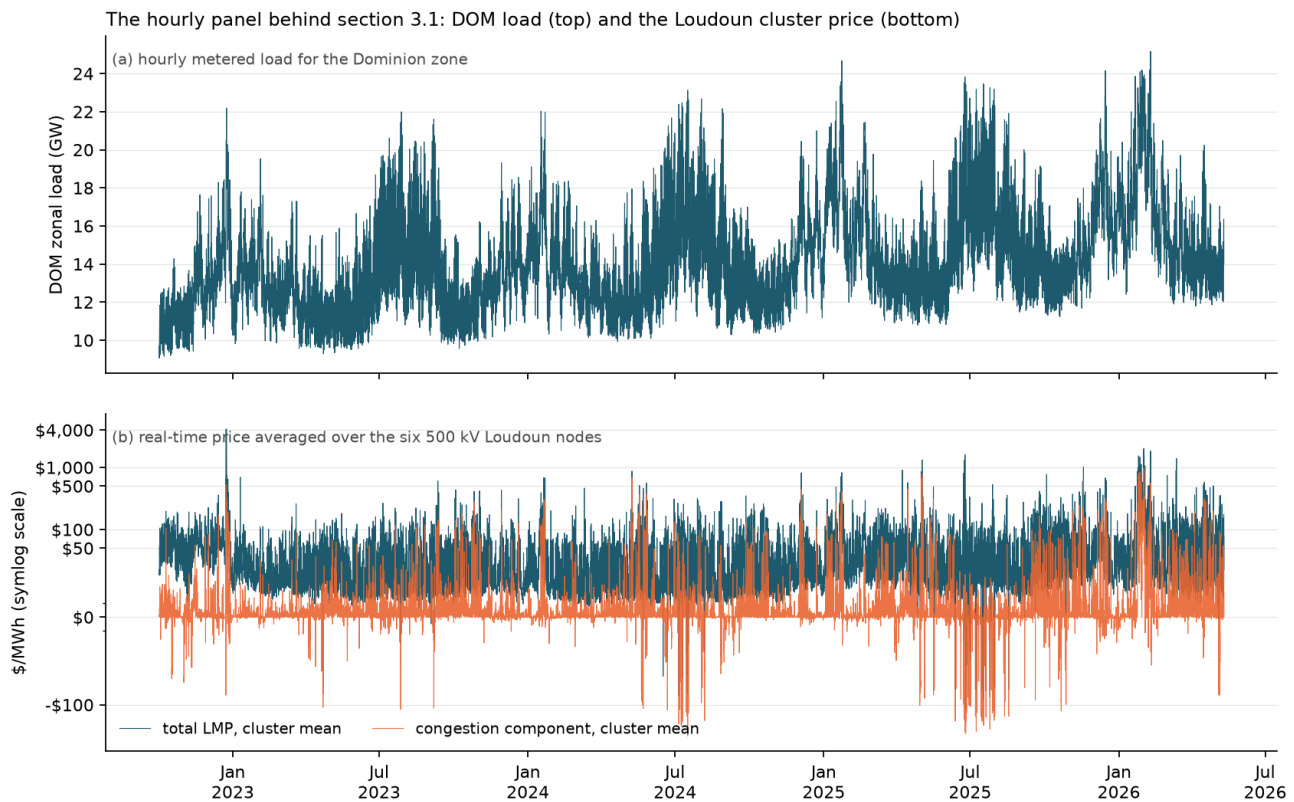


Figure 3. The hourly panel behind this section: DOM zonal load (top) and the real-time price averaged over the six Loudoun nodes (bottom), 2022-10-02 to 2026-05-10. The price axis is a symmetric log scale (linear between $-\$50$ and $\$50$, logarithmic beyond), so both the everyday \$20 to \$60 range and the \$1,000-plus spikes are visible at once. Drawn by `scripts/plot_hourly_series.py`.

Is there a threshold? Threshold regression (TAR, Hansen's method) fits two separate regimes on either side of a cut in Z and searches for the cut that fits best. It put the cut at 4.3927 MW/min on the cluster and on both control nodes alike, with the bootstrap p-value at its floor. However, it turned out that the response

curve is smooth and monotone (it rises steadily, with no kink), and the estimator was just finding the steepest point of a smooth curve. Moreover, as design choices such as the filter, resolution, and definition of Z changed, the estimated cut also moved. Thus, it was determined that there is no single MW/min threshold, and that the original proposal's question was a dead end.

How much does the top of the price distribution move with Z? Quantile regression estimates how a chosen percentile of prices, rather than the average, responds to Z. At the 90th, 95th, and 99th percentiles ($\tau = 0.90, 0.95, 0.99$, where τ is the percentile written as a fraction) the congestion slopes were 0.393 [0.325, 0.462], 0.578 [0.428, 0.761] and 0.358 [-0.075, 1.194] dollars per MWh per MW/min of Z. On the total LMP at $\tau = 0.95$, the slope was 2.334 [1.85, 2.73], about four times the congestion slope, because PJM's reserve-shortage penalty lands in the system energy component. However, there was nothing Loudoun-specific in it: the cluster, OX, and BRISTERS all sat at 0.57 to 0.61.

How heavy is the price tail? A generalized Pareto distribution (GPD) is fitted to the prices above a high threshold; its shape parameter ξ says how heavy the tail is (larger is heavier, zero is exponential). On the cluster's total LMP, $\xi = 0.851, 0.706, 0.275$, and 0.024 at the 90th, 95th, 99th, and 99.5th percentile thresholds: a heavy tail that becomes essentially exponential at the extreme.

Does high volatility make the tail heavier? (the proposal's mechanism) Exceedances above the 95th percentile were split into a high-Z half and a low-Z half, and then their two tail shapes were compared. The difference was -0.180 [-0.371, -0.044] ($n = 789$ per half, congestion): the high-volatility half has the *lighter* tail, the opposite of the proposal's prediction.

Tail-risk curves inside the filter. Figure 4 plots the probability that the LMP exceeds \$X given the decile of Z (the hours sorted into ten equal groups from the calmest tenth to the most volatile tenth), for 10 deciles, 5 thresholds and 7 nodes, inside the proposal's own filter ($n = 2,027$). The probability is exactly 0.000 at \$250, \$500, \$1,000 and \$2,000 in every decile and at every node, and only 10 hours exceed \$100. The filter that isolates the volatility signal removes the very price events the proposal was about.

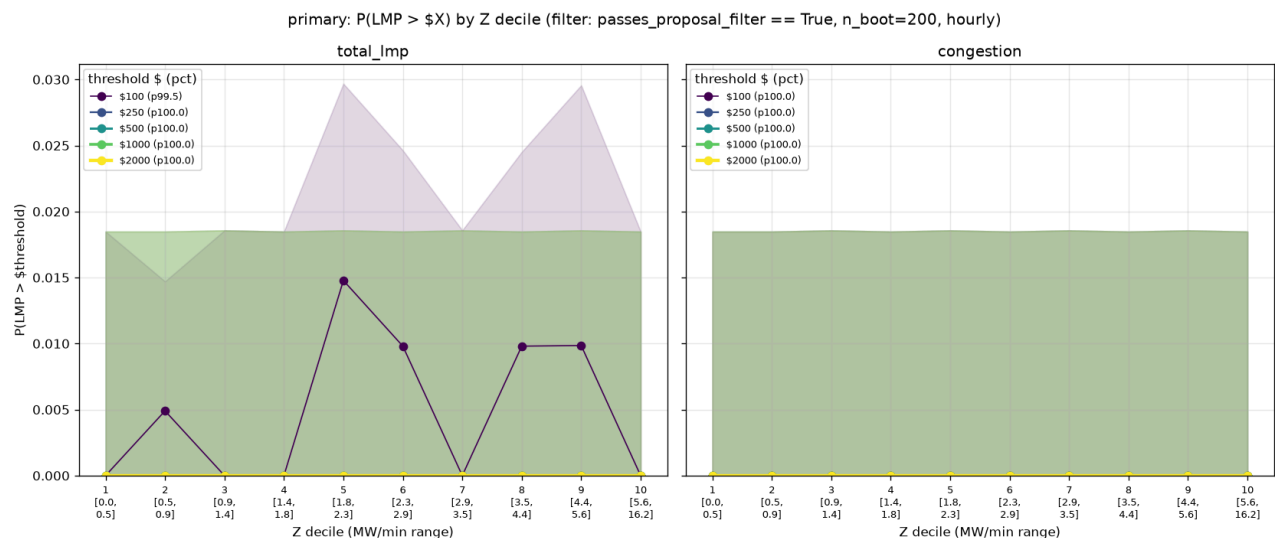


Figure 4. Probability of LMP exceeding \$X by decile of Z inside the proposal's filter, Loudoun cluster. Each line is a price threshold, and the horizontal axis runs from the calmest tenth of hours (decile 1) to the most volatile tenth (decile 10). Every line above \$100 is flat at zero: inside the filter there are no large price events left to explain.

Tail-risk curves on the full panel. Figure 5 repeats the exercise on the full hourly panel. The top decile of Z runs 13.4 to 36.4 MW/min. The probability that total LMP exceeds \$250 given the top decile is 0.019 on the cluster and 0.047 at Ashburn TX1; the congestion ratio between the top and bottom deciles is about 1.3 (0.7 on the DOM zonal price).

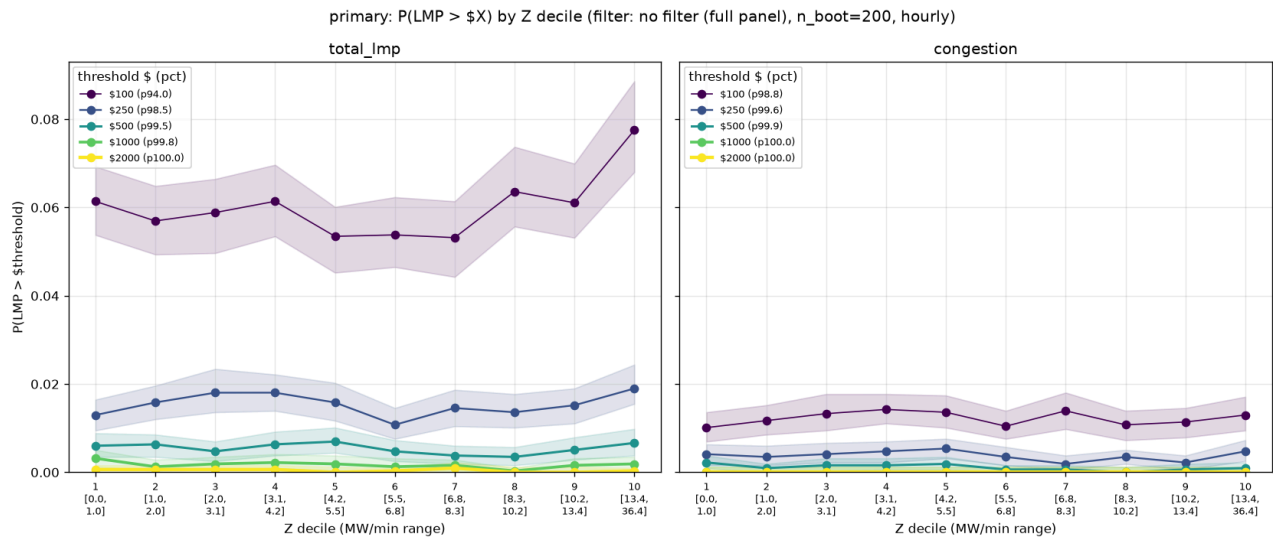


Figure 5. The same curves on the full hourly panel (no filter, 200 bootstrap resamples), Loudoun cluster: total LMP (left) and congestion (right). The shaded bands are the bootstrap intervals. Flat across deciles at every threshold, with a small lift in the top decile on the total LMP only.

Additionally, more tests were run and left out of this record as more detail than the question needed. Each is written up in full in `decisions.md`.

- A year-effects split of the quantile slopes.
- A robustness battery on the tail-shape comparison.
- Spec B, a continuous version of the tail-shape test.
- The same test on the three price components separately.

Where things stood after the hourly analysis. The honest summary was that progress had been very slow, and that three problems were on the table.

- Resolution: the question is about volatility, so 5-minute data makes more sense than hourly. However, PJM keeps 5-minute load for only 30 days (six months in its archive) and publishes no load at individual nodes at all, only the DOM aggregate; the fix was to switch to gridstatus.io, whose 5-minute DOM load dates back to February 2023.
- The filter: inside the proposal's isolation hours, the maximum LMP was \$177/MWh and the average about \$25, while outside it the maximum was above \$4,000. The filter was removing exactly the high-price events the question was about, so the filter was then dropped.
- The premise: the non-linear phase transition came from the literature, and the smooth curve above suggested it might simply not exist in PJM.

Full record: `decisions.md` 2026-05-11, 05-13, 05-14 (Strategy C, conditional-Z, Spec B, #2, #3, #4) and 05-15 (#6, #9).

3.2 Five-Minute Companions and the Extended Panel

Zooming in to 5-minute data, still from PJM Data Miner 2. The only 5-minute DOM load held by Data Miner 2 was PJM's 30-day archive, so the joint 5-minute load and price window was 27.2 days (6,855 five-minute stamps, 845 inside the proposal's filter). Comparing six months of 5-minute LMP (582,780 rows, 11 nodes) with the published hourly LMP produced a standalone finding: hourly publication hides a large share of 5-minute price spikes, and the hidden share *rises* with severity (Table 3). A spike is "hidden" when a 5-minute price exceeded the threshold but the published hourly average for that hour did not.

Table 3. Five-minute price spikes hidden by hourly publication (PLEASANT VIEW, 2025-11-12 to 2026-05-14).

Threshold	5-min intervals above it	Hour-buckets containing one	Hidden by the hourly value	Hidden fraction
\$50	17,293	2,448	839	0.343
\$100	7,950	1,329	604	0.454
\$250	3,932	739	428	0.579
\$500	1,539	279	162	0.581
\$1,000	413	81	48	0.593

The two-sided 5-minute test, on gridstatus.io 5-minute data. One year of gridstatus.io data (2025-06-24 to 2026-06-24, 105,019 rows, three nodes), with three pre-registered tests: the quantile regression at $\tau = 0.95$, the tail-shape median split, and the decile curves. "Pre-registered" means the test, its data window, and the rule for reading its results were written into decisions.md before looking at the data, so the outcome could not be steered after the fact. Surprisingly, all three came back weaker than hourly: the quantile regression slopes at $\tau = 0.95$ all span zero (cluster congestion -0.0031 $[-0.0295, +0.0007]$); the tail-shape median split was -0.034 $[-0.100, +0.033]$, the same direction as hourly; and the decile curves were flat (congestion top-to-bottom decile ratio 1.0). From this analysis, there was no confirmatory evidence for a volatility-to-tail mechanism at native resolution.

The extended 5-minute panel (still gridstatus.io, but pulled further back: February 2023 to June 2026, about 351k rows) is the panel behind the figure set below and behind the July reframe. Four findings came out of it, and each has its figure.

The premise is weak. DOM load grew 28.0% from 2023 to 2026 (full years; the first draft's 21.5% compared a half-year with a full one). The 90th percentile of the 5-minute ramp moved from 21.91 to 28.38 MW/min, with a trend p-value of 0.158 (not significant), and normalized by the load of the day, it *fell* every year, from 0.1850% to 0.1596%. A typical 5-minute ramp moves 0.32% of zonal load. Figure 6 shows the three series together.

Figure 6 — The premise: load grew, volatility did not

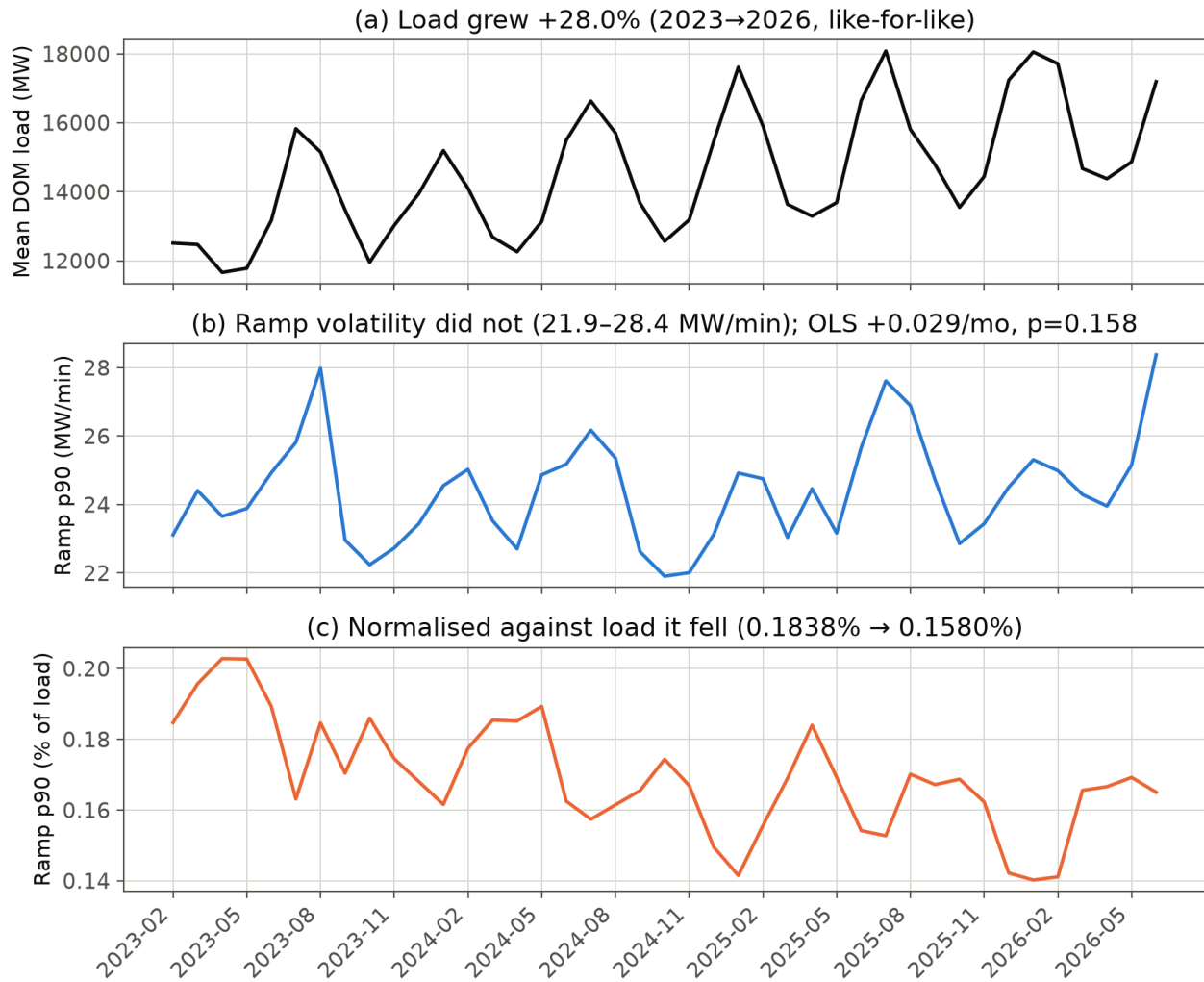


Figure 6. The premise. Monthly mean DOM load (rising), the 90th percentile of the 5-minute ramp in MW/min (flat; trend $p = 0.158$), and the same ramp as a share of load (falling by year, 0.1850% to 0.1596%). If data-center growth were making load spikier, the ramp panels would rise with the load panel; they do not.

Congestion is driven by load level. Intervals with congestion above \$500 sit at the 99.1st percentile of load but only the 45.9th percentile of ramp; the correlation of congestion with load is +0.188 and with the absolute ramp +0.008. By load decile, the system energy price rises smoothly from \$17.22 to \$61.80; congestion behaves like a switch, with a median of about \$0.30 for nine deciles and a 95th percentile that jumps from \$8.14 to \$254.36 in the tenth. Holding load fixed, the 95th percentile of congestion *falls* across ramp deciles, from \$103.39 to \$67.71. Figure 7 is the picture of this, and the centerpiece of the whole record.

Figure 7 — Congestion tracks load level, not ramp volatility

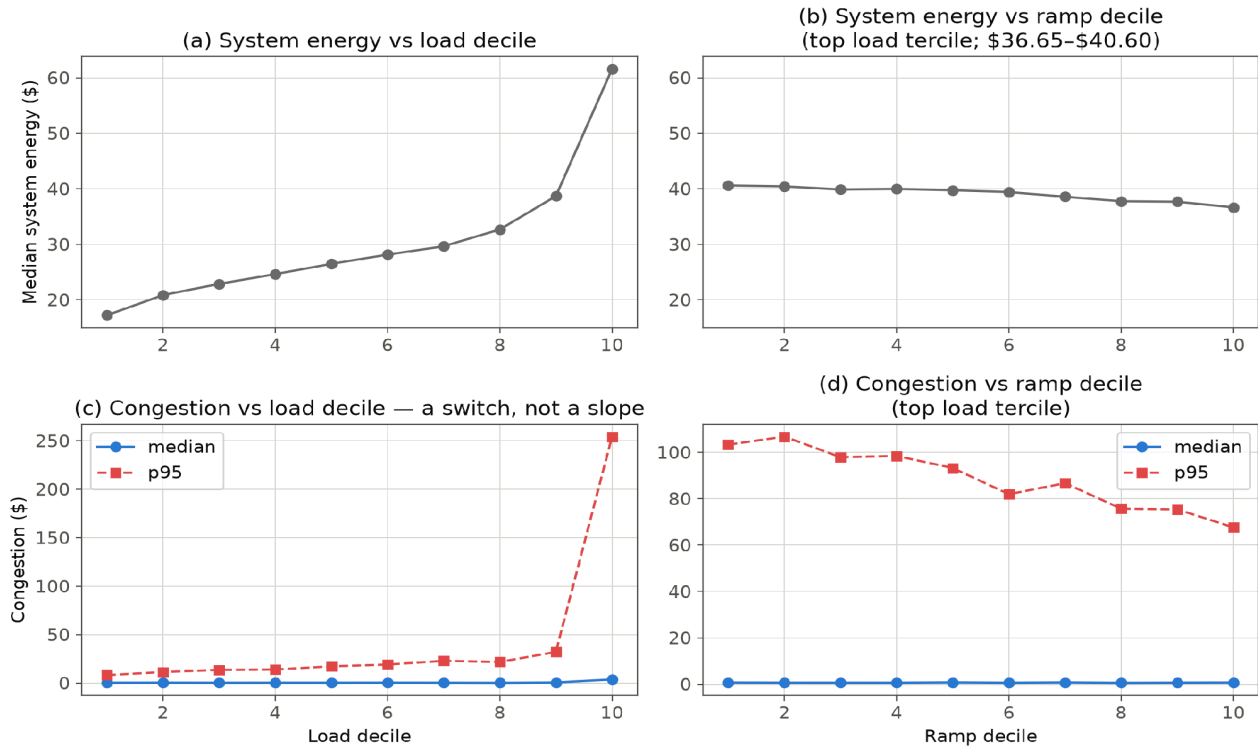


Figure 7. The centerpiece. Left: system energy and congestion by load decile (the intervals sorted into ten equal groups from lowest to highest load). Right: the same by ramp decile within the top third of load, so that load level is held roughly fixed. System energy rises smoothly with load; congestion switches on in the tenth load decile; neither responds to ramps.

The effect size is small. Across the whole observed range of Z (about 29.3 MW/min), the $\tau = 0.95$ slope implies a shift in the 95th-percentile congestion price of \$1.91 pooled, or \$4.44 in 2025, against the \$71.20 it would take to reach \$100. Above $\tau \approx 0.97$, nothing is measurable. Figure 8 shows the implied shift at every quantile.

Figure 8 — Effect size across quantiles, both specifications

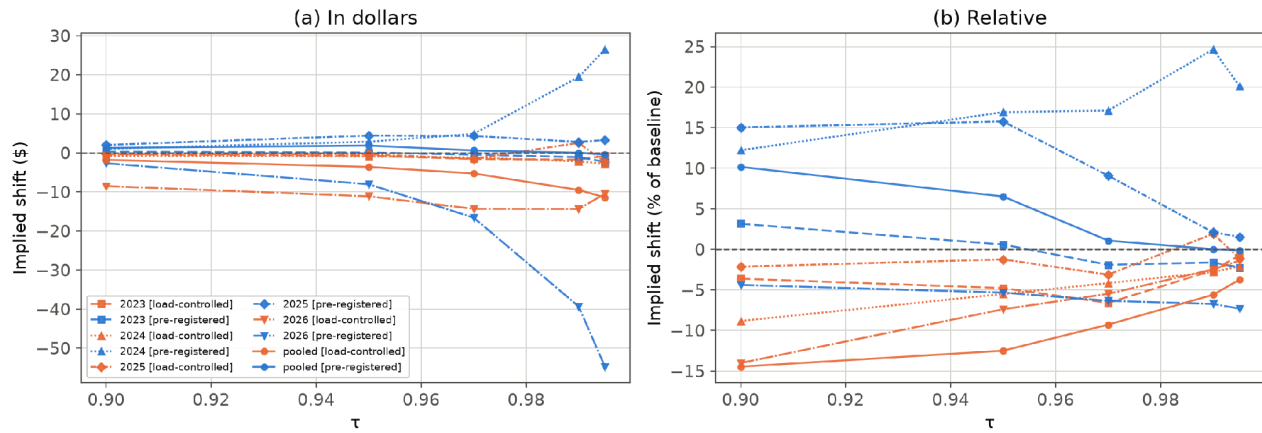


Figure 8. The congestion shift implied across the full range of Z , by quantile τ , in dollars (left) and as a percentage of the baseline quantile (right). The shaded bands are bootstrap intervals; where they cross zero, nothing is measurable.

The no-filter tail risk (Figure 9). With 1,000 bootstrap resamples over 351,371 rows, the probability of congestion above \$100 by decile of Z is flat at 1.78 to 1.95%, top-to-bottom ratio 0.9797. The test resolves $\pm 19\%$ on that ratio against a predicted 2 to 5% lift, so this is a non-result, not a refutation.

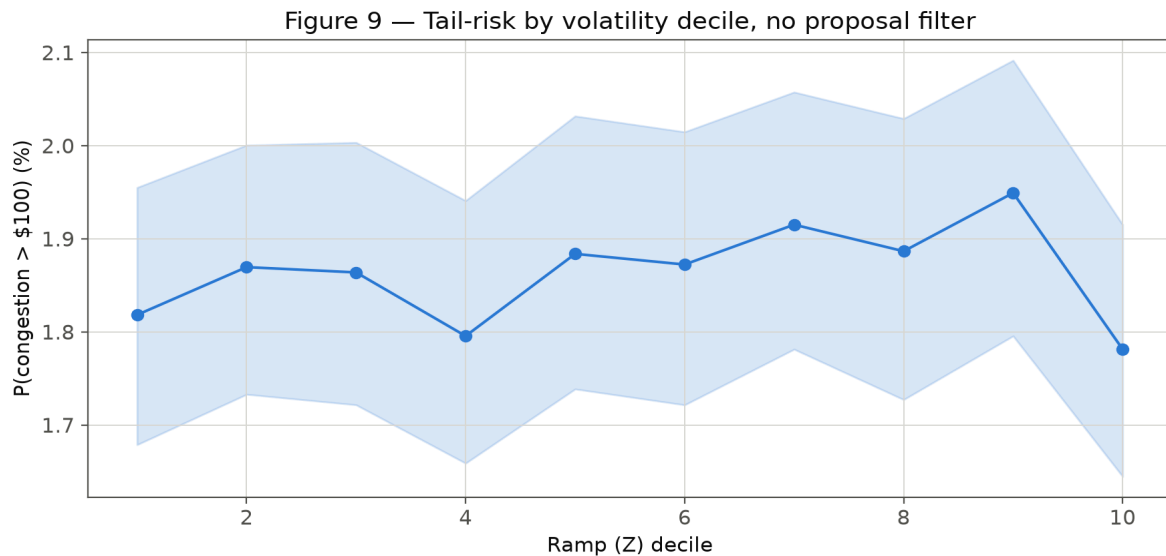


Figure 9. Probability of congestion above \$X by decile of Z, unfiltered 5-minute panel.

Figure 10 is the fuller version of Figure 9: the same decile curves for the Loudoun cluster mean at all five dollar thresholds (\$100, \$250, \$500, \$1,000 and \$2,000), pooled over the whole February 2023 to June 2026 window, for the total LMP (left) and the congestion component (right). It is included because Figure 9 shows the \$100 threshold in detail, and whether the flatness holds at the larger thresholds, where the proposal's phase transition would have lived, deserves its own look. It does: every curve is flat across the deciles of Z, within its bootstrap band, and the higher thresholds are simply rarer (the \$1,000 line sits near 0.2% of intervals and the \$2,000 line on the axis), not more volatility-dependent.

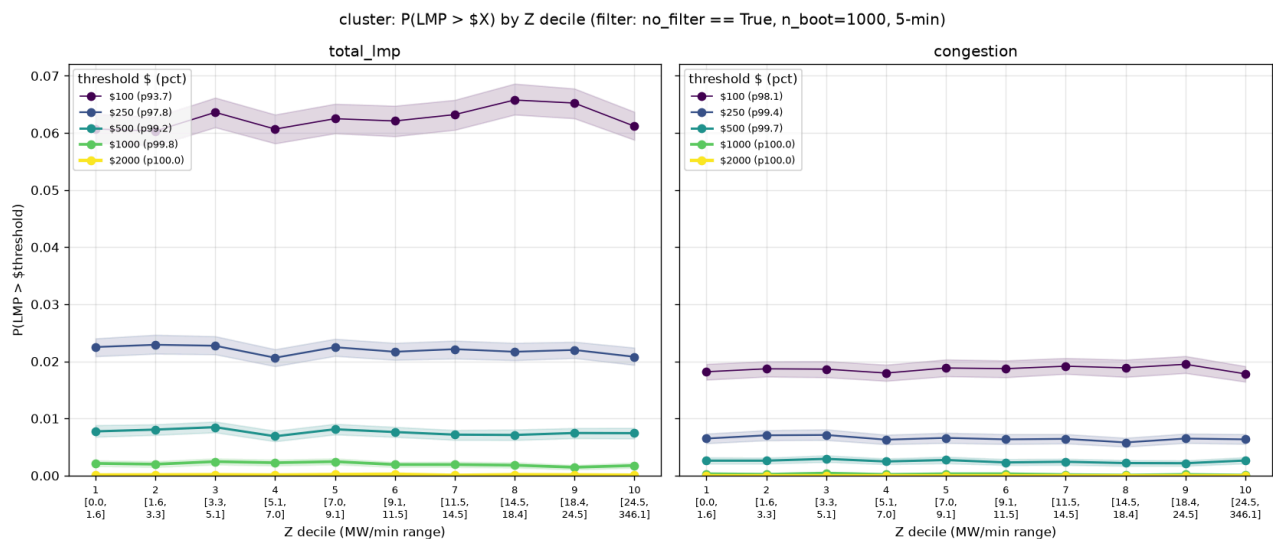


Figure 10. The cluster-mean decile curves at all five thresholds, pooled 2023 to 2026 window, total LMP (left) and congestion (right). The legend gives each threshold's percentile in the price distribution.

Full record: [decisions.md 2026-05-15 \(#8\)](#), [2026-07-19](#), [2026-07-29](#), [2026-07-30 \(extended-panel interpretation\)](#), [2026-08-08 \(figure set\)](#).

3.3 The 2026 Escalation: What Changed, and What It Is Not

Congestion in the Loudoun pocket rose sharply in 2026, even though 2026 was only halfway done at the time of this analysis (Table 4).

Table 4. Five-minute intervals with cluster congestion above each threshold, by year.

5-minute intervals with congestion above	2023	2024	2025	2026 (half-year)
\$100	479	804	1,505	3,768
\$250	59	206	399	1,620
\$500	0	73	128	681
\$1,000	0	7	13	62

Figures 11 and 12 show the same escalation month by month. Figure 11 counts the intervals above \$100 in each month, and next to it counts the intervals above a threshold that moves with the times (the trailing 12-month 99th percentile), so that a month is judged against its own recent past rather than against 2023. Figure 12 splits the counts by threshold.

Figure 11 — Large congestion events per month

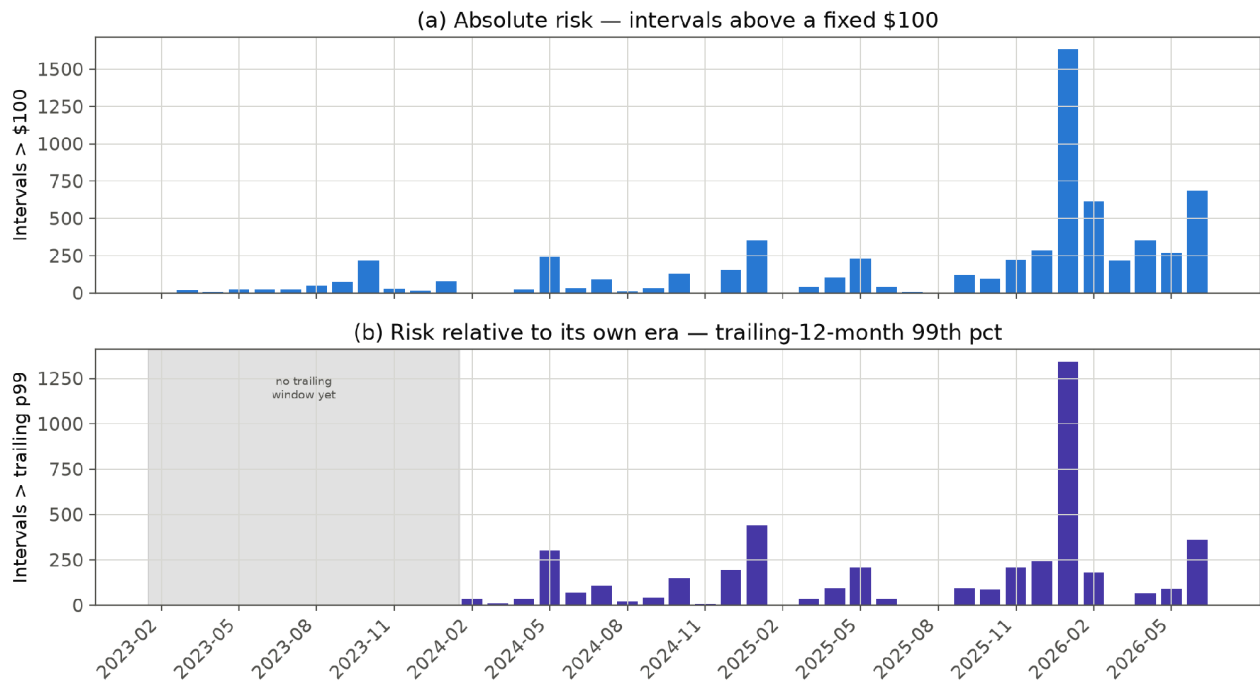


Figure 11. Intervals with congestion above \$100 per month (absolute) and intervals above a trailing 12-month 99th percentile (relative to their own era). The divergence between the two in 2026 is the regime shift: prices are not only higher than 2023, but they are also higher than the preceding year.

Figure 12 — Severity escalation, by month

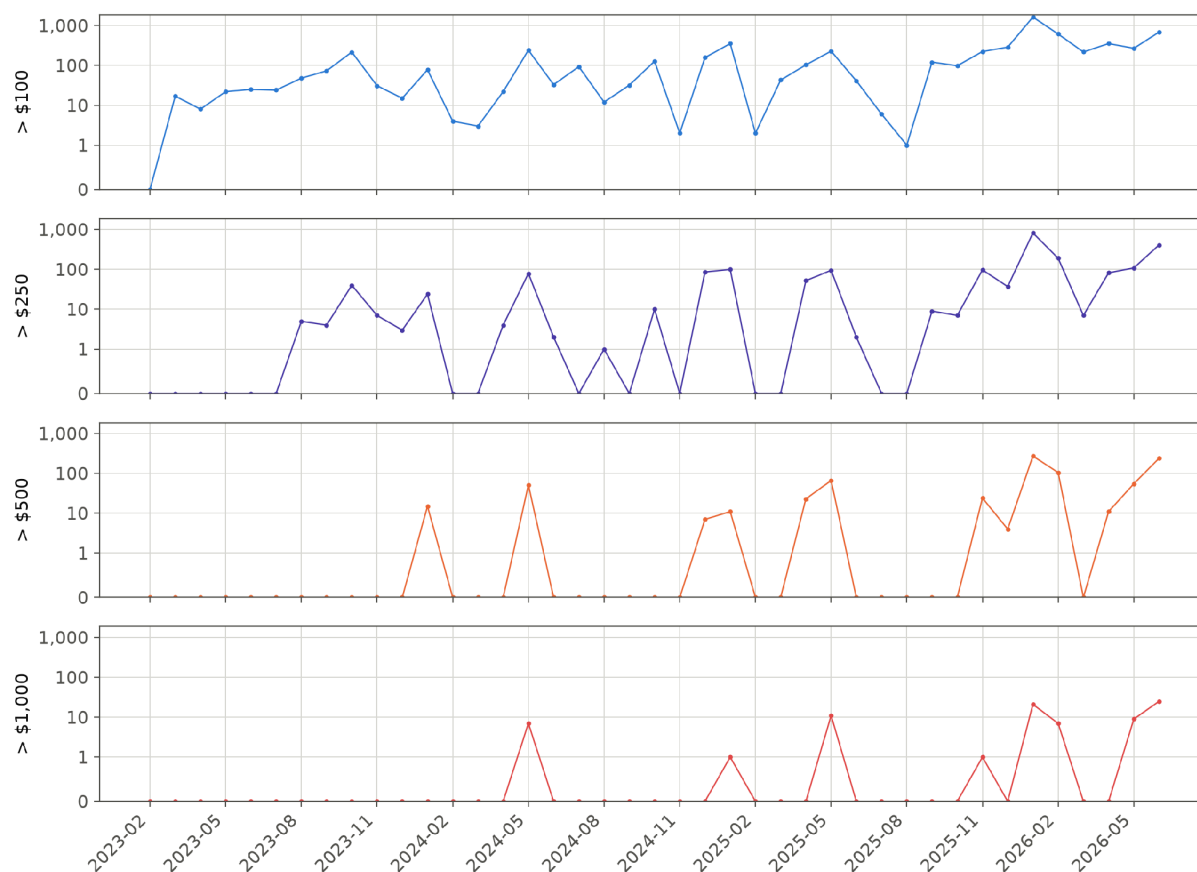


Figure 12. Severity by threshold and year, in monthly buckets, 2026 partial.

The 90th percentile of congestion by year is \$9.56, \$8.81, \$13.46, and \$60.76 (2026 is 6.4 times 2023). Within a fixed 20 to 22 GW load bin, the probability of congestion above \$100 runs 1.89%, 5.58%, 5.02%, and 35.84%: the same load now produces \$100-plus congestion seven times as often. Applying 2023's price-given-load relationship to each later year's load distribution, load growth alone explains 85.1% of 2024's exceedances, 59.2% of 2025's and 12.7% of 2026's. Figure 13 shows both calculations.

Figure 13 — What actually changed in 2026

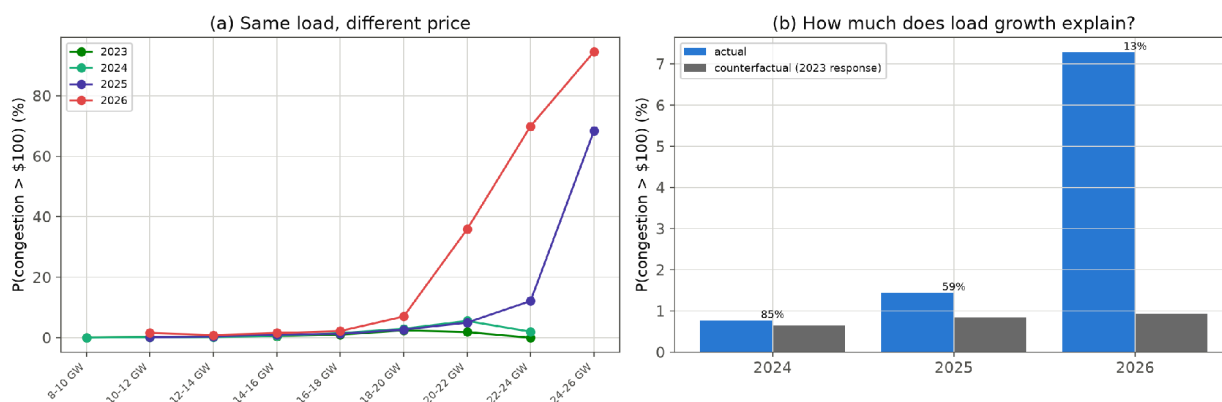


Figure 13. Same load, different price: the probability of congestion above \$100 within fixed 2 GW load bins by year (left), and the share of each year's exceedances explained by load growth alone (right). The caveats (2026 is a half-year; the 2023 baseline has few high-load intervals) are printed in the figure.

But the change is not local. It is a January 2026 step in *both* price components: the 90th percentile of congestion went from \$20.46 to \$231.29, and the 90th percentile of system energy went from \$86.32 to \$292.19. Desk research (note F) ranks Winter Storm Fern (2026-01-21 to 2026-01-30) as the best single explanation of a simultaneous step. Fern brought 18 to 19 GW of forced generator outages, against 12 to 13 GW in the comparable January 2025 event, and about \$798M of gas-electric misalignment uplift (payments to generators whose gas costs were not covered by the energy price). Spot prices ran above \$3,000/MWh PJM-wide and above \$1,800/MWh in Dominion's territory, PJM called pre-emergency demand response specifically for localized transmission constraints in the BGE (Baltimore Gas and Electric), Dominion, and Pepco zones, and the Department of Energy (DOE) issued a §202(c) emergency order. Additionally, the congestion spike of 2026-01-31 to 2026-02-01 coincides with a PJM-requested DOE order directing data centers onto backup generation. Figure 14 shows all three price components over the full panel; the system energy panel is PJM-wide, so its 2026 rise is not a Northern Virginia phenomenon.

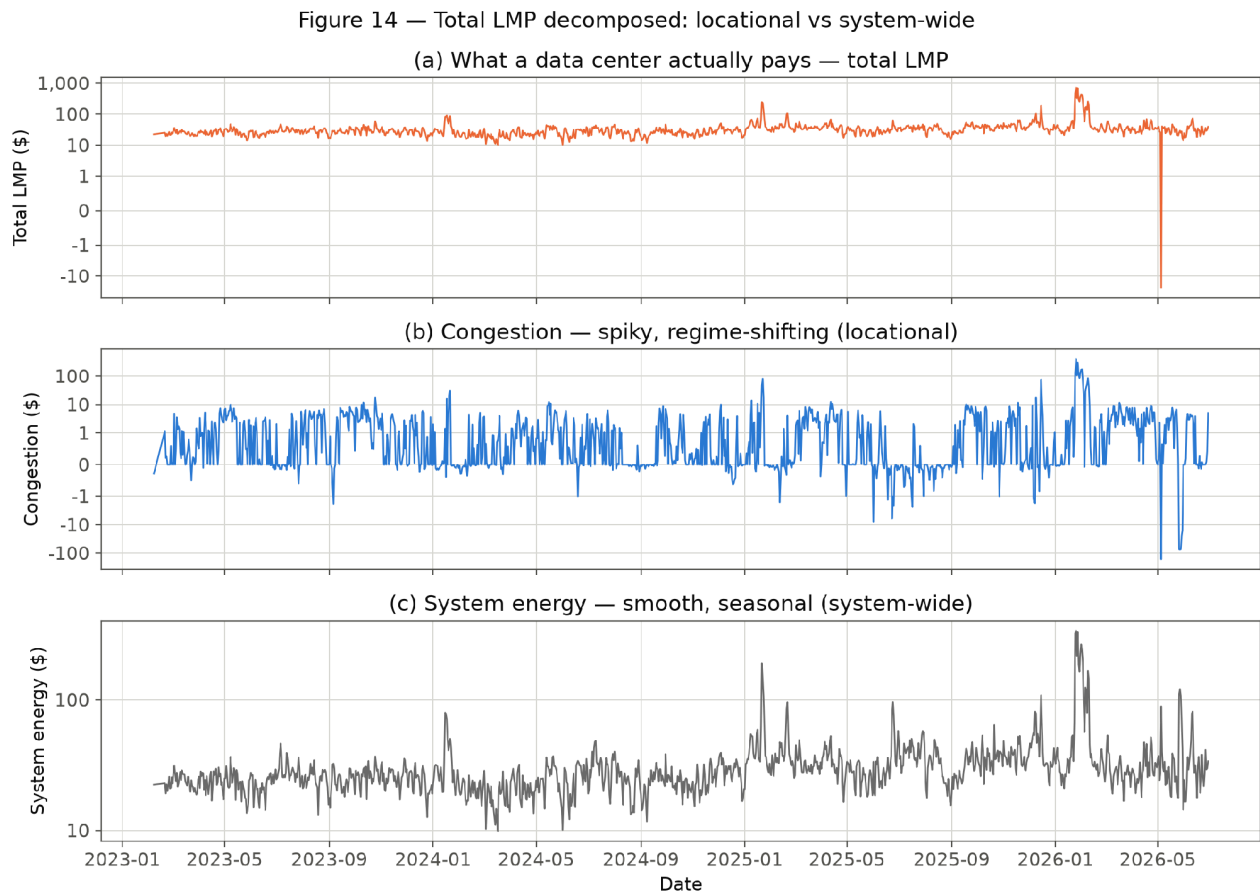


Figure 14. Cluster total LMP, congestion, and system energy over the full 3.4-year panel (symlog scale). System energy is smooth and seasonal; congestion is spiky and shifts regime. Panel (c) is PJM-wide, so its 2026 rise is not a Northern Virginia phenomenon.

*Full record: `decisions.md` 2026-07-30 (three entries), 2026-08-08;
 research-notes/F-jan2026-driver.md, H-event-catalog.md.*

3.4 Location: Ashburn Against the Rest

Ashburn TX1, the 35 kV distribution-side node in the data-center corridor, behaves differently from every transmission node. A transmission node sits on the high-voltage backbone (500 kV here) that moves bulk

power across the region, while a distribution-side node sits on the lower-voltage network (35 kV) that delivers power to a local area, so its price reflects local conditions that the backbone averages away. On the common window (2024-08-06 to 2026-05-10, $n = 15,432$), 4.78% of Ashburn's hours have congestion above \$100 against 1.60% at SKFFSCRK, and its 99th percentile is \$610.03 against \$95.92: roughly a threefold contrast. Ashburn correlates only 0.25 to 0.48 with every other series, while the cluster, both controls, and SKFFSCRK correlate with each other at 0.83 to 0.94. Figure 15 shows the exceedance rates and the correlation matrix.

However, Ashburn is a 35 kV distribution-side node, while everything compared to it sits on 500 kV transmission lines, so the two are not very comparable. Not a lot of analysis was done after that, so comparing Ashburn to another 35 kV distribution-side node outside of a data-center-heavy location is an open direction to explore.

Figure 15 — Location matters even where volatility does not

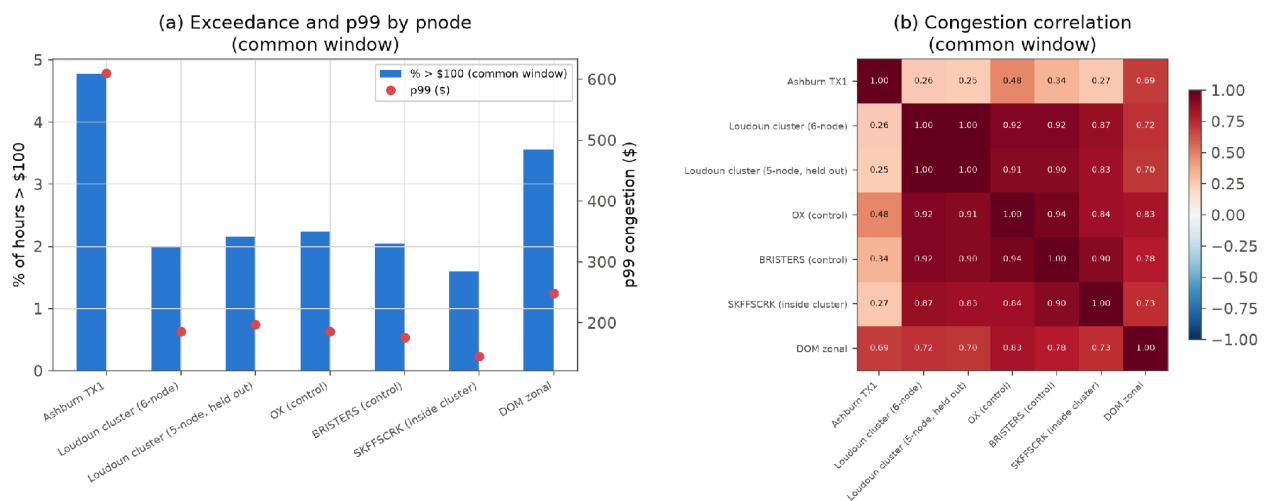


Figure 15. Exceedance frequency and 99th percentile by node (a) and the correlation matrix (b) on the common window. Ashburn TX1 stands apart on both. SKFFSCRK is labeled "inside cluster", never "control".

Full record: `decisions.md` 2026-07-30, 2026-08-07 (SKFFSCRK ruling), 2026-08-08; `research-notes/B-loudoun-geography.md`.

3.5 The One Verified Event, and the Event Catalog

2024-07-10, 19:05 Eastern. NERC's incident review describes about 1,500 MW of data-center load across about 60 facilities dropping to backup power after a lightning-arrestor failure on the Ox to Possum 230 kV line (six faults in 82 seconds, voltage down to 0.25 to 0.40 per unit, that is, 25 to 40% of normal). A one-week supplemental 5-minute pull (4 requests) reproduced it in the project's own series: DOM load fell 1,478.6 MW in five minutes, a gradient of 295.7 MW/min (the maximum in the pulled week; for scale, the hourly analysis's 90th-percentile Z is 13.4), and the system energy price fell from \$136.76 to \$56.70. That endpoint matches gridstatus.io's external report to the penny. This is the one place in 3.4 years (found in this project; similar events that were missed could still exist) where a large, *verified* load loss and a large price response coincide. It also served as the positive control for the load-artifact screen: three apparently larger excursions that reverted within minutes (−1,710, −1,630, and −1,575 MW) moved system energy by at most \$13.30, two of them *upward*. Figure 16 shows the event in the project's own data.

Figure 16 — The 2024-07-10 data-center trip

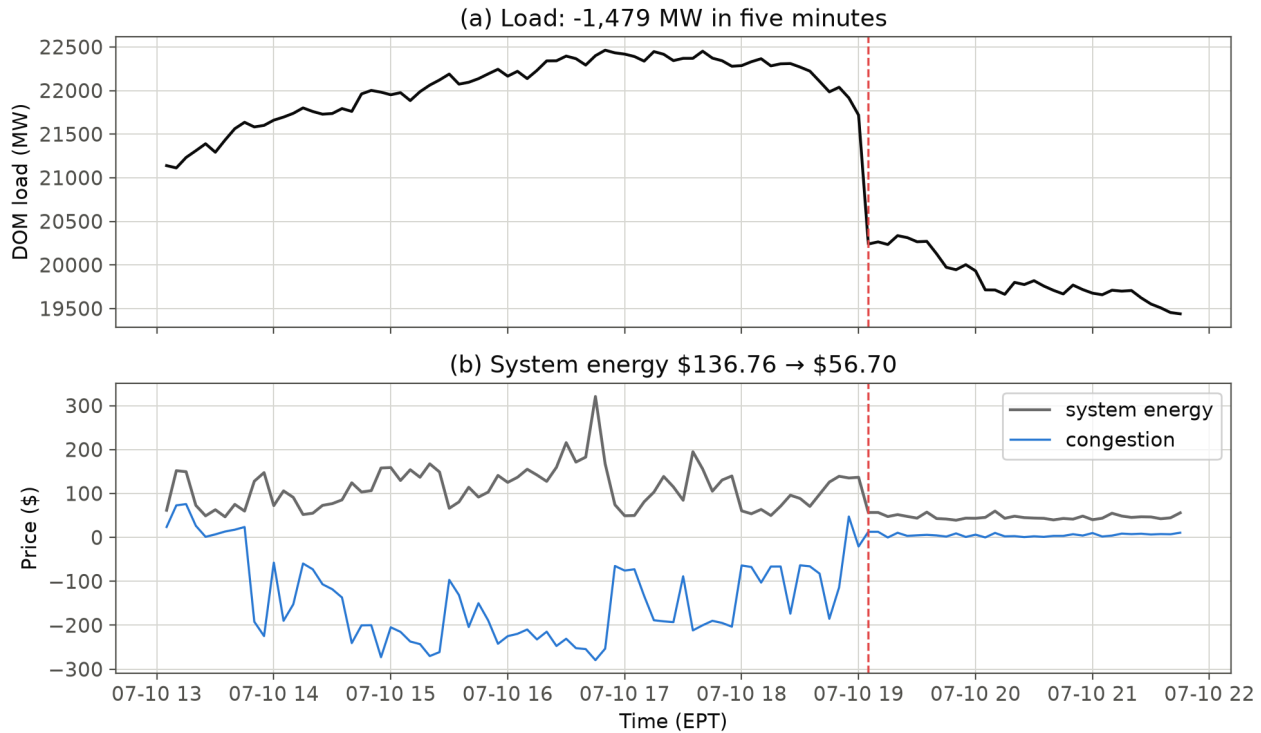


Figure 16. The 2024-07-10 ride-through event in the project's 5-minute series: load, gradient, and the system-energy price response. The congestion component is negative for most of that afternoon, and that is real, not a plotting error: PJM measures congestion relative to a load-weighted reference bus, so a node on the cheap side of a constraint that is binding somewhere else in PJM gets a negative congestion price. The cluster mean ran between -57 and -280 \$/MWh from 14:00 to 18:30, and all four cluster nodes were negative; PJM's own paper notes that the congestion and loss components can each be positive or negative [11].

The event catalog (note H) and the event-correlation scan. Additional research was also conducted to find external events that could cause further grid stress: thirty dated PJM and DOM grid-stress and market events from February 2023 to July 2026, built from Monitoring Analytics, PJM, NERC, and DOE primary records, with a year-by-year census of PJM-declared alerts. The panel's own flag for synchronized-reserve events (37 events, 39 event-hours) turned out to be an unused ground-truth signal, and only 2 of the 39 event-hours fall inside the proposal's filter. Ranking days by congestion and searching each one externally corroborated seven events or clusters: Winter Storm Elliott (2022-12-23 and 2022-12-24, cluster total LMP about \$4,130 against the \$3,700 cap), the January 2025 all-time winter peak (143,714 MW), a second January 2024 cold-weather alert, a tornado outbreak on 2024-05-26 (the only price event without an elevated gradient), and the cold wave of 2026-01-24 to 2026-02-09. Eight candidate dates stayed unexplained after equal effort, the largest being 2025-05-01 to 2025-05-03 at \$1,020. A naive scan of the largest gradients mostly re-finds the 9 AM daily ramp (31 of the top 50 rows). The catalog also records NERC's Level 3 "Essential Action" alert on computational loads (2026-05-04), the all-time PJM peak of 168,158 MW on 2026-07-02, and the 2026-07-22 Ashburn event in which about 3.1 GW of data-center load transferred to backup power in about 30 seconds, the largest on record, just outside the panel window. Note that no statistical analysis was done here; this was research and context exploration.

Full record: `decisions.md` 2026-07-21 (three entries), 2026-07-30, 2026-08-08;
 research-notes/H-event-catalog.md; plans/2026-07-21-subq3-event-catalog-scan.md.

3.6 ERCOT Stage 1

The question. Do the DOM findings hold in a different market? ERCOT was the first out-of-market check. The panel is 83,975 hourly rows from 2017-01-01 to 2026-07-31: load for the eight weather zones plus the ERCOT total (nine series), and prices at fifteen settlement points. Figure 17 shows the weather zones.

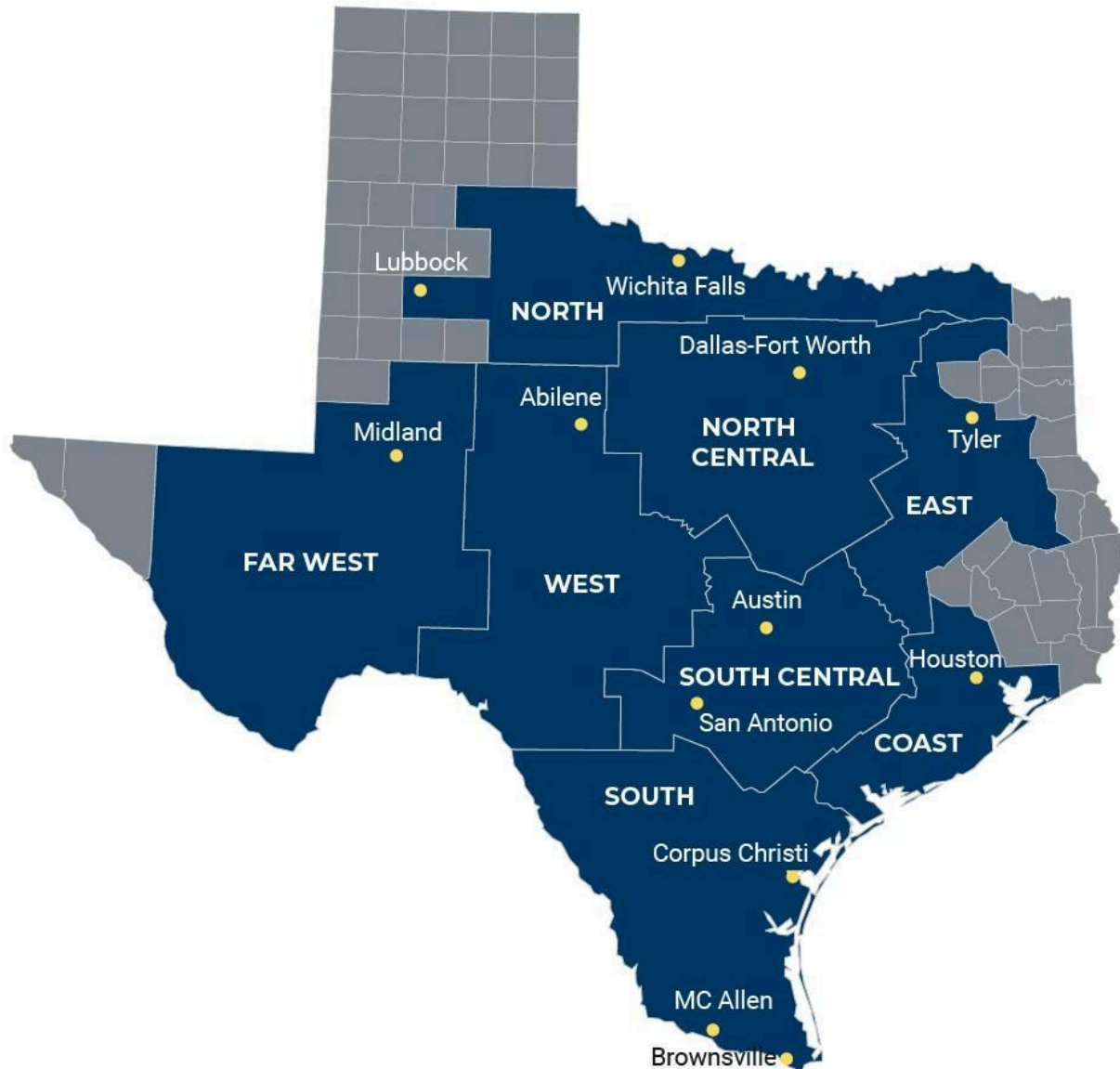


Figure 17. ERCOT's eight weather zones, from ERCOT's own map [107]. Far West holds the Permian Basin oil and gas load that drives its growth; North Central holds Dallas-Fort Worth, and Coast holds Houston.

"Stage 1" is the project's name for the level-versus-volatility check that every market after DOM went through: for each zone, how much of the price is explained by the zone's load level, and how much by its load volatility? Table 5 gives the ERCOT growth and volatility numbers behind Figures 18 and 19.

Table 5. ERCOT: change from 2017 to 2025 in mean load, raw ramp, and normalized ramp, for the system and four zones.

Zone	Mean load 2017 to 2025	Raw ramp	Normalized ramp	95th pct normalized
ERCOT	+36.7%	+8.4%	-20.7%	-23.3%
FWEST	+211.0%	+156.0%	-17.7%	-1.0%
NORTH	+106.5%	+93.4%	-6.3%	-10.7%
SOUTH	+27.9%	+3.4%	-19.2%	-18.4%
EAST	+29.9%	+29.6%	-0.3%	-2.6%

Figure 18 plots the normalized ramp (the mean hour-to-hour change in load divided by the mean load, so a dimensionless number) for every zone and year. Every line drifts down over the decade, the ERCOT-wide line by 20.7%, and Far West sits far below the rest because its load is dominated by flat industrial demand. This is the figure that says growth was being mistaken for volatility: load got bigger, and its movements did not keep up.

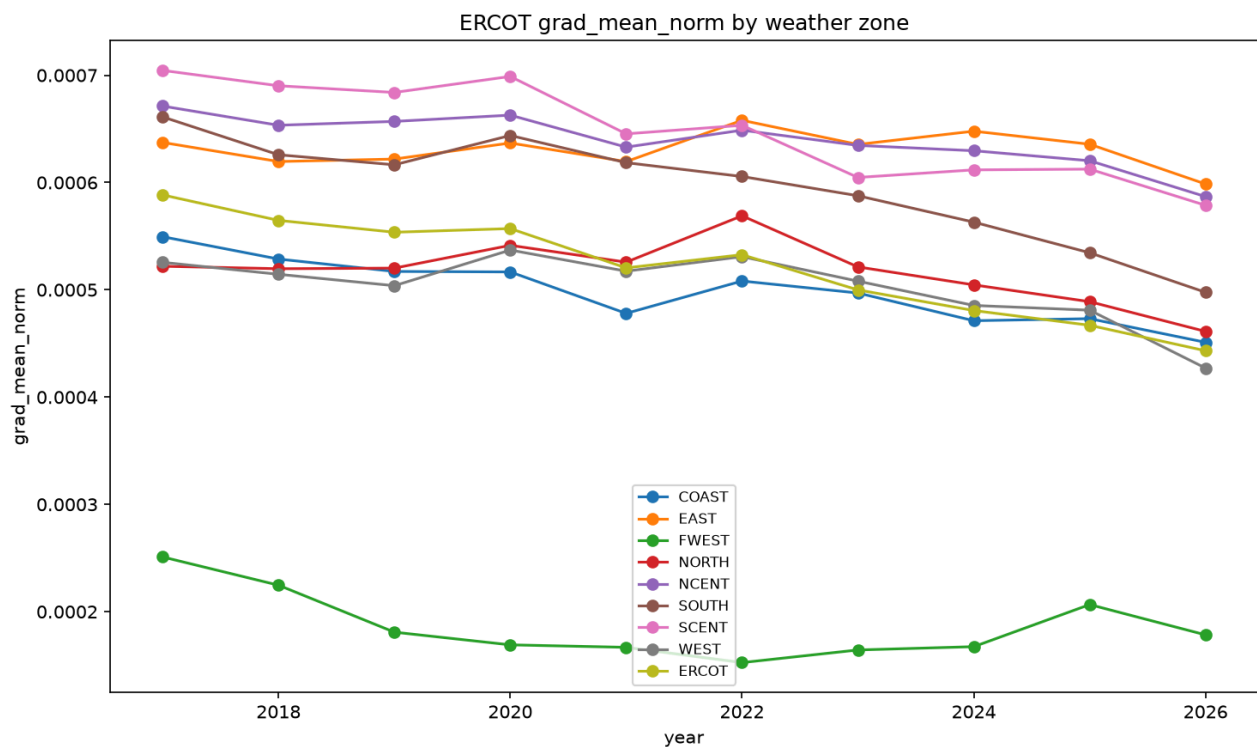


Figure 18. ERCOT: normalized ramp (grad_mean_norm, mean absolute hour-over-hour change in MW per minute divided by mean load) by weather zone and year, 2017 to 2026. Every zone falls; the ERCOT total (yellow) falls 20.7%.

Figure 19 plots the mean load itself, and is the other half of the same story: the ERCOT total grew 36.7% and Far West more than tripled, so any statistic that is not normalized by load would read as "more volatile" simply because there is more load.

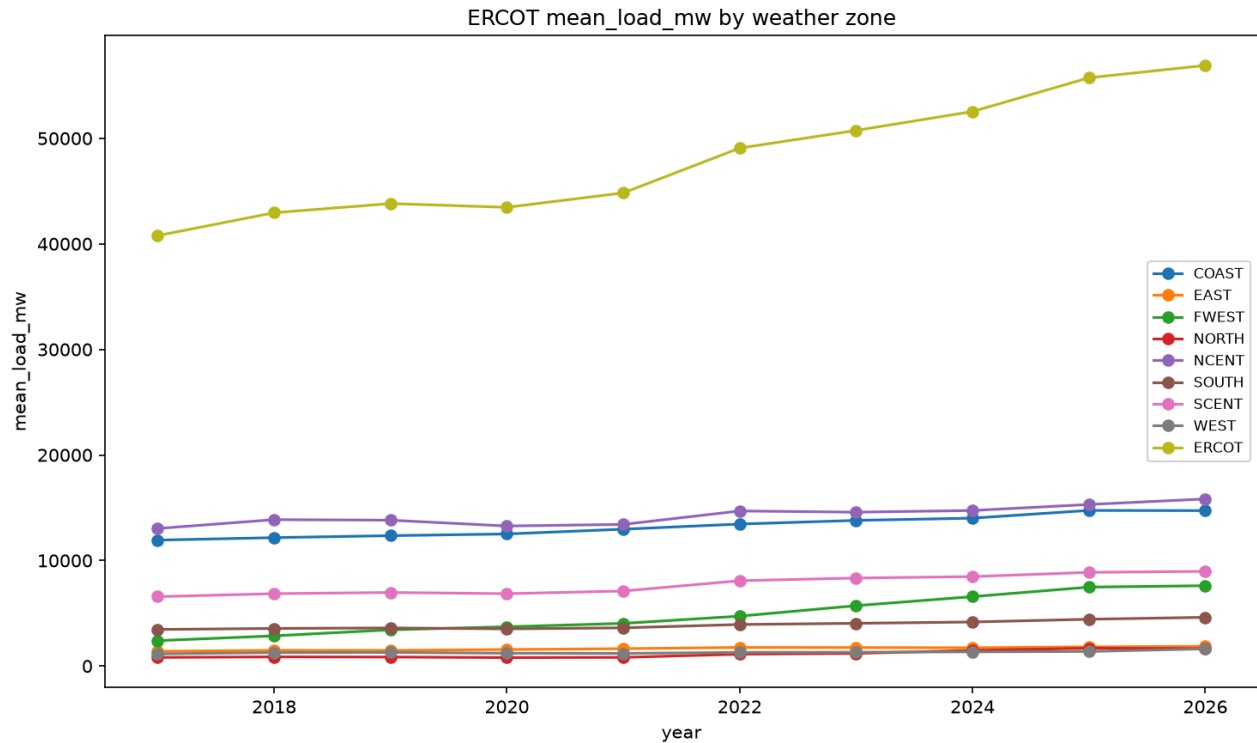


Figure 19. ERCOT: mean load in MW by weather zone and year, 2017 to 2026. The ERCOT total (top line) grows 36.7% over the window; Far West (green) grows 211%.

Thus, two conclusions were made from the ERCOT analysis.

- Normalized volatility (the ramp divided by the load of the day) falls in all nine series. Raw ramps grow, Far West's by 156%, but more slowly than load. Growth was being mistaken for volatility.
- Load level beats load volatility as a predictor of price in 132 of 135 zone-by-settlement-point pairs (standardized regression coefficients; for ERCOT against the hub average, level +0.185 and volatility -0.036). Volatility's coefficient is *negative* in 8 of 9 zones. Far West inverts, and that is the wind confound (electrification load in the Permian coincides with high wind and low prices), not a data-center signal.

Full record: `decisions.md 2026-08-07 (ERCOT Stage 1); specs/2026-08-07-ercot-...;`
`plans/2026-08-07-ercot-diagnostic-ERRATA.md.`

3.7 Cross-ISO Stage 1: Eight Markets, Eleven Panels

The question. Is "level beats volatility" a DOM fact or a general one? The same descriptive horse race was run in every market with a public archive: a standardized regression of each zone's price on its own load level and its own absolute load gradient, with no time controls, at each market's maximum window and again at a locked common window (2023-01-01 to 2025-05-01). The machinery is shared (`src/surg/diagnostics/stage1.py`), and each market has one fetch script and one diagnostic driver. Table 6 summarizes every panel; "level wins" counts the zone-price pairs in which load level had the larger standardized coefficient, and R^2 is how much of the price variation the two variables explain together (0 is none, 1 is all).

Table 6. The cross-ISO Stage 1 results, one row per panel.

Market / panel	Price depth	Load growth (span)	%/yr	Normalized volatility change	Level wins (common window)	Median R ²
DOM (5-min, reference)	~3.4 yr	+28.0% (2023 to 2026)	+8.58	0.1850% to 0.1596% (falling)	see §3.2	n/a
ERCOT (hourly)	~9 yr	+36.7% (2017 to 2025)	+3.99	-20.7%	132/135 (97.8%)	0.004 to 0.056
NYISO, merged (10 zones)	~25 yr	-4.5% (2002 to 2025)	-0.20	-12.6%	213/220 (96.8%)	0.175
NYISO, split (11 zones)	~21 yr	-6.5% (2006 to 2025)	-0.35	-8.7%	235/242 (97.1%)	0.163
CAISO, full depth (4 zones)	~2.3 yr	+31.4% (2010 to 2025)	+1.84	-9.8%	8/8	0.194
CAISO, modern (6 zones)	~2.3 yr	+2.4% (2019 to 2025)	+0.40	-3.5%	11/12 (91.7%)	0.108
IESO (Ontario)	~22 yr	-8.5% (2004 to 2024)	-0.44	-1.1%	11/11	0.119
MISO (6 LRZ groups)	2023 to 2026	+3.8% (2023 to 2025)	+1.88	-5.6%	36/36	0.332
ISO-NE (8 zones, control)	2016 to 2026	-5.2% (2016 to 2025)	-0.59	+9.9%	64/64	0.274
SPP (17 zones)	2017 to 2026	+13.2% (2016 to 2025)	+1.39	-14.2%	289/289	0.245
Italy (7 zones, ENTSO-E)	2015 to 2025	n/a	n/a	falls in 5 of 6, Sardinia +4.7%	36/36	n/a

Finding 1: level beats volatility essentially without exception. Between 91.7% and 100% of zone-price pairs in eleven panels, twelve with Italy. It is the most robust cross-market regularity in the project.

Finding 2: "normalized volatility always falls" no longer holds, and it failed in the control market. ISO-NE came out at +9.9%, with 5 of 8 zones rising. But decomposing it, the raw gradient rose only 3.7% (and in only 3 zones) while load fell 5.2%: mostly a shrinking-denominator effect. Only Vermont (raw +38.9%) and Maine (+18.9%) rose in absolute terms. The working explanation was that these are the states with the least data-center growth and the least grid investment; the follow-up in [§3.8](#) then traced Vermont's rise to rooftop solar (its midday load floor has collapsed) and Maine's partly to the same. Sardinia later added a second exception.

Finding 3: the control market behaves exactly like the treated ones. ISO-NE, whose operator says New England "has not experienced similar growth so far, and only a small amount is expected in the coming decade", returns 64 of 64 with a *higher* median R² than most treated markets. That cuts against reading the pattern as anything data-center-specific: on this evidence it is a property of how power systems price load.

The figures that follow show, for each market, the same two plots as Figures 18 and 19 for ERCOT: the normalized ramp by zone and year on top, and the mean load by zone and year underneath. They are the evidence behind the "normalized volatility change" and "load growth" columns of Table 6, and the thing to look for in each pair is whether the top plot falls while the bottom plot rises.

MISO (Figure 20) is the shortest panel, 2023 to 2026, because MISO retains its market reports only for the current year plus three. Normalized volatility falls in five of the six LRZ groups and rises only in LRZ4 (Illinois); load grows 3.8%. Level wins 36 of 36 pairs, with the caveat from [§2](#) that LRZ3_5 and LRZ4 share a hub and so are not independent.

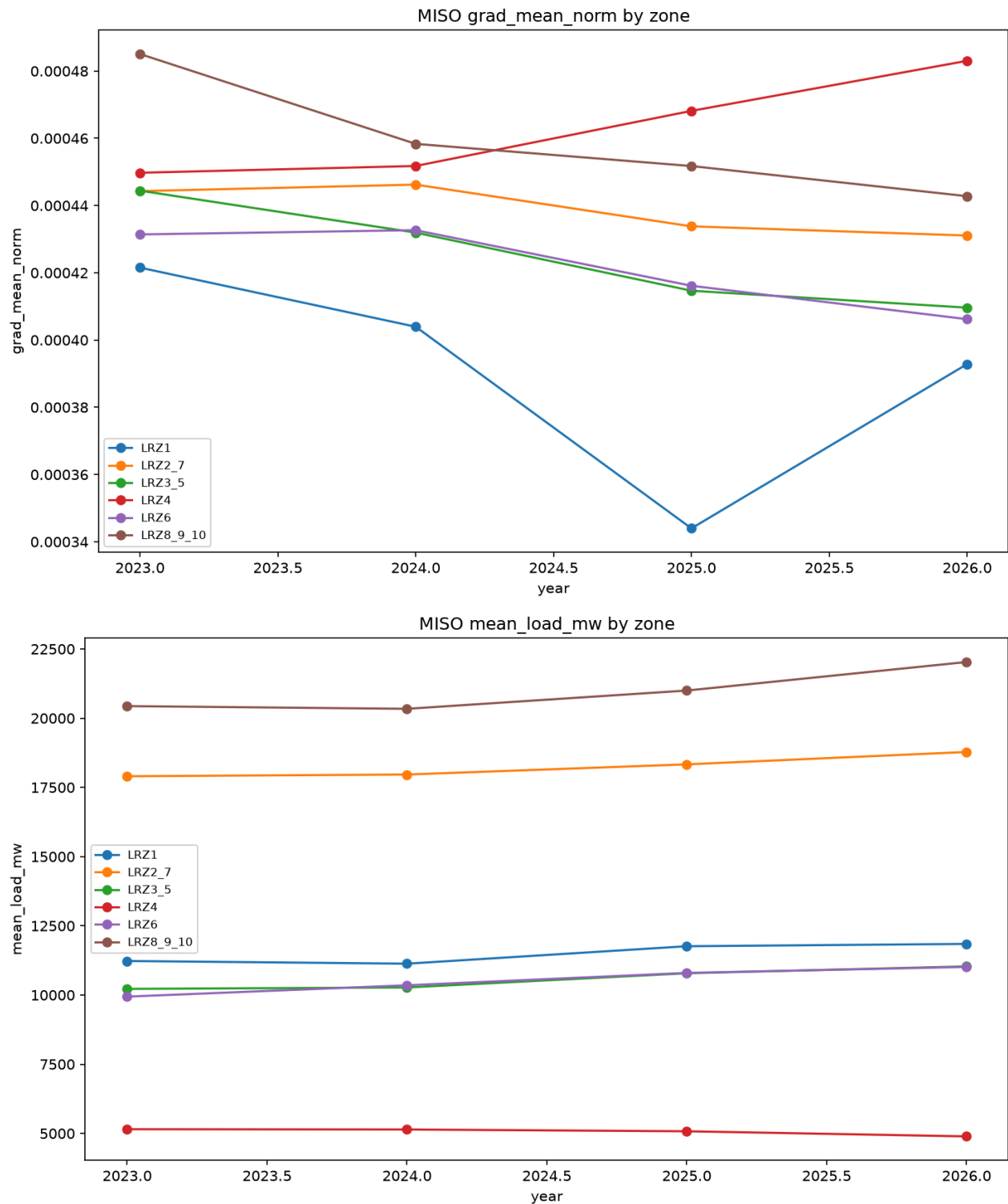


Figure 20. MISO: normalized volatility (top) and load level (bottom) by LRZ group and year, 2023 to 2026 (LRZ3_5 and LRZ4 share ILLINOIS.HUB).

SPP (Figure 21) covers seventeen control zones in the most wind-penetrated region in North America. Load grows 13.2% over 2016 to 2025, and normalized volatility falls 14.2%; level wins every one of the 289 pairs. It is the cleanest instance of the pattern in a market with almost no data-center story of its own.

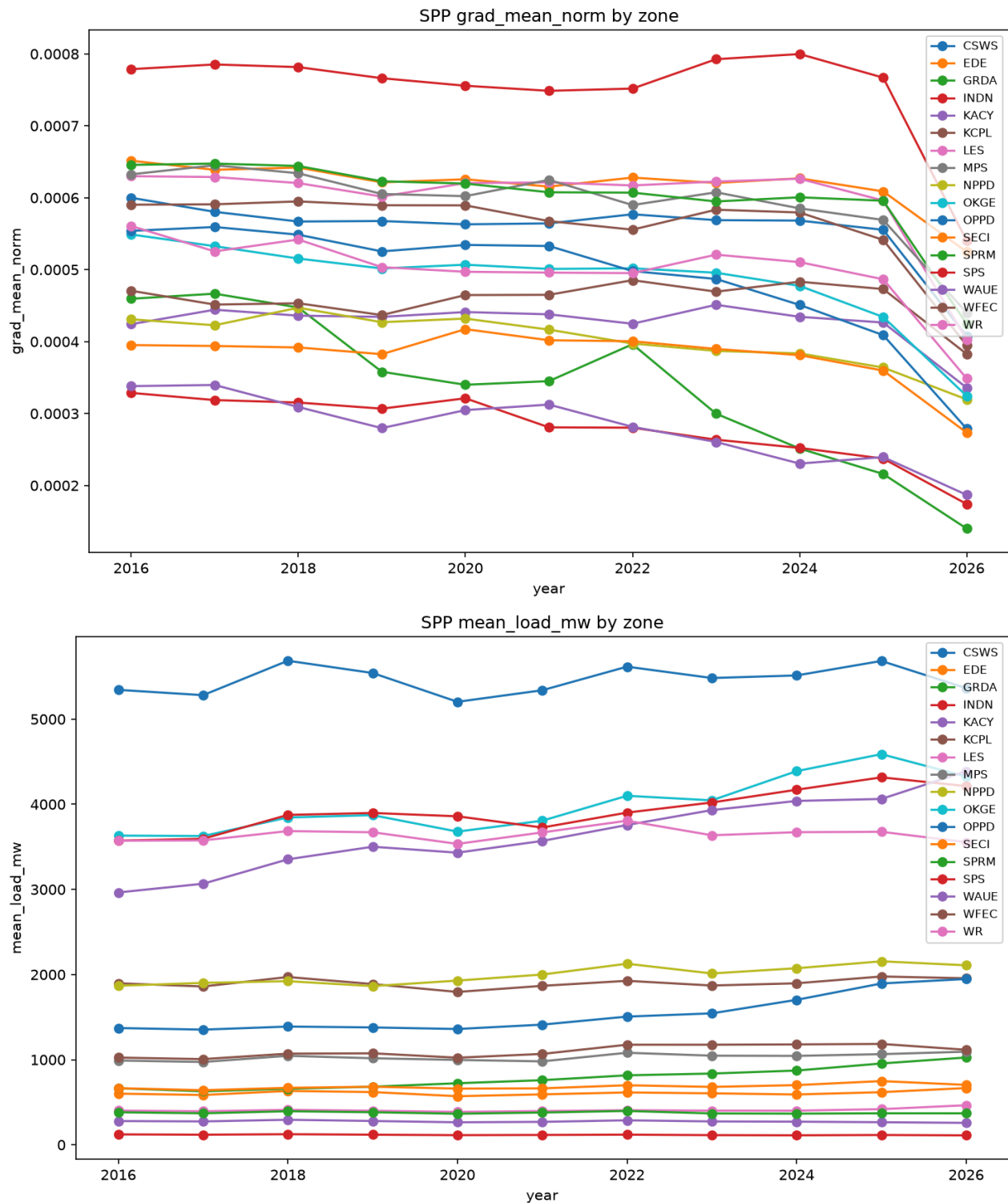


Figure 21. SPP: normalized volatility (top) and load level (bottom) by control zone and year, 2017 to 2026.

ISO-NE (Figure 22) is the control market, and the one panel in which the top plot rises. Vermont's normalized volatility climbs steadily from 2017, Maine's and Rhode Island's jump after 2023, and the Massachusetts zones fall; meanwhile, load falls 5.2%, so much of the rise is a shrinking denominator. [§3.8](#) traces the Vermont rise to rooftop solar.

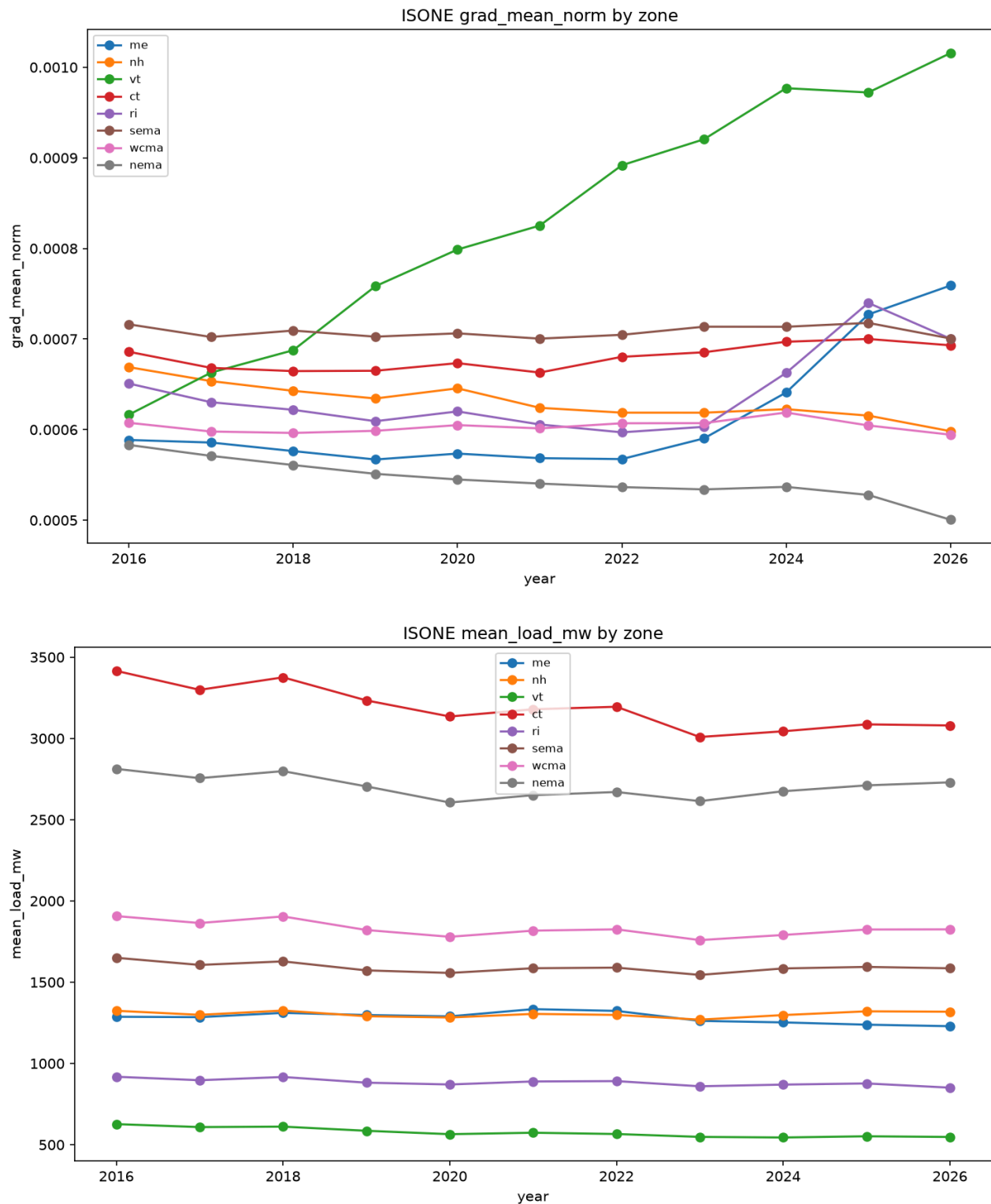


Figure 22. ISO-NE, the control market: normalized volatility (top) and load level (bottom) by zone and year, 2016 to 2026. Normalized volatility rises in five zones (Vermont most of all) while load falls; §3.8 says what that turned out to be.

NYISO (Figure 23) is the longest panel, a quarter century on the merged 10-zone roster. Normalized volatility falls in most zones from 2001 to about 2020 (Millwood most dramatically) and turns up again after 2021 in several, while load falls 4.5% over the span; level wins 213 of 220 pairs. The recent upturn is worth a follow-up that this project did not do.

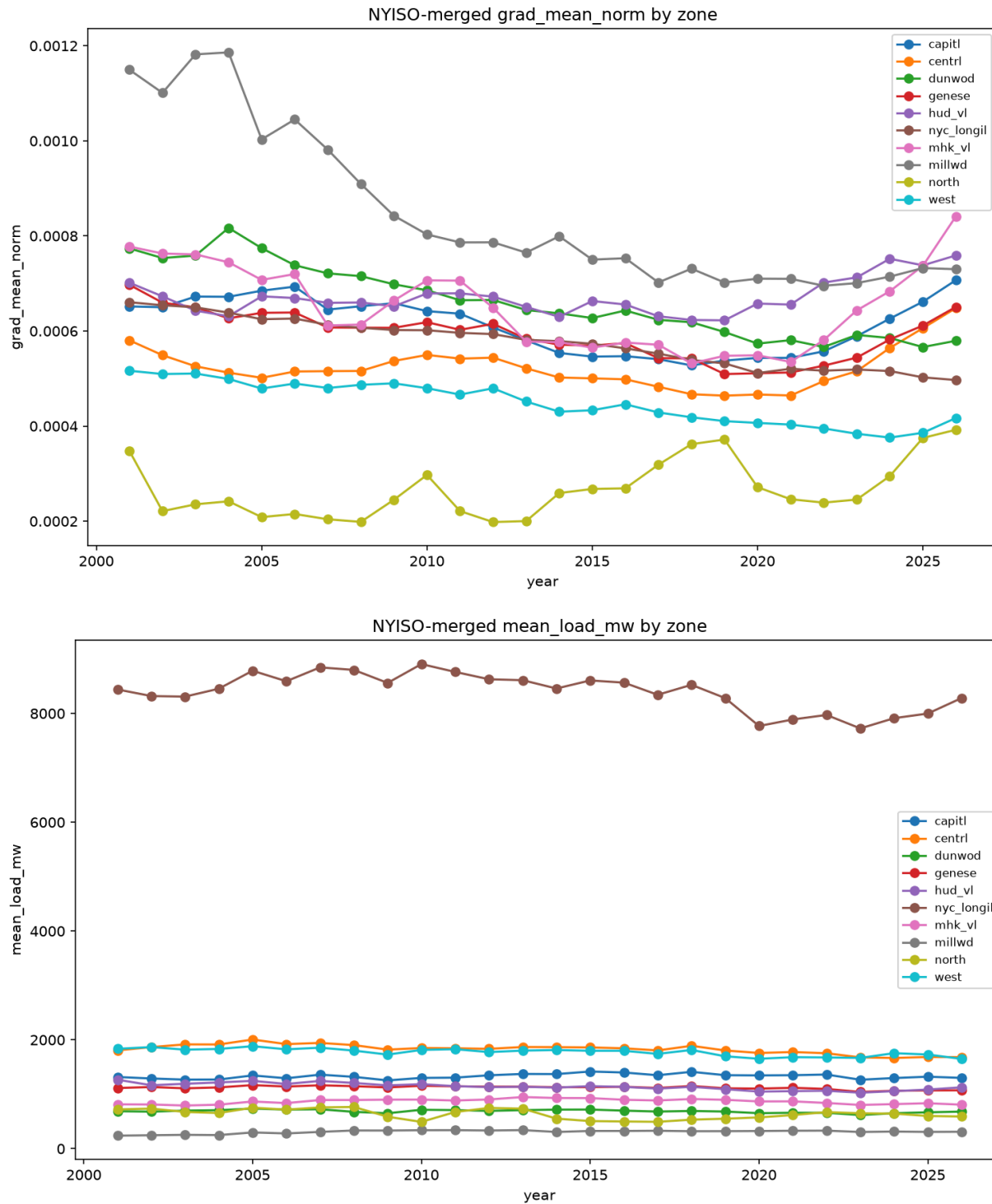


Figure 23. NYISO (merged 10-zone roster): normalized volatility (top) and load level (bottom) by zone and year, 2001 to 2026.

CAISO (Figure 24) is the modern six-area panel, because CAISO's price archive only reaches back to April 2023. Load is nearly flat (+2.4% over 2019 to 2025) and normalized volatility falls 3.5%; level wins 11 of 12 pairs. Silicon Valley's data-center cluster is invisible here because Santa Clara is served by a municipal utility outside CAISO's load areas.

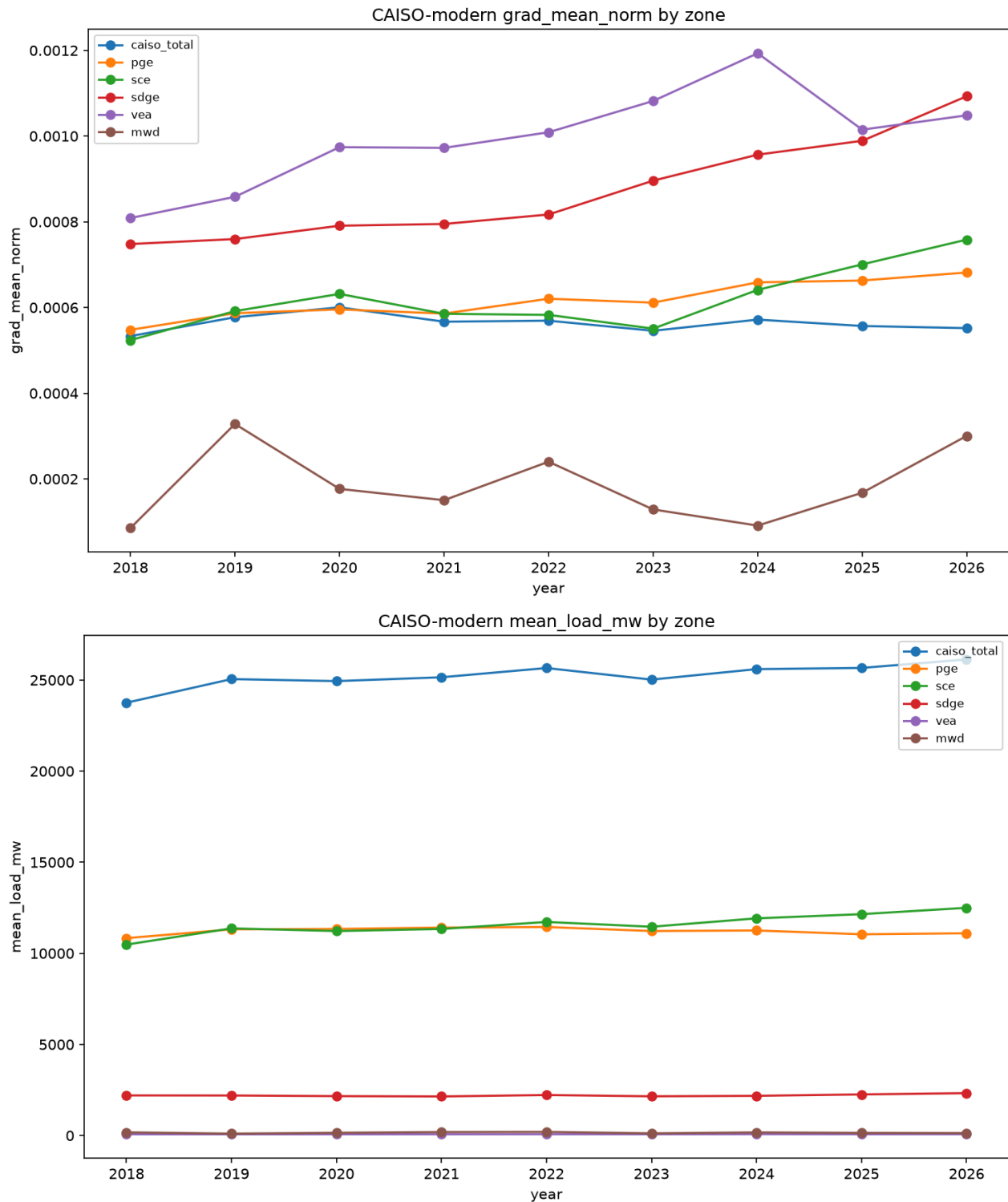


Figure 24. CAISO (modern panel): normalized volatility (top) and load level (bottom) by utility area and year, 2018 to 2026. Price depth only from April 2023.

IESO (Figure 25) is Ontario, whose growth story is electric vehicles rather than data centers. Load falls 8.5% over 2004 to 2024, and normalized volatility is essentially unchanged (-1.1%); level wins 11 of 11 pairs.

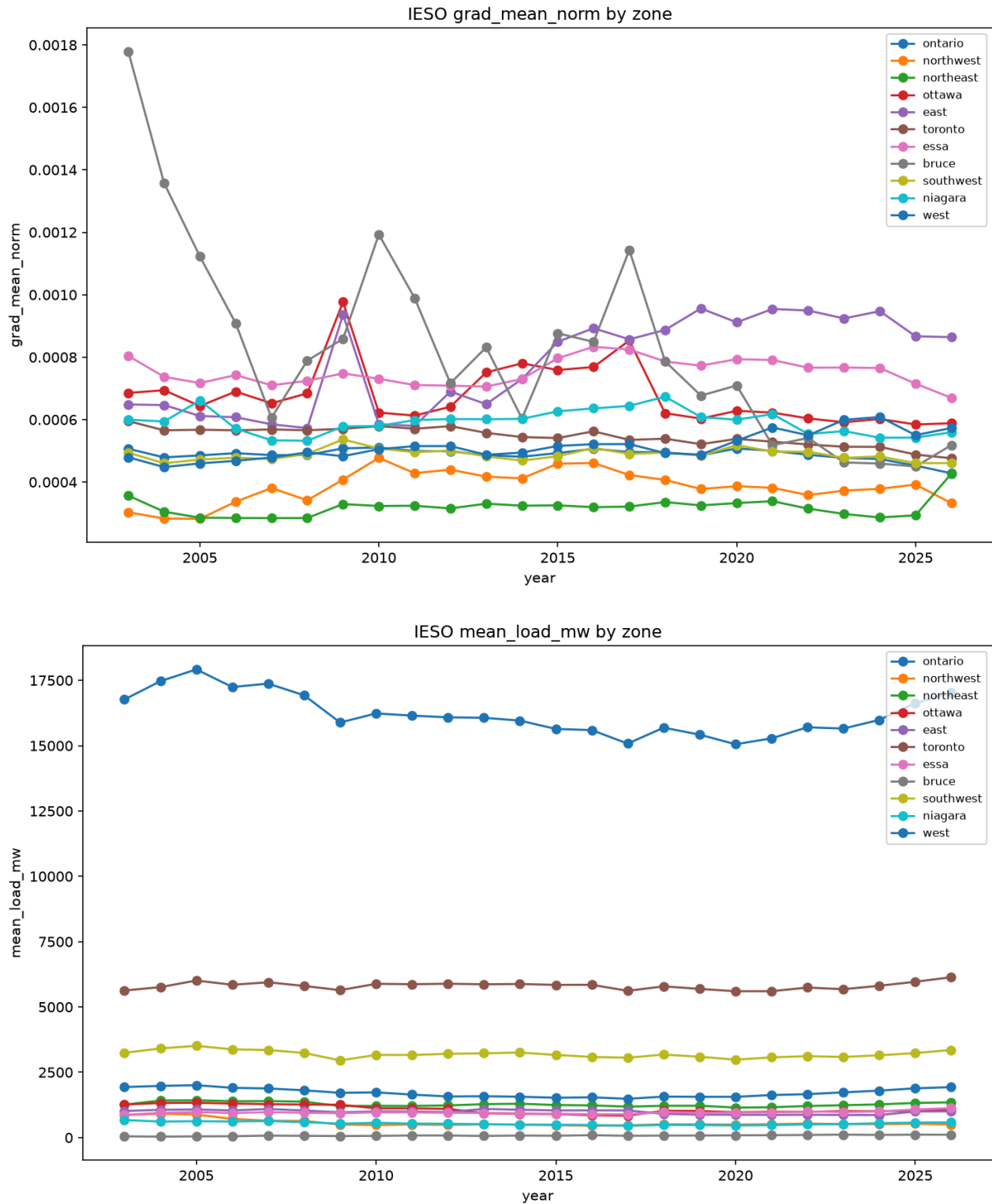


Figure 25. IESO, Ontario: normalized volatility (top) and load level (bottom) by zone and year, 2003 to 2026. An EV-driven growth story, not a data-center one.

Italy (Figure 26) is a robustness check rather than a finding: the same statistic on six European bidding zones from the ENTSO-E panel of [§3.10](#). Normalized volatility falls in five of the six, with the South and Sicily spiking in 2022 and 2023 before falling back; Sardinia (the large island west of mainland Italy, which is its own bidding zone) is the exception, rising 4.7%. Level wins 36 of 36 pairs.

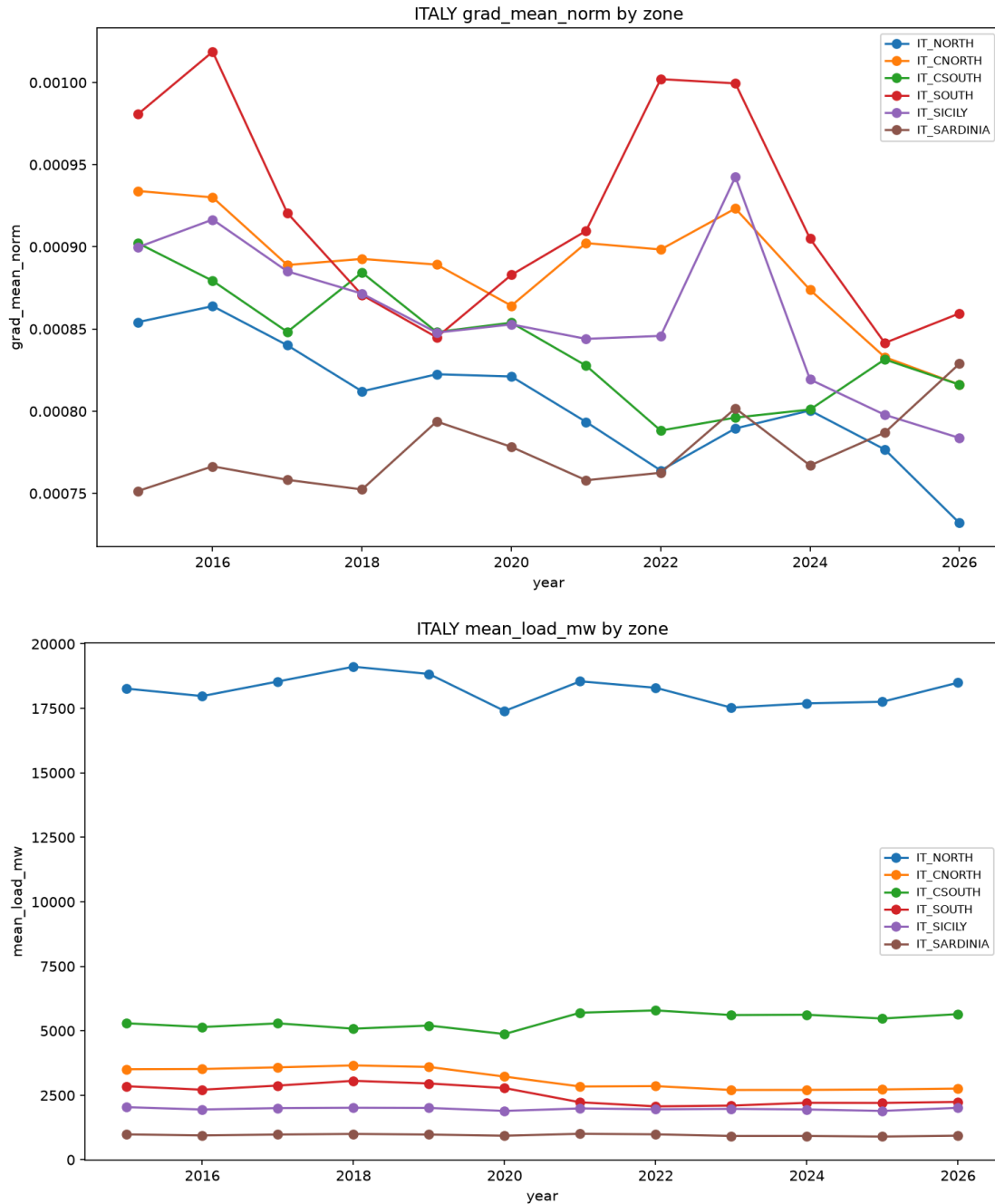


Figure 26. Italy: normalized volatility (top) and load level (bottom) by bidding zone and year, 2015 to 2026. Six bidding zones (Calabria excluded: a bidding zone only from 2021).

Full record: [decisions.md](#) 2026-08-09 (four entries), 2026-08-10 (capstone), 2026-08-12 (Italy);
[sources/availability/cross-iso-data-availability-summary.md](#);
[plans/2026-08-09-cross-iso-stage1-plan-a-trio.md](#), ...-08-10-...-plan-b.md.

3.8 ISO-NE Follow-Ups: The Canadian Tie, Solar, and NECEC

The question. Following the results for ISO-NE from the last section, the project then pivoted to finding whether New England's rising volatility comes from its connection to Canada and its increased use of renewable energy sources, which may be more volatile. The answer was a resounding no.

1. The Canada-tie hypothesis fails on three independent grounds.

- Mechanism: imports over an interface (a transmission connection to a neighboring grid) are supply, and never enter metered demand.
- Geography: ISO-NE's complete list of external interfaces lands in Maine (New Brunswick, 1,000 MW), Massachusetts (the Phase I/II HVDC link, 2,000 MW at Sandy Pond; HVDC is high-voltage direct current, the technology used for long transmission lines) and Vermont (Highgate, 225 MW); Rhode Island has no external interface of any kind and rises anyway, and the largest line lands where volatility *fell* in all three Massachusetts zones (−3.2%, −4.8%, −12.7%).
- Timing: there is no step change inside the window.

2. A pre-registered test for a distributed-solar fingerprint. If rooftop solar is behind the rise, the midday hours should have gained volatility and the night hours should not. The rising zones (Vermont, Maine, Rhode Island) do show the midday gradient up 66% to 126% between 2016-17 and 2024-25 with night flat, but so do all five falling zones (up 32% to 49%), so the test is uninformative on the question asked. What it established instead is region-wide: every ISO-NE zone gained volatility in the solar-day hours (11 to 19) and lost it in the late evening and morning shoulders, and a zone's annual sign is decided by which side wins. The seasonal clause of the pre-registration failed as written.

3. An exploratory mechanism scan, not a result. The share of days whose minimum load falls between 10 AM and 4 PM (the "duck curve" migrating into the daytime: the dip in metered load when rooftop solar is producing) went from 1.4% to 68.2% in Vermont, 0% to 36.7% in Maine, and 0% to 0.5% in Rhode Island; its rank correlation with the volatility trend is +0.707. Floor erosion (the annual minimum divided by the mean) is −91.7% in Vermont, −20.1% in Maine, and −38.5% in Rhode Island, with a rank correlation of −0.952 and no overlap between rising and falling zones (partly by definition). Vermont's annual minimum fell from 420 to 31 MW: at midday the meter has stopped seeing the load. The rising trio is not one phenomenon. Vermont is solar, Maine partly, and Rhode Island's stable overnight core shrank with its minimum still at 3 AM.

4. The NECEC dose check. The New England Clean Energy Connect line, a 1,200 MW HVDC link from Québec to Lewiston, Maine, began commercial operation on 2026-01-16. It runs as a firm block of about 1,059 MW (89% of its rating, with almost no hour-to-hour variation), which is dispatchable reservoir hydro, not variable renewables. The legacy Hydro-Québec interfaces fell from +558 to +27.5 MW at the same time: a gigawatt of Canadian injection *relocated* from Massachusetts and Vermont into Maine. Data-quality find: EIA-930's ISO-NE to Hydro-Québec interchange series omits NECEC.

5. A pre-registered price test for NECEC. The test, its result, its robustness checks, and its caveats are laid out one at a time below, because this is the one place in the ISO-NE work where a pre-registered prediction partly failed.

The test. If the new line changed anything, it should show up in Maine's price relative to its neighbors. The "basis" is that relative price: Maine's day-ahead price minus the average of Connecticut's and Rhode Island's, so a negative basis means Maine is cheaper than the rest of New England. For February to June, Maine's basis went from −1.88 \$/MWh in 2025 to −4.82 in 2026, and Maine averaged \$56.41/MWh against \$60 to \$63 everywhere else.

Could it be chance? To judge whether a change of that size could have happened by chance, the same before-and-after comparison was run on 45 "placebo" cells (other zones and other periods, where NECEC should have had no effect); none of the 45 moved as much as Maine did ($p = 0.0217$, the smallest value the design can produce). The result holds on medians and in all 24 hours of the day, and the Massachusetts zones, which lost a gigawatt of Canadian injection to Maine, moved the opposite way, as predicted (+0.34, weak). After the fact: Maine's basis is positive on every day before 2026-01-16 and negative on every day after (2026-01-01 to 2026-01-15: +1.45; 2026-01-16 to 2026-01-31: -14.54).

The part that failed. The pre-registered *volatility* prediction (that the basis would get calmer, or not change) failed: the mean absolute daily change in the basis went from 0.87 to 1.96, more than in any of the 45 placebo cells. But the typical hour got *calmer* (median 0.345 to 0.310), so the extra movement is in the tails, consistent with an export limit that binds now and then.

The caveat and the honest formulation. Maine's basis was already falling in 2025 (April 2025 was -4.52 with no NECEC). The honest formulation: NECEC did not create Maine's negative basis; it roughly tripled it. It does not support the original hypothesis. A constant was added to a congested network; nothing variable arrived. Figure 27 shows the basis by year against the placebo distribution.

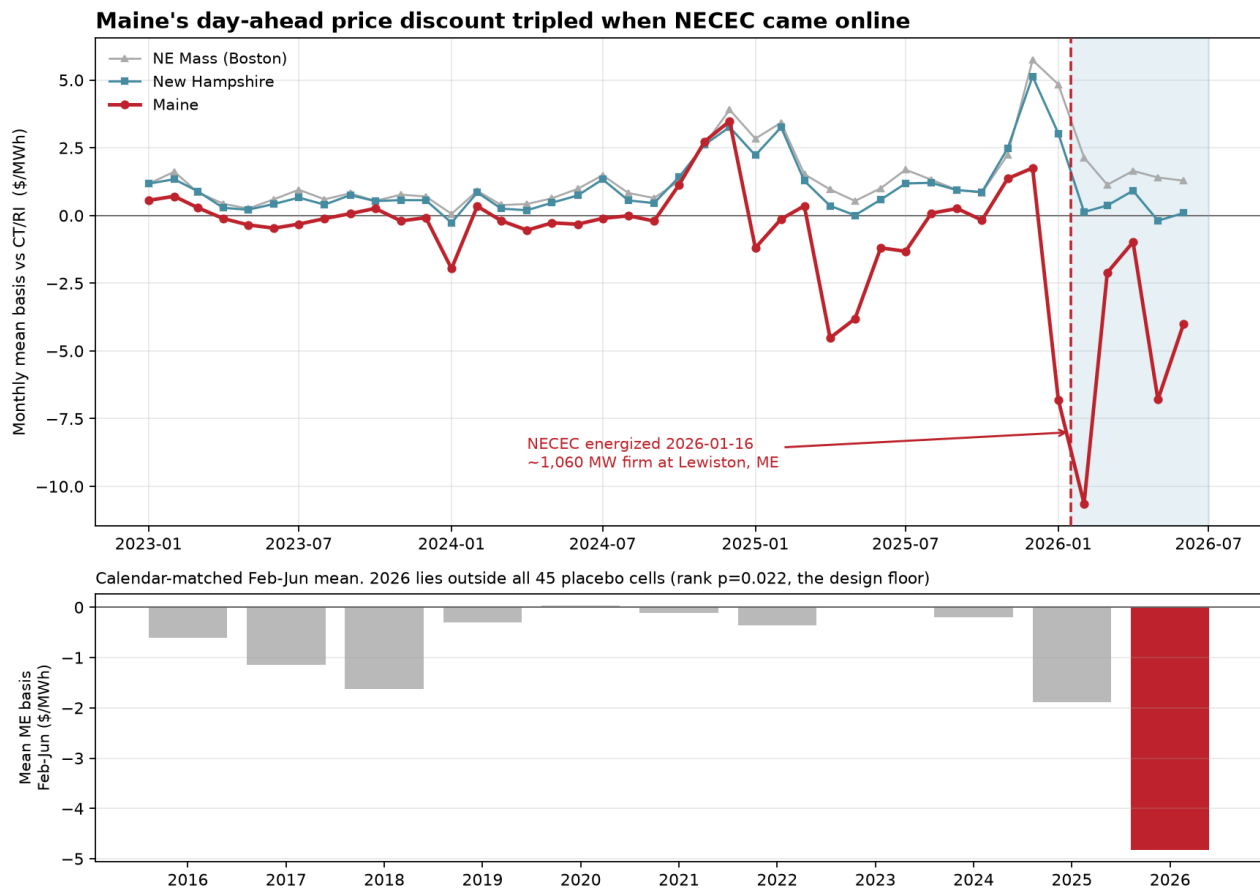


Figure 27. Maine's day-ahead basis against the Connecticut/Rhode Island reference by year, February to June, with the 45-cell placebo distribution. The 2026 point sits outside every placebo.

Full record: `plans/2026-08-11-isone-canada-tie-verification.md`,
`...-isone-der-fingerprint-prereg.md`, `...-isone-volatility-mechanism-scan.md`,
`...-necec-dose-check.md`, `...-isone-necec-price-prereg.md`; `scripts/necec_price_test.py`.

3.9 Is Data-Center Load Actually Spiky? UKPN Facility Profiles (Note J)

The question. Everything up to here used regional aggregates in which data centers are one unlabeled part. UK Power Networks publishes the first facility-level data in the project: 96 anonymized distribution-connected data centers in London, the East, and the South East, as a half-hourly utilization ratio (observed apparent power divided by contracted import capacity), 2023 to 2026. No MW, no location, no price; 71 sites survive screening. Every statistic is a within-site ratio so that the contracted-capacity denominator cancels. Table 7 sets the UKPN numbers against its closest comparator.

Table 7. Flatness of the 71 UK data-center sites against matched comparators (medians across sites, annual basis).

Quantity (median across 71 sites, annual)	UKPN	Matched comparator
Diurnal spread (busiest hour divided by quietest hour, local time)	1.050	1.467 (ISO-NE, 27 zone-years)
Load factor against own 99th percentile / against own realized peak	0.836 / 0.723	0.88 colocation, 0.94 hyperscale (EPRI; resolution unstated)
Coefficient of variation (σ/μ)	0.094	n/a
Normalized hourly ramp	0.000240 (1.44%/h)	0.000574 (3.44%/h, 60 ISO zones)

Figure 28 shows the distributions behind the table: one panel per statistic, with the 71 sites as a histogram, the ISO-NE comparator marked, and the value for the aggregate of all sites marked separately.

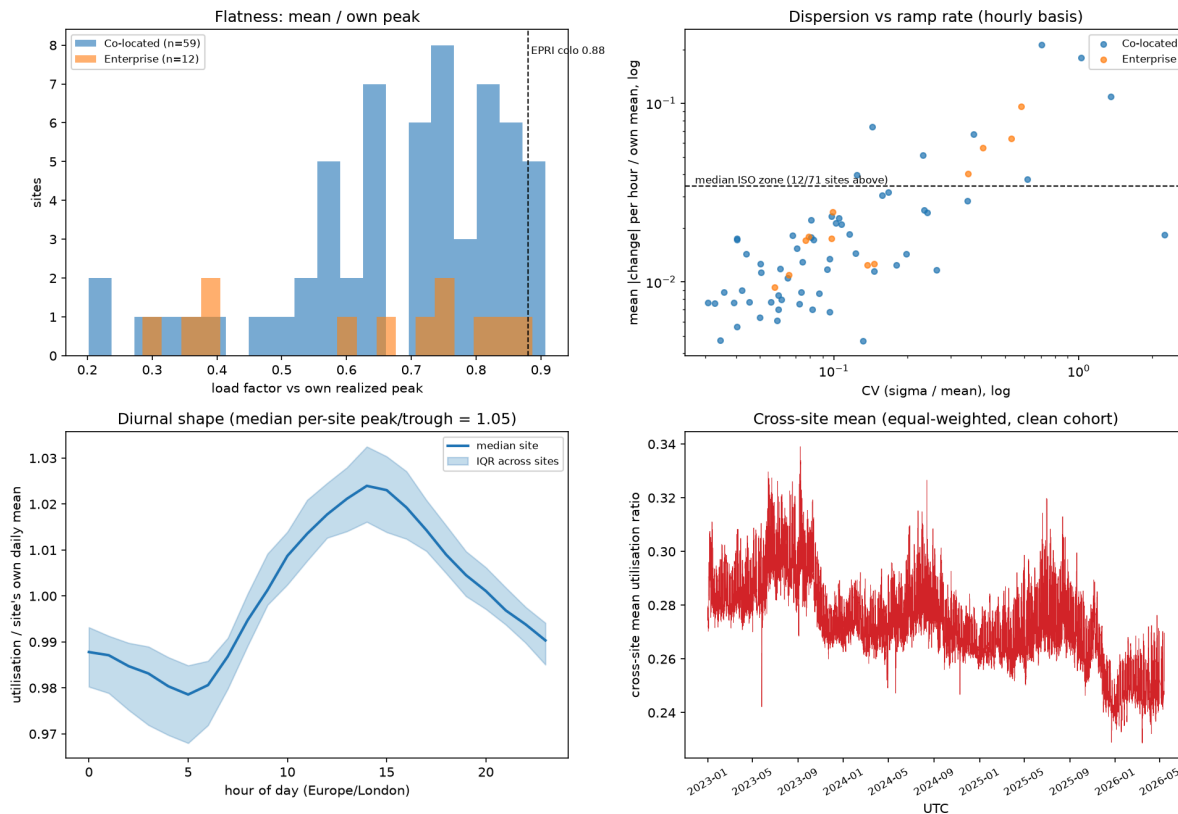


Figure 28. Note J. Distributions of the within-site flatness metrics across 71 UK data-center sites, the matched ISO-NE comparator, and the cross-site aggregate. The sites cluster far to the flat side of the ISO zones on every metric.

Thus, a few conclusions can be made from the UKPN dataset.

- There is almost no daily cycle: a 5% swing between the busiest and quietest hour, with 77.5% of sites below 1.10, while an ISO zone's system demand swings about nine times more across the day. Whatever the sites are doing, they are not following human activity as much as other sources.
- The sites do not move together. Each site on its own wanders by about 9% around its average over a year (a coefficient of variation, the standard deviation divided by the mean, of 0.094 for the median site), but the average across all sites wanders by only 3.4 to 4.5%, and the hour-to-hour ramps of that average are 0.31 times the typical site's in all three years. In other words, roughly two-thirds of what any one site does is cancelled out by the others: their ups and downs are idiosyncratic rather than synchronized. For a price question, that is the finding that matters, because the grid sees the sum, not the sites.
- There is no trend toward spikier on the balanced 71-site panel. EPRI (the Electric Power Research Institute) had reported, from 4 metered facilities, that data centers run close to their own peak nearly all the time; the same qualitative picture holds here on 71 sites. Whether the numbers match (EPRI's 88 to 94% against 72 to 84% here) cannot be settled, because it depends on what counts as the "peak": using each site's 99th-percentile hour instead of its single highest hour moves the answer by 11 percentage points, more than the gap between the two studies.
- Limits: these are probably conventional colocation and enterprise sites, not hyperscale AI training; the denominator is *contracted* capacity; and half-hourly resolution cannot see the seconds band.

Full record: [research-notes/J-ukpn-flatness.md](#); [sources/ukpn-api-constraints.md](#); [scripts/ukpn_flatness.py](#).

3.10 Europe: Ireland, the Netherlands, and Solar (Notes K and L)

Why Ireland. Its data-center exposure is *measured* (CSO table MEC02), the first time in the project after a summer of geographic proxies. Data-center consumption went from 291 to 1,991 GWh (6.84 times), all metered consumption grew 1.28 times, and non-data-center consumption was flat at 1.02 times: essentially every megawatt of Irish demand growth over eleven years was data centers, and their share went from 4.43% to 23.65%. The Netherlands, which matched on renewable share, was designed as a placebo (a country with the same kind of data but no data-center growth), and it turned out to be a low-dose control instead (CBS: 1.48% to 4.58%, tripled), which converted the design into a dose-response test: did the country with the larger data-center "dose" change more?

Note K: the shape changed, and the control changed too (2015 to 2025). Table 8 compares the two countries' hourly load series at the two ends of the window.

Table 8. Ireland against the Netherlands, 2015 to 2025: the change in each load-shape statistic, written as a multiplier (×0.719 means the 2025 value is 71.9% of the 2015 value).

Hourly panel	Ireland	Netherlands
mean load	×1.298	×1.196
raw mean $ \Delta\text{load} $ (MW/min)	×0.933	×0.854
$\text{vol_norm} = \Delta\text{load} \div \text{mean load}$	×0.719	×0.714
daily peak ÷ trough (median)	1.694 to 1.399	1.669 to 1.363
night floor (daily min ÷ mean)	0.722 to 0.823	0.715 to 0.840
$\text{corr}(\text{raw } \Delta\text{load} , \text{Irish DC share}), n = 44 \text{ quarters}$	−0.261	−0.268

The normalized decline is mostly a denominator effect (raw volatility fell 6.7% while load grew 30%), the ISO-NE artifact a second time and in the opposite direction. The mechanism is shared: the day flattened by filling in the night, with the largest change at 3 AM in both countries. Per percentage point of dose, the Netherlands moved 2.5 to 3.9 times *more*, so there is no dose ordering either way. Figure 29 shows the flattening directly: the average load at each hour of the day, scaled to the daily mean, in 2015 and in 2025. The night fills in, in both.

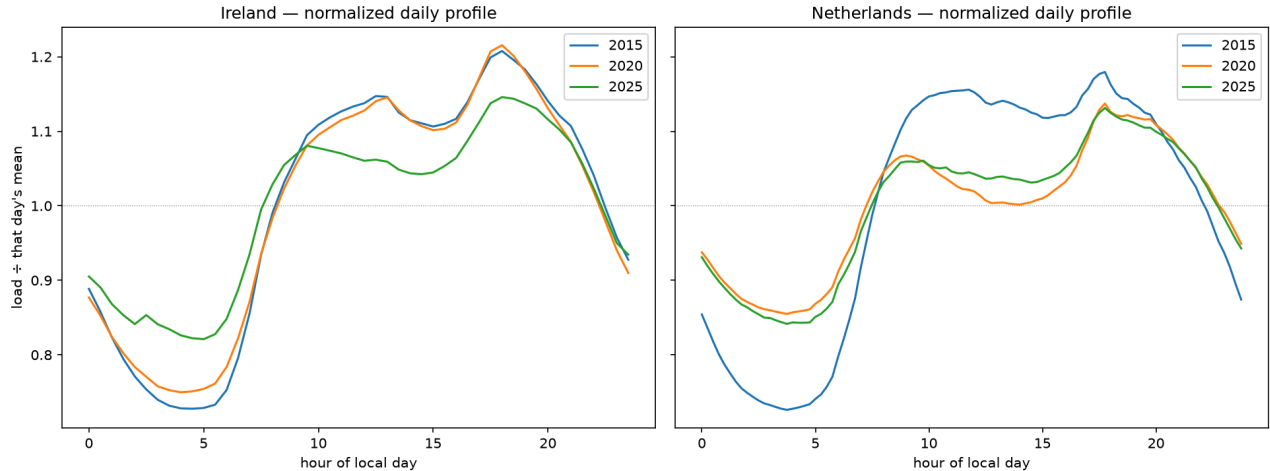


Figure 29. Note K. Normalized hour-of-day load profiles, Ireland and the Netherlands, 2015 against 2025. The night fills in, in both.

Moreover, Figure 30 tracks five yearly statistics of the load shape for both countries, 2015 to 2026. Each is computed from the hourly load series of one country in one year:

- `vol_norm` is the average hour-to-hour change in load, divided by that year's average load: how much the load moves, as a share of how big it is. This is the project's standard volatility measure.
- `mean_abs_grad` is the same average hour-to-hour change in raw MW per minute, without dividing by anything. It is plotted next to `vol_norm` because a ratio can fall simply because its denominator (load) grew, and this panel shows whether the numerator moved at all. The Dutch line sits higher only because the Dutch system is about 3.4 times larger.
- `pt_ratio` is the daily peak divided by the daily trough, for the median day of the year: how much bigger the busiest hour is than the quietest. A value of 1.7 means the peak is 70% above the trough; a flatter day gives a number closer to 1.
- `load_factor` is the year's average load divided by its single highest hour: how fully the system is used on average. Higher means flatter.
- `night_floor` is the daily minimum divided by the daily mean, for the median day: how far load falls at night. Rising toward 1 means the night is filling in.

Read together: in both countries the day got flatter (`pt_ratio` down, `night_floor` up), the normalized volatility fell, and the raw MW/min movement barely changed. The two countries move the same way, which is the point of note K.

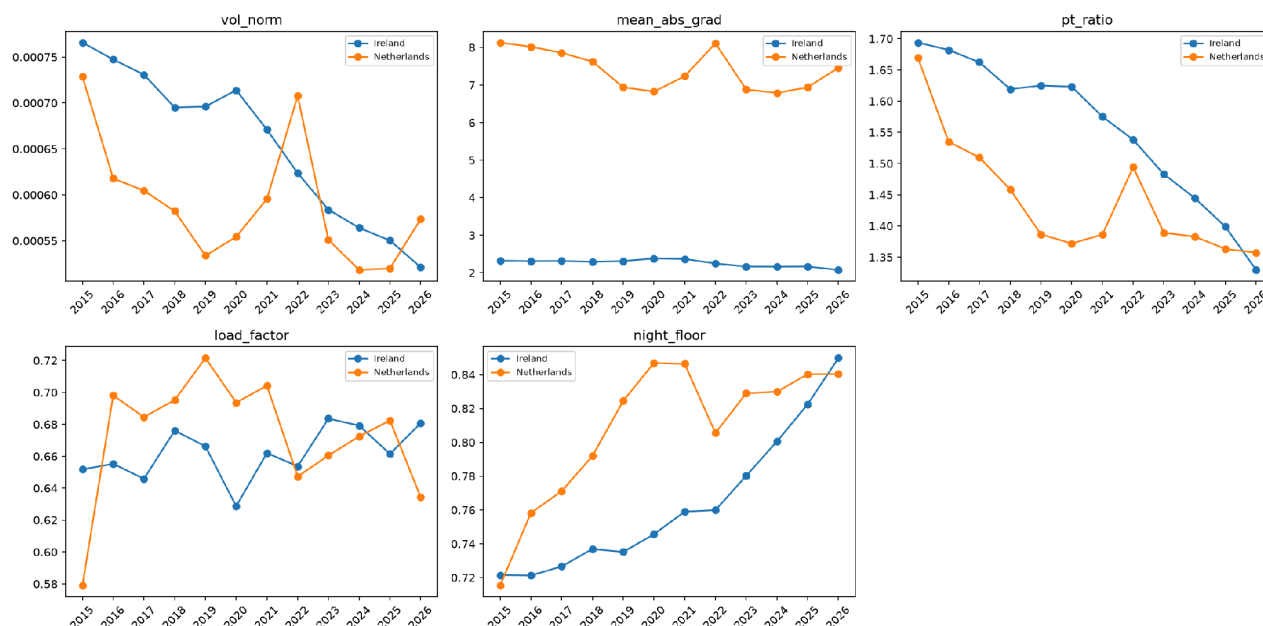


Figure 30. Note K. The five annual shape statistics defined above for Ireland and the Netherlands, 2015 to 2026.

Note L: the Dutch series breaks, and Ireland's flattening is seasonal.

Note L is the follow-up that checked whether note K's Ireland-against-the-Netherlands comparison survived two problems it had not anticipated: a definitional break in the Dutch series and a seasonal pattern in the Irish flattening.

- First, the Dutch load series changes definition in April 2023 (a midday-concentrated +1,395 MW step, absent in every other zone and year, found by comparing April against March in that year versus every other year and every other zone), so note K had compared a pre-break Irish endpoint with a post-break Dutch one. Re-run on the break-free 2015 to 2022 window, vol_norm is $\times 0.814$ for Ireland against $\times 0.971$ for the Netherlands: the matched null on vol_norm dies. What survives, cleaner, is the denominator finding (Irish raw volatility -3.2% against load $+18.9\%$) and the raw-numerator null ($r = -0.078$ for Ireland against -0.133 for the Netherlands, $n = 32$). Ruling: no consistent dose ordering in either direction, no detectable movement in raw volatility in either country, the Irish load-shape change still cannot be attributed to data centers.
- Second, Ireland's winter midday deviation went from +313 to +320 MW, unchanged across eleven years, while its summer midday deviation fell from +435 to +193 MW (-56%). Data centers run flat year-round and cannot produce a seasonal signature; rooftop solar can.
- Third, across 12 zones, the summer-minus-winter signature tracks installed solar with a rank correlation of $+0.714$ ($n = 7$). That is suggestive only, because ENTSO-E's installed-capacity feed covers about 100% of national PV in the Netherlands and about 3% in Finland; Finland (the lowest dose) does not move, and both Danish zones move against. Spain moves the solar way against its own cooling confound (summer -62% against winter -38%).

Figure 31 plots the summer-minus-winter signature against installed solar for every zone and year, and Figure 32 shows the seasonal daily profiles for Ireland that produced the second bullet.

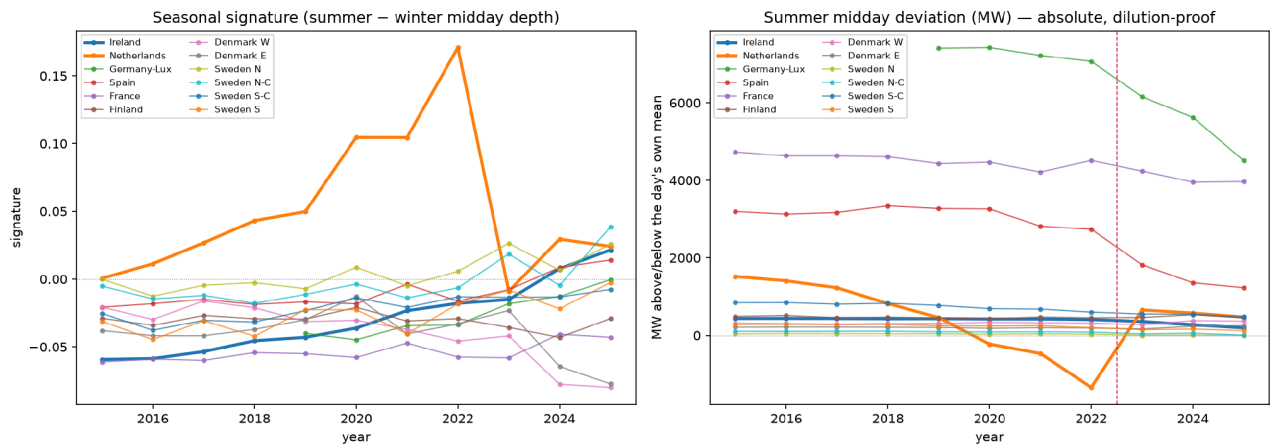


Figure 31. Note L. Summer-minus-winter midday signature by zone and year against installed solar capacity; Finland flat, Denmark against.

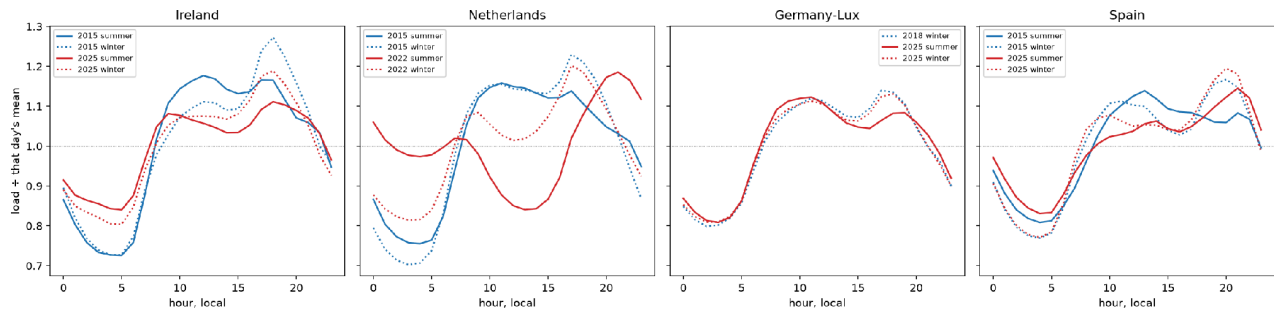


Figure 32. Note L. Seasonal diurnal profiles: Ireland's winter curve is unchanged in absolute MW; its summer midday has hollowed out.

It should be noted that an independent re-derivation of every headline number from the raw parquet files caught four defects in the original work (capacity years off by one; the break detector run on a different day sample; a corruption guard blind to intra-day corruption; a rank-correlation robustness check used as reassurance it could not provide). All are recorded and fixed; the load-only findings were untouched.

Full record: [decisions.md 2026-08-12](#) (four entries); [research-notes/K-..., L-...;](#) [plans/2026-08-12-entsoe-ireland-design.md](#); [src/surg/analysis/entsoe_seasonal.py](#).

3.11 Pecan Street and XFRA: Is There Panel Headroom for a Home Compute Node? (Note M)

The question. Professor Wei asked about SPAN and NVIDIA's XFRA program: a 16-GPU, roughly 12.5 kW always-on compute node plus a 15 kWh battery in new houses, in a 100-home pilot with PulteGroup. Does the "idle residential capacity" exist, and does it survive the hottest afternoon? The data is the Pecan Street dataset from 73 real homes (Austin 25, New York 25, California 23) at 1-minute and 1-second resolution, with whole-home draw reconstructed through the panel. Table 9 gives the answer for three electrical service sizes (the amperage rating of the home's main panel).

Table 9. Share of homes whose panel could fit a 12.5 kW firm node at the home's single worst minute of the year, by service rating (Austin / New York / California; the National Electrical Code's 80% continuous-load derating applied, meaning a circuit may carry only 80% of its rating continuously).

Node size	100 A service	150 A	200 A
12.5 kW firm, at the home's single worst minute of the year	0.04 / 0.16 / 0.48	0.84 / 0.91 / 0.91	1.00 / 1.00 / 1.00

The answer is decided by the electrical service rating, which the free tier never records. At 200 A, every home fits the node at its worst minute; at 100 A, that collapses to 4 to 48%. Excluding the homes enrolled in a load-changing intervention program (which leaves 9 Austin, 25 New York, and 12 California homes) leaves the 200 A result unchanged at 1.00 in all three cities and softens Austin's 150 A figure only slightly, from 0.84 to 0.78, so the headline is not an artifact of the intervention homes. The data is excellent and decisive, but the one variable that determines the answer is the one nobody measured: the project's binding constraint in miniature.

Homes barely touch their limits. The typical annual peak minute is 7 to 12 kW against about 38 kW usable on a 200 A panel, and most minutes sit below 5 kW. Two-thirds of Austin homes set their annual peak on a summer afternoon near 5 PM, and headroom never approaches zero. Fifteen-minute files understate true 1-minute peaks by about 20% (peak-shaving ratio 0.78 to 0.80). Figures 33 to 35 show, for each city, the distribution of whole-home draw and when each home's annual peak minute falls; Figure 36 shows one ordinary day in one home, with the 12.5 kW node drawn on top of it for scale.

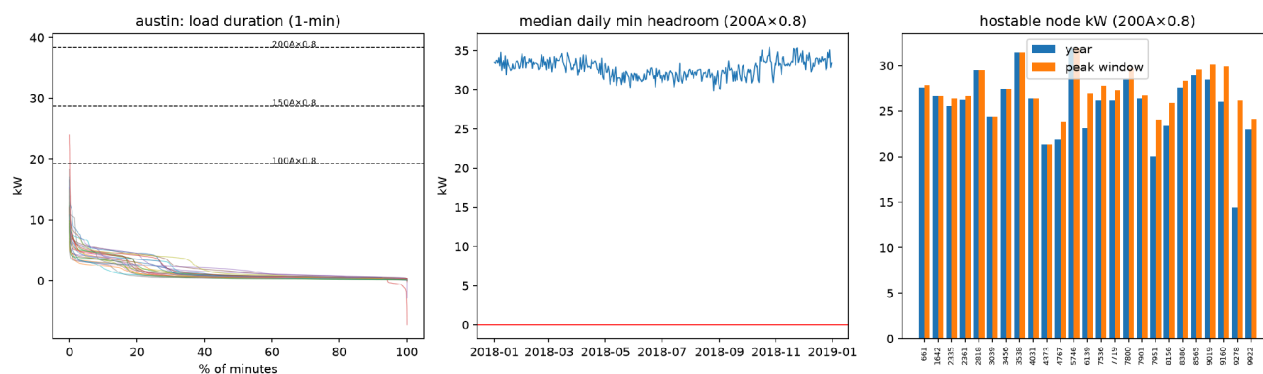


Figure 33. Note M. Austin: distribution of whole-home draw (left) and the coincidence of each home's annual peak minute with hour of day and season (right).

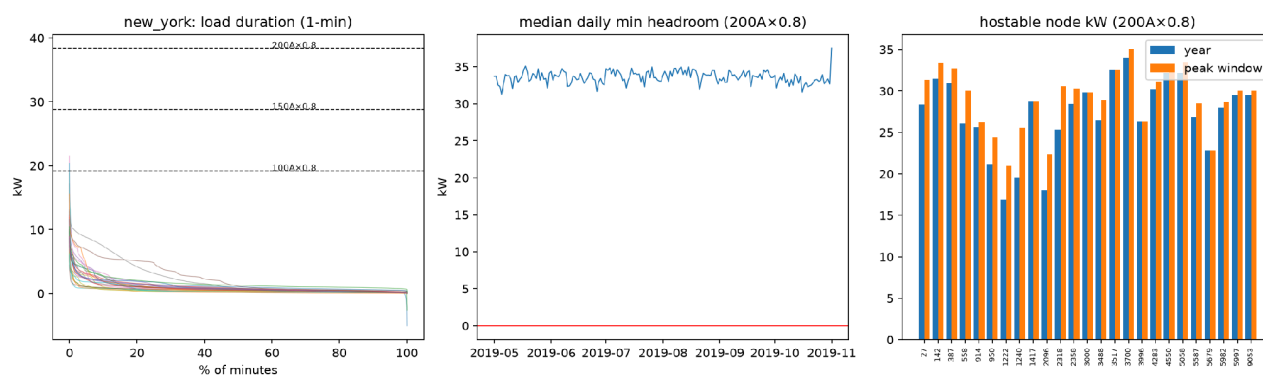


Figure 34. Note M. New York (six months, no winter): distribution of whole-home draw (left) and the coincidence of each home's annual peak minute with hour of day and season (right).

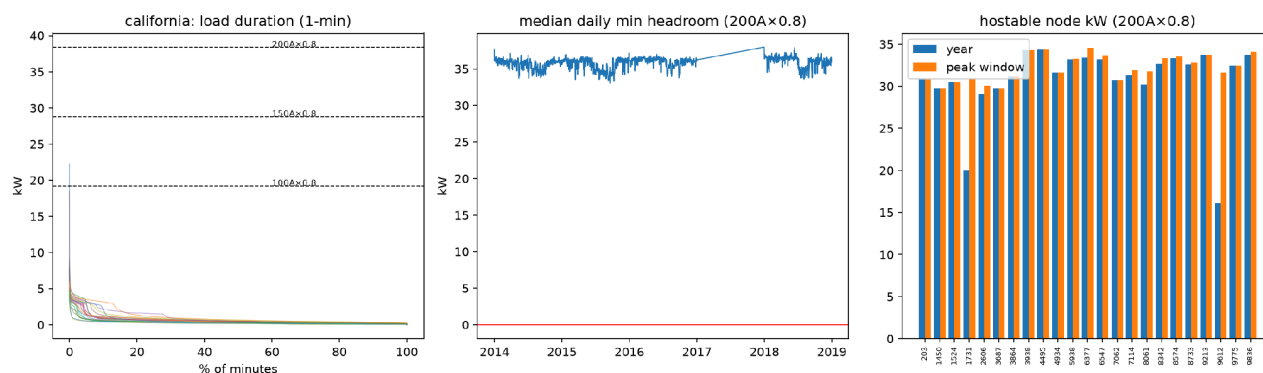


Figure 35. Note M. California (in fact San Diego; 1-minute data only): distribution of whole-home draw (left) and the coincidence of each home's annual peak minute with hour of day and season (right).

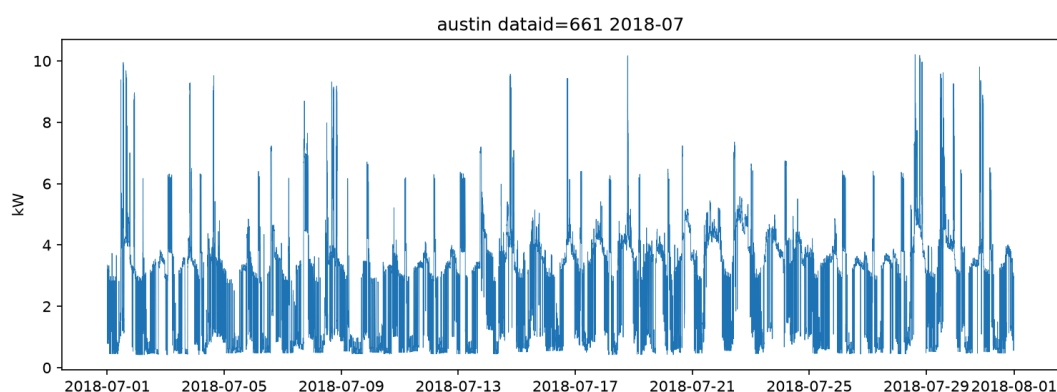


Figure 36. Note M. One Austin home, one day, 1-minute: what a 12.5 kW firm node would sit on top of.

The node is firm, not oscillating. The white paper (p. 15) keeps the node running and curtails the battery first and EV charging second, so XFRA is a capacity question, not a power-quality one. If a node did pulse at full amplitude, each swing would be about seven times a home's largest everyday step.

Household 1-second volatility is tiny and self-canceling. The median second-to-second change is about 2 W against 12,500 W for the node, and across N homes it cancels as $\sqrt{N}$ ($\sigma = 0.396$ kW for one home): pooling four times as many homes halves the relative wobble, because the homes' small movements are independent of each other. However, while the $\sqrt{N}$ result applies to residential homes, it does not generalize to all workloads' load profiles ([§3.12](#)). Figure 37 shows the cancellation as more homes are pooled, and Figure 38 shows the power spectral density (how much of the 1-second signal's variation sits at each frequency) of the whole-home draw.

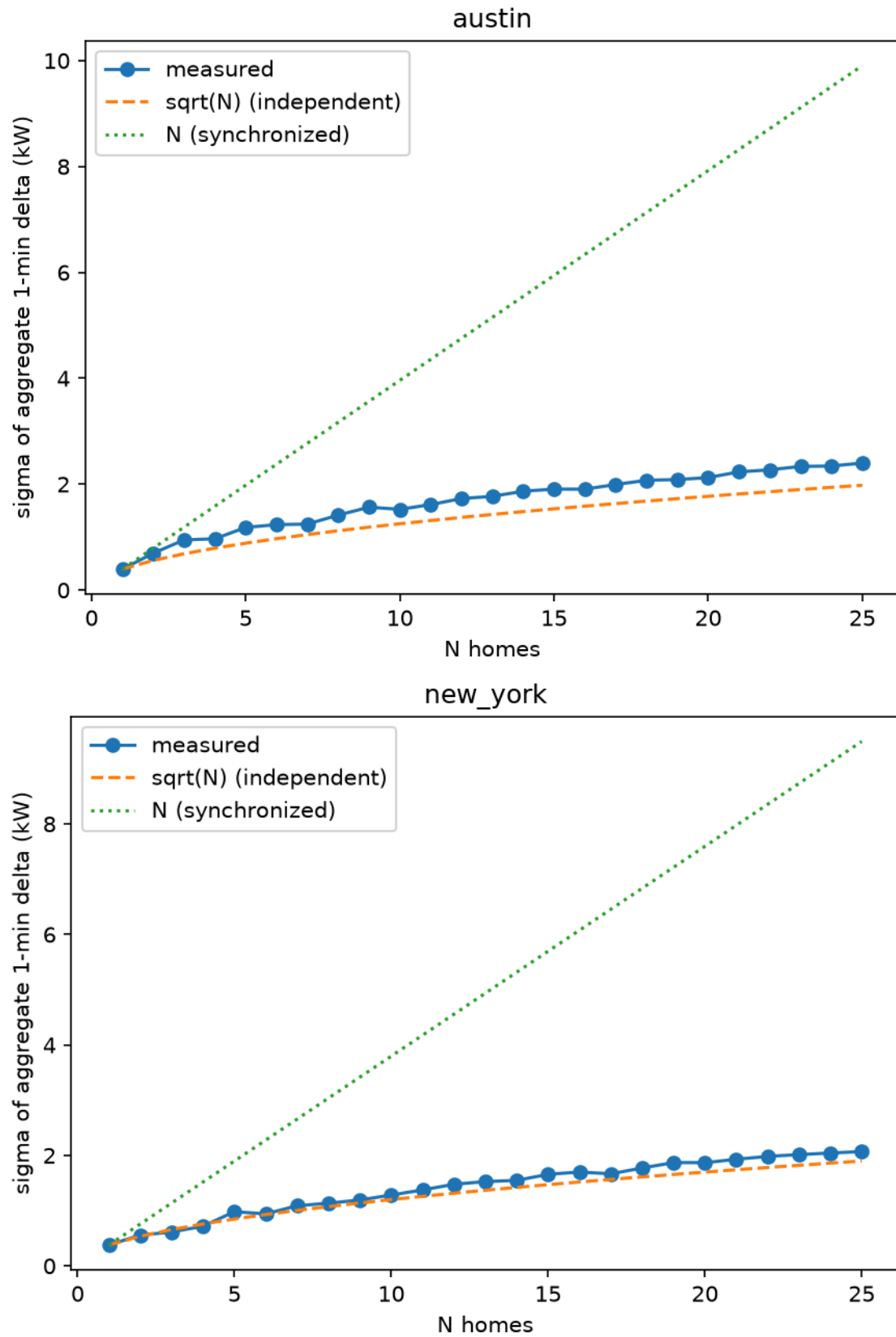


Figure 37. Note M. Aggregate 1-minute volatility against the number of homes pooled, Austin (top) and New York (bottom): $\sqrt{N}$ cancellation.

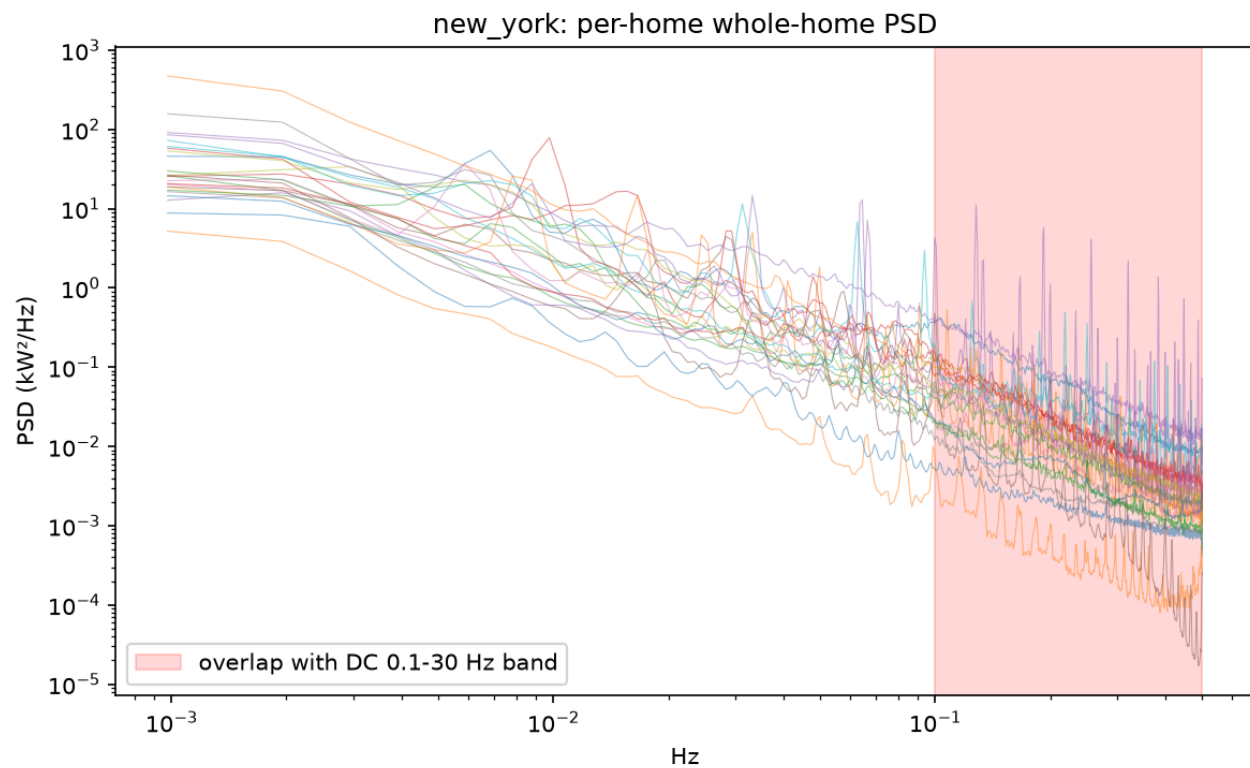
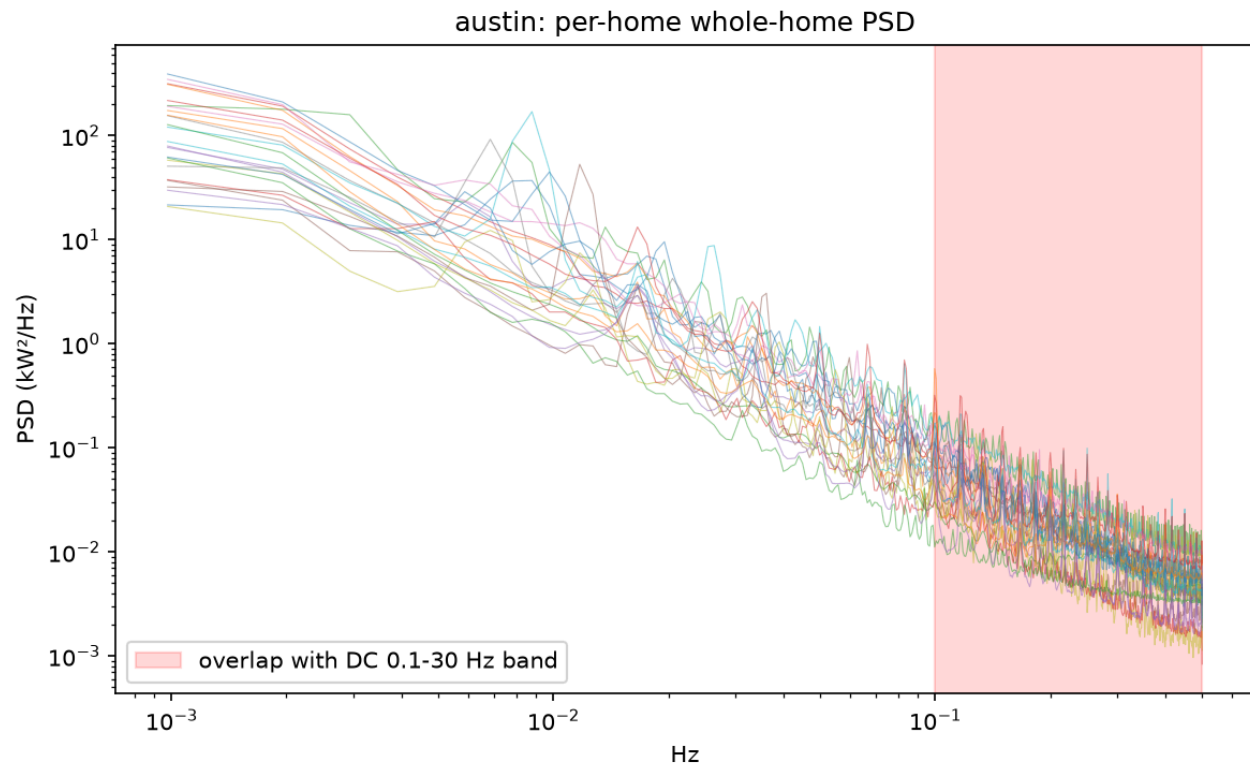


Figure 38. Note M. Power spectral density of 1-second whole-home draw, Austin (top) and New York (bottom). The Nyquist frequency, the highest frequency a 1-second sample can resolve, is 0.5 Hz.

Subtracting residential load from system totals cannot isolate data-center load, for four independent reasons: the sample is volunteers, it is not representative, the residual is all commercial and industrial load, and the data pre-dates modern AI hyperscaler build-outs.

Limits. The data is 2018 to 2019 vintage (before heat pumps, induction, and Level-2 EV charging spread); tilted to solar and EV owners; New York has six months and no winter; and 1-second sampling reaches only about 11% (linear) or 34% (log) of the measured 0.2 to 3 Hz band.

*Full record: [research-notes/M-pecanstreet-xfra-headroom.md](#);
[specs/2026-08-14-pecanstreet-xfra-headroom-design.md](#);
[sources/pecanstreet-access-constraints.md](#); [scripts/pecanstreet_*.py](#) (24 tests).*

3.12 Where the Sub-Second Claim Comes From, and Why Hourly Load Is Flat (Note N)

The question. After analyzing the Pecan Street data, a deeper dive into the load signatures of AI workloads (model training, fine-tuning, inference) was executed. One source, the NERC Large Loads Task Force (LLTF) slide deck, claimed that workloads caused load fluctuations at the second and sub-second level, up to tens of MW at a time. This prompted a deeper dive into this claim, described below.

Where "10 to 20 MW several times per second" comes from. Table 10 is the provenance ladder: every source found that supports the claim (within a tight timeframe; more sources definitely exist), from least to most substantiated.

Table 10. The provenance of the sub-second volatility claim.

Source	What it is	What it actually shows	Weight
#1	Tesla's slides at the NERC Large Loads Task Force workshop , with Musk's line from a podcast that AI loads shift "10 to 20 MW several times per second", and a Google technical lead's statement that batch-synchronous ML workloads on dedicated clusters show "power fluctuations in the tens of megawatts"	Nothing published and no data released, and Tesla is selling the Megapack battery that supposedly fixes it	Lowest
#2	arXiv 2508.14318 , <i>Power Stabilization for AI Training Datacenters</i> (Microsoft, OpenAI, and NVIDIA, August 2025, about 50 authors; not a neutral study)	Production telemetry: each training iteration has a compute phase with GPUs at maximum power and a communication phase at low power, and the switching is the swing; FFT energy (the share of the signal's variation at each frequency) concentrated at 0.2 to 3 Hz; amplitude grows in proportion to GPU count; GPUs are more than half of server power; jobs can span a majority of a data center or several on one grid. Every figure is normalized 0 to 1 with no MW axis, and no raw data is released	The best quantification that exists, and it is the operators' own
#3	The rs-7943457 dataset (York University and IESO; preprint): 32 real training runs on 8-GPU H100 and B200 nodes, per-GPU power at an effective ~10 Hz (the file says 50 Hz; the sensor refreshes every ~103 ms)	None of the sessions oscillate in the 0.2 to 3 Hz band; the cycle repeats every 8.5 to 15.9 seconds (0.06 to 0.12 Hz), but each transition inside it is a step of 2 to 4.5 kW completed in about a second, the abrupt change the claim describes at the scale of one node. Cross-GPU coherence inside the node is 0.994 to 0.995, independent confirmation that GPUs in a synchronized job move together	One node, one parallelism strategy, small models (frontier models are trillions of parameters)

Source	What it is	What it actually shows	Weight
#4	EPRI's Powering Intelligence (annual and peak use)	A footnote: "Despite relatively constant levels of output across minutes and hours, abrupt and large changes in load at second and sub-second timescales (that result from coordinated computing tasks stopping and starting) can have significant implications for operational reliability of the grid", with a link to an EPRI report that costs \$25,000. Unfortunately, the EPRI report's \$25,000 price tag exceeded the budget of this project by \$25,000	The duty-cycle numbers (94% and 88%) are metered; the sub-second claim is a footnote

NERC's May 2026 Reliability Guideline cites source #2 by name (footnotes 28 and 40) as the basis for its mitigations, so the regulatory framework rests on operator-supplied data, perhaps increasing its credibility.

The three filters between a GPU and a meter. Three things stand between a swing inside a server and what a grid meter records, and they do not all work in the same direction.

- **Averaging.** Any meter reports an average over its interval, and an average over a window of length T shrinks a fluctuation at frequency f by a factor of at least πfT . At the center of the measured band, a 250 MW campus swinging ± 112.5 MW leaves 0.05 MW of ripple in an hourly meter, 0.02% of the campus. Every dataset in this project would show a flat hourly load even if the industry claim were true in full; this is arithmetic, not a finding.
- **Coherence,** which does *not* filter for training. GPUs in a synchronized training job all switch at the same moment, so their swings add up: the aggregate amplitude scales with the number of GPUs N , and the $\sqrt{N}$ cancellation that holds for homes ([§3.11](#)) does not apply. This was measured inside a real 8-GPU node: the power of the eight cards moves together with a correlation of 0.994 to 0.995, a third and independent line of evidence for coherence. For *inference* (serving a trained model to users), requests arrive unsynchronized, and arXiv 2608.01250 finds the ramp-rate risk shrinking as $1/\sqrt{N}$. Whether neighboring campuses swing in step with each other has never been measured in the field; arXiv 2606.13853 assumes a correlation of about 0.4 in simulation.
- **On-site absorbers.** The uninterruptible power supply (UPS) or a battery can supply the swing so the meter never sees it, and the non-GPU load in the building (cooling, storage, networking) dilutes whatever gets through (although the impact is minimal; discussed in further detail below).

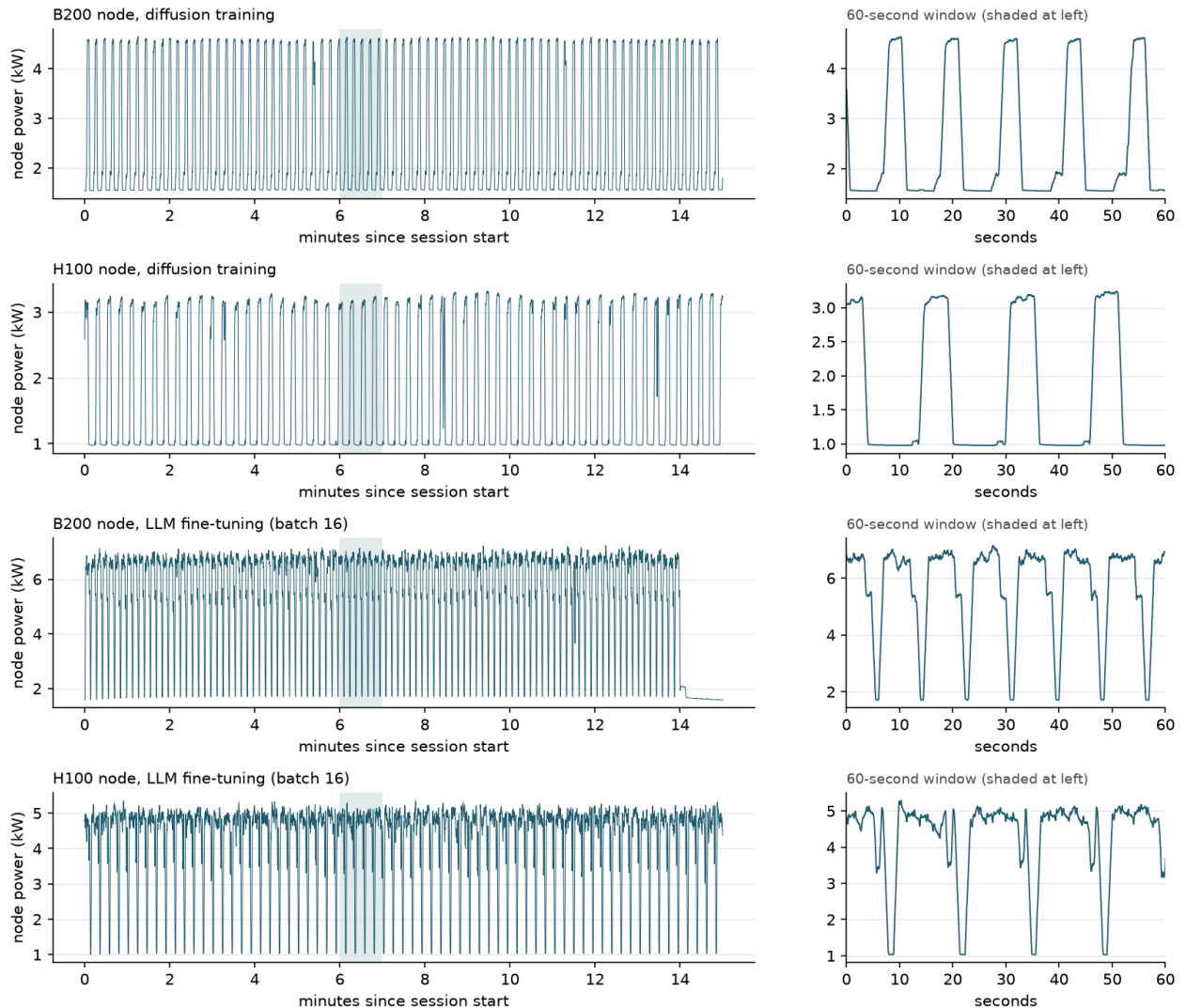
Additional relevant questions about why fluctuations don't seem to appear, and their respective answers.

- Is a battery fast enough to absorb second/sub-second fluctuations in load? Yes, easily. The measured band swings with periods of 0.33 to 5 seconds; a battery's power electronics respond in tens of milliseconds, and a full grid-forming response takes about 250 ms. Tesla's Megapack does exactly this at xAI's Colossus.
- Does data center facility HVAC cancel out fluctuations? No. A power usage effectiveness (PUE, total facility power divided by the power that reaches the computing equipment) of 1.10 (Google, Q2 2026) or 1.08 (Meta, 2023) means that cooling and everything else outside the servers is only about 9% of the load, and cooling has minutes-scale thermal inertia, so it cannot follow swings at 0.2 to 3 Hz.
- What does regulation say? NERC's White Paper 2 (March 2026) admits that no allowable-oscillation limit exists in any NERC standard and that standard instrumentation cannot characterize the fast band. ERCOT proposed a limit on 2026-02-19 ("load power shall not repetitively exceed 10 MW change in a sliding 5-second window"), a flat 10 MW peak-to-peak cap across the whole measured band, against which a frontier-scale cluster would be non-compliant by up to an order of

magnitude. Under such a rule, the room for oscillation at a given point on the grid becomes a first-come, first-served locational resource, structurally like congestion.

Additional analysis from the rs-7943457 telemetry (York University and IESO). Figure 39 shows what the four baseline sessions look like in the time domain, before any spectral analysis: the power drawn by the whole 8-GPU node, over the full 15-minute session on the left and over one 60-second window on the right. The cycle is plain to the eye: a rise to full power and a fall back toward idle every 8 to 16 seconds. What the repetition rate hides is how fast each transition is: the drop and the recovery inside every cycle are steps of 2 to 4.5 kW completed in about a second (0.7 to 1.8 s from 10% to 90% of the swing; Table 11), so the traces do contain the abrupt second-scale changes that EPRI's footnote and the operators describe, repeated slowly rather than oscillating at 0.2 to 3 Hz.

Power drawn by one 8-GPU training node, summed over its eight GPUs (rs-7943457 telemetry)



Per-GPU power sensors polled every 20 ms (the sensor itself refreshes about every 103 ms). Each session is about 15 minutes. The rise-and-fall cycle every 8 to 16 seconds is the training loop's duty cycle; the cycle is slow, but each drop and recovery inside it is a step of about a second (scripts/aidc_edges.py).

Figure 39. Node power (the sum of the eight GPUs' power sensors) over time for the four baseline rs-7943457 sessions: full session (left) and a 60-second window from its steady state (right, shaded on the left). Diffusion training switches cleanly between about 1.5 and 4.5 kW; LLM fine-tuning idles for shorter fractions of each cycle. The repetition is slow, but each drop and recovery is a step of about a second. Drawn by scripts/plot_aidc_timeseries.py.

Four sessions were analyzed (H100 and B200, diffusion and LLM fine-tuning of 1B to 8B models), with node power summed over the 8 GPUs, trimmed, detrended, and analyzed with a Welch power spectral density and a periodogram cross-checked against the autocorrelation (three standard ways of measuring which frequencies carry the variation). Table 11 gives, for each session, the dominant cycle period, how much of the variation falls in each frequency band, and how large and how fast each transition inside the cycle is: the node's low and high power levels, and the median time each drop and recovery takes to cover the middle 80% of that swing (10% to 90%).

Table 11. The project's own spectrum and edge speed: dominant period, share of variance by band, and the size and speed of each transition (medians over the session's cycles), four rs-7943457 sessions.

Session	Dominant period	Variance below 0.1 Hz	Variance 0.1 to 2 Hz (NERC)	Variance 0.2 to 3 Hz (measured)	Low to high (kW)	Fall / rise, 10 to 90% (s)	Ramp rate, fall / rise (kW/s)
B200 diffusion	11.4 s	72.9%	26.3%	5.4%	1.6 to 4.6	0.84 / 1.56	2.9 / 1.5
H100 diffusion	15.9 s	75.6%	24.3%	7.3%	1.0 to 3.2	0.82 / 0.92	2.2 / 1.9
B200 LLM (batch 16)	8.5 s	0.4%	99.5%	36.8%	2.4 to 6.9	1.79 / 0.74	2.0 / 4.8
H100 LLM (batch 16)	13.5 s	45.7%	54.2%	31.2%	2.0 to 5.0	0.70 / 0.68	3.5 / 3.6

Out of all of the sessions observed, none sustains an oscillation in the 0.2 to 3 Hz operator band; all repeat more slowly (0.06 to 0.12 Hz), but one node at one scale cannot say how frequency scales with job size. What every session does reproduce is the abruptness of the transitions: a drop of 2.2 to 4.5 kW in 0.7 to 1.8 s and a recovery in 0.7 to 1.6 s, 2 to 5 kW/s for a single node, which is the "coordinated computing tasks stopping and starting" of EPRI's footnote at the scale of one server. Because the GPUs in a synchronized job switch together (the 0.994 to 0.995 coherence above), steps of that speed add up across nodes rather than cancelling, and the megawatt swings the operators report are what such steps look like at the scale of thousands of nodes. So the telemetry partially supports the operator claim rather than contradicting it: it shows the fast edges at 8-GPU scale, but not a sustained 0.2 to 3 Hz oscillation.

Scaling the node to a training job. The claim is about jobs of tens of thousands of GPUs, so the last step is to ask what the node's transitions would look like at that scale (`scripts/aids_scale.py`). Divided by its eight cards, the node's step is 0.28 to 0.56 kW per GPU, 61 to 69% of the node's high level. If every GPU in a job stepped at the same moment, which is what the 0.994 within-node coherence suggests and what the operators' account requires, the step would grow in proportion to the GPU count, and Table 12 gives the result for four job sizes taken from public descriptions of real training clusters. A 10 MW step takes about 18,000 to 36,000 synchronized GPUs and a 20 MW step about 36,000 to 72,000, the size of the jobs that arXiv 2508.14318 describes, so the operators' 10 to 20 MW sits inside the range the node telemetry implies for them (the paper's upper figure of over 100 MW would take roughly 180,000 to 360,000 GPUs at these per-GPU steps). The ramp rate scales the same way, 7 to 22 MW/s at 36,000 GPUs, so a single step of that size would exceed ERCOT's proposed limit of 10 MW in any 5-second window on its own. Three caveats bound the arithmetic. The sessions are small models running below the cards' rated power (0.4 to 0.86 kW per GPU at the high level, against 0.7 kW for an H100 and 1 kW for a B200), so a frontier job's per-GPU step could be larger; coherence across nodes is assumed, not measured; and the number is the load before any UPS or battery absorbs it, which is the quantity the claim is about.

Table 12. The node's per-GPU step scaled to a synchronized training job of N GPUs (perfect coherence assumed; per-GPU step 0.28 to 0.56 kW and ramp rate 0.19 to 0.61 kW/s from the four sessions in Table 11; each range is the spread across sessions).

GPUs in the job	What that is	Step	Ramp rate
16,384	Llama 3 405B pre-training [108]	4.5 to 9.2 MW	3.2 to 9.9 MW/s
24,576	One of Meta's two 2024 H100 clusters [109]	6.8 to 13.8 MW	4.7 to 14.9 MW/s
50,000	A round number	13.8 to 28.0 MW	9.7 to 30.3 MW/s
100,000	xAI's Colossus, first phase [110]	27.6 to 56.0 MW	19.3 to 60.6 MW/s
17,800 to 36,200	GPUs needed for a 10 MW step	10 MW	
35,700 to 72,400	GPUs needed for a 20 MW step	20 MW	

Full record: research-notes/N-subsecond-provenance-and-filters.md (\$0 to 11); scripts/aide_psd.py; scripts/aide_edges.py; scripts/aide_scale.py; plans/advisor/2026-08-24-advisor-meeting-agenda.md.

3.13 Desk Research Digest

The notes below collect additional, scattered desk research. It provides context rather than findings, and some of it is only loosely tied to the main question.

A, primary-source verification. The trade-press claims about the FERC (Federal Energy Regulatory Commission) co-location order, the Virginia SCC (State Corporation Commission) rate cases, and PJM's transmission plan are all confirmed by the primary records: 193 FERC ¶ 61,217 stands with its compliance process still open, the SCC created the GS-5 class for customers of 25 MW and up, and PJM's board approved \$11.84B of new projects, including a 3,000 MW underground HVDC line into Loudoun.

B, Loudoun geography. Goose Creek substation is the terminus of two concurrent import builds, and SKFFSCRK, a 500 kV switching station about 150 miles away on the Hampton Roads peninsula, still tracks the Loudoun cluster at $r = 0.870$, so congestion in the pocket is network-wide rather than local.

C, capacity and forecast. PJM's capacity auction clearing price went from \$28.92/MW-day to \$269.92 in a single year and has held at \$325 to \$333 since, the RTO (regional transmission organization, PJM's whole footprint) is 6,600 to 6,800 MW short of its reliability requirement, and DOM's import capability sits below its requirement while the Chanceford to Doubs line is delayed.

D, JLARC and LBNL. JLARC finds that data centers currently pay the full cost of serving them and projects +\$14 to \$37 a month for a Dominion residential customer by 2040, while LBNL (Lawrence Berkeley National Laboratory), Brattle, and E3 back the "rates aren't rising" claim and the Harvard Electricity Law Initiative and the PJM market monitor rebut it.

E, flexible load. Curtailable data-center load is mostly announced rather than demonstrated: Emerald AI showed a 40% reduction in under 60 seconds at five sites, EPRI's DCFlex has nine sites and no quantified results yet, Google has 1 GW of demand response under contract with no dispatched event reported, and the 2026-07-22 Ashburn event moved 3.1 GW.

F, the January 2026 driver. Winter Storm Fern ranks first as the explanation of the simultaneous January 2026 step in both price components ([§3.3](#)).

G, control nodes. A candidate list of control buses outside DOM, with the structural finding that the good control sites are exactly where data centers are now siting (Mountaineer dropped, Conemaugh at elevated risk).

H, the event catalog. Thirty dated PJM and DOM stress events from February 2023 to July 2026 ([§3.5](#)).

External context. PJM's compliance proposal makes new loads above 50 MW ineligible for netting, 9.8 GW of nuclear and small modular reactor (SMR) co-location is committed nationally, and CRS (Congressional Research Service) report R48646 finds that the states with the largest data-center growth generally saw prices fall from 2019 to 2025 except where the grid is constrained, which is the DOM story stated nationally.

I, Professor Wei's links. Across the five links from the 2026-08-10 meeting (PJM's machine-learning forecasting deck, EPRI's *Powering Intelligence* and headroom framework, the NERC task-force deck, and arXiv 2601.02275), not one claims an energy-price effect from load volatility; every stated consequence is reliability.

EU-0 to EU-5. The method does not port to Europe, because the day-ahead coupling algorithm prices congestion inside a bidding zone at exactly zero and the EU facility database is legally closed, but the scoping produced UKPN's facility profiles ([§3.9](#)), Ireland's measured dose ([§3.10](#)), and the solar question that note L went on to test.

The JLARC napkin projection. With JLARC's 1.95, 2.23, and 2.90 times scenarios and the quantile-regression slopes, the \$850 benchmark was uninformative, and the formal projection layer was scrapped after the pivot away from the original proposal's question.

4. Papers, Reports and Primary Sources Read

Sources are grouped by what they were used for in the order they were read: the original proposal's references, the PJM and Virginia-specific material, the cross-ISO sources, the European sources, and finally the additional resources considered. Each row contains one reference, what it is about, what was found in it, and what was taken away from it. The bracketed numbers point to the full citations in §6, and each source links to its document.

Table 13. Framing and mechanism (the proposal's references).

Ref	Source	What it is about	What was found	What was taken away from it
[1]	JLARC, <i>Data Centers in Virginia</i>, Report 598 (December 2024)	Virginia's legislative audit of the data-center industry: demand forecasts to 2040 under 1.95×, 2.23×, and 2.90× scenarios, and rate-impact projections	It treats data-center load as flat baseload on annual and monthly data; finds data centers currently pay the full cost of serving them; projects +\$14 to \$37 a month for a Dominion residential customer by 2040	The growth scenarios and the 0.26 data-center share of DOM load for the (later abandoned) projection, and the "chronic state" half of the original question
[2]	Y. Li and Y. Li, <i>AI load dynamics: a power electronics perspective</i> (arXiv 2502.01647)	AI training load from the power-electronics side, with GPU-level measurements	It measures a single workstation, not a facility; the megawatt claims are extrapolation	The "spiky load" premise. Its evidence is single-GPU, not facility-scale (§3.12)
[3]	X. Chen et al., <i>Electricity demand and grid impacts of AI data centers</i> (arXiv 2509.07218)	A survey of AI data-center electricity demand and grid impacts	A review of the same operator claims, with no new measurement	Premise and framing for the proposal
[4]	PJM Manual 11, <i>Energy & Ancillary Services Market Operations</i> (revision 137)	PJM's rules for the energy and reserve markets, including the reserve demand curves	The \$850 and \$300 steps sit on every reserve demand curve (section 4.3.3); congestion is priced on line loadings (section 2.2); load volatility has no route into the scarcity channel	The price-formation model in §0 and the ORDC figure
[5]	PJM Manual 12, <i>Balancing Operations</i> (revision 57)	How PJM keeps generation and load matched in real time, and how reserve shortages are declared	Shortages are declared per reserve product and location, which is what triggers the ORDC penalty	The scarcity-channel half of §0
[6]	PJM Manual 03, <i>Transmission Operations</i> (revision 71)	How transmission constraints are monitored and controlled	The constraint-control actions that produce congestion prices	The congestion-channel half of §0
[7]	Monitoring Analytics, <i>2023 State of the Market Report for PJM</i>, Volume 1	The independent market monitor's annual report on PJM for 2023	The Pleasant View to Ashburn and Goose Creek constraints, with DOM congestion costs already rising	The choice of Loudoun nodes (§0)
[8]	Monitoring Analytics, <i>2024 State of the Market Report for PJM</i>, Volume 1	The same report for 2024	The same constraints named again, and the census of PJM alerts for 2024	The choice of Loudoun nodes (§0) and the event catalog (§3.5)
[9]	Monitoring Analytics, <i>2025 State of the Market Report for PJM</i>, Volume 1	The same report for 2025	DOM congestion costs and the census of PJM alerts for 2025	The event catalog (§3.5)

Ref	Source	What it is about	What was found	What was taken away from it
[10]	Monitoring Analytics, 2026 Quarterly State of the Market Report for PJM: January through March	The monitor's quarterly report covering January 2026	The monitor's own diagnosis of the January 2026 price step	§3.3 and the event catalog (§3.5)
[11]	PJM, Formation of LMP and its system energy component during reserve shortage events (March 2023)	How the reserve-shortage penalty enters the price, worked through a real shortage day	The penalty is a step function that lands in the uniform system energy component; the congestion and loss components can each be positive or negative	The reason the tail model was dropped as the primary test on day one and threshold plus quantile regression took its place; the reading of the negative congestion in Figure 16 (§3.5)
[12]	W. Hogan and S. Harvey, Locational marginal prices and electricity markets (2022)	The economics of locational marginal pricing	LMP = system energy + congestion + marginal loss	The three-component price definition used throughout
[13]	P. Hines et al., Cascading failures in power grids (IEEE Potentials, 2009)	Self-organized criticality in power grids	Grids behave linearly up to a stress limit and then cascade	The "phase transition" framing of the proposal, which the price data did not support
[14]	Z. Yang et al., distribution LMPs for a data-center park (2023)	An optimization model of a data-center park under distribution-level LMPs	Demand-side resources can be scheduled against distribution prices	Background on distribution-level pricing; not used in the analysis
[15]	J. Lindberg et al., locational marginal CO₂ (2021)	Using locational marginal emissions to shift hyperscale load between regions	Geographic load shifting is a real lever for hyperscalers	Background on load as something operators move; not used in the analysis
[16]	B. E. Hansen, Inference when a nuisance parameter is not identified under the null hypothesis (1996)	Testing for a threshold whose location is unknown, when there may be none	The bootstrap p-value for the threshold test	The p-value in the TAR test (§3.1)
[17]	B. E. Hansen, Sample splitting and threshold estimation (2000)	Estimating a threshold and its confidence interval	The sample-splitting estimator	The TAR estimator that found, and then un-found, the threshold (§3.1)

Table 14. PJM and Virginia policy and markets (notes A to H, external context).

Ref	Source	What it is about	What was found	What was taken away from it
[18]	FERC, Order on Show Cause Proceeding, 193 FERC ¶ 61,217 (2025-12-18)	FERC directing PJM to rewrite its rules for co-located (behind-the-meter) load and generation	PJM's tariff was found unjust and unreasonable; the 50 MW netting threshold is a PJM convention chosen to capture hyperscale loads	The answer to "why would transmission costs fall on homeowners": co-located load still leans on the grid for backup and reliability, and fixed costs are recovered per billed MWh (note A)
[19]	FERC, orders on PJM's compliance filings: Docket ER26-1088 (2026-04-16) and	FERC's rulings on PJM's first two attempts at the new rules	Both filings were rejected in part; the process is still open	A shelf-life caveat on all metered-load data:

Ref	Source	What it is about	What was found	What was taken away from it
	Docket EL25-49-002 (2026-06-18)			what counts as load is still being redefined
[20]	PJM, thirty-day compliance filing (2026-01-20, Docket ER26-1088) and sixty-day compliance filing (2026-02-23, Docket ER26-1479) in response to the EL25-49 order	PJM's proposed co-location rules	New loads above 50 MW would be ineligible for netting; netted behind-the-meter generation is invisible in metered load	Note A and the external-context digest (§3.13)
[21]	Virginia SCC, biennial review order PUR-2025-00058 (2025-11-25)	Dominion's rate review	A new GS-5 class for customers of 25 MW and up, with 85% and 60% minimum demand charges from 2027-01-01	The state's own answer to who pays (note A)
[22]	Virginia SCC, Rider T-1 final order PUR-2026-00056 (2026-07-31)	Dominion's transmission rider	The residential rider cut from \$2.90 to \$0.94 a month, and 90 days for Dominion to produce a direct-connect cost policy	Note A
[23]	PJM, 2026 PJM load forecast report (January 2026)	PJM's long-term load forecast	DOM growing at 6.5%/yr against 1.6%/yr RTO-wide	Note C
[24]	PJM, 2025/2026 base residual auction report	The capacity auction for 2025/26	A clearing price of \$269.92/MW-day, up from \$28.92 the year before; DOM cleared at a \$444.26 zonal cap	Note C
[25]	PJM, 2026/2027 base residual auction report	The capacity auction for 2026/27	\$329.17/MW-day under the FERC-approved price collar (ER25-1357)	Note C
[26]	PJM, 2027/2028 base residual auction report (December 2025)	The capacity auction for 2027/28	\$333.44/MW-day; the collar masked a DOM-specific adder that binds in the uncapped simulation (\$542.83 against \$529.80 RTO-wide); the RTO 6,623 MW short of its reliability requirement	Note C
[27]	PJM, 2028/2029 base residual auction report (July 2026)	The capacity auction for 2028/29	\$325.00/MW-day under the extended collar (ER26-1556); the RTO 6,831 MW short; DOM's import capability below its requirement while the Chanceford to Doubs line is delayed	Note C
[28]	PJM Transmission Expansion Advisory Committee (TEAC), white paper to the Board on the 2025 RTEP window (approved 2026-02-12)	The transmission plan approved by PJM's board	\$11.84B of new baseline projects, including the roughly \$2.3B, 185-mile, 3,000 MW underground HVDC line into Loudoun (June 2032)	Note C
[29]	PJM, PJM reviews January cold weather operations (Inside Lines, 2026)	PJM's review of Winter Storm Fern	18 to 19 GW of forced generator outages, about \$798M of gas-electric misalignment uplift, spot	Winter Storm Fern as the best explanation of the January 2026 step (§3.3)

Ref	Source	What it is about	What was found	What was taken away from it
			prices above \$3,000/MWh PJM-wide	
[30]	PJM, System performed well during Winter Storm Gerri (Inside Lines, 2024)	PJM's review of the January 2024 cold-weather event	A cold-weather alert with a far smaller outage total than Fern	The January 2024 entry in the event catalog (§3.5)
[31]	PJM, PJM serves load through record-breaking July heat (Inside Lines, July 2026)	PJM's review of the July 2026 heat	The all-time PJM peak of 168,158 MW on 2026-07-02	The event catalog (§3.5)
[32]	NERC, 2026 Summer Reliability Assessment	The reliability regulator's summer outlook	Large loads treated as a planning obligation rather than an emerging risk	The regulator's own framing (note A)
[33]	NERC, 2025 Long-Term Reliability Assessment	The ten-year reliability outlook	The resource-adequacy picture that note C's PJM shortfalls sit inside	Note C
[34]	NERC, incident review of the 2024-07-10 large load loss	The regulator's review of the one verified event	About 1,500 MW across 60 facilities dropped to backup power after a 230 kV fault	The verified event (§3.5)
[35]	NERC, Level 3 alert on computational loads (2026-05-04)	NERC's "Essential Action" alert, issued with the May 2026 Reliability Guideline [88]	Large loads escalated to the regulator's highest alert level	The event catalog (§3.5)
[36]	Congressional Research Service, R48646, Data Centers and Their Energy Consumption: FAQ (May 2026)	A primer for Congress on data-center energy use, prices, and data collection	States with the largest data-center growth generally saw prices fall from 2019 to 2025, except where the grid is constrained; the Energy Information Administration's (EIA) data collection failed twice; the Clean Cloud Act would give the EPA and the EIA collection authority	The national statement of the DOM story, and the "data hidden even from government" conclusion (§5)
[37]	A. Shehabi et al., LBNL, 2024 United States Data Center Energy Usage Report (December 2024)	The national data-center energy estimate	LBNL's national numbers; LBNL itself flags PJM as the exception to the flat-rates picture	Note D
[38]	R. Wiser et al., LBNL and Brattle, Retail Electricity Price Trends and Drivers: Data Update, 2026 edition (March 2026, updated July 2026)	Retail price trends and what drives them	Backs the "rates aren't rising" claim nationally	Note D
[39]	E3, Understanding the Drivers of Rising Electricity Rates and the Role of Data Centers (May 2026)	The same question, funded by the Data Center Coalition	Backs "rates aren't rising"	Note D
[40]	E. Martin and A. Peskoe, Harvard Electricity Law Initiative, Extracting Profits from the Public (March 2025)	The rebuttal: how ratepayers pay for Big Tech's power	Rebuts the "rates aren't rising" claim, as does the PJM market monitor	Note D; JLARC's numbers are the ones used
[41]	T. Norris et al., Duke, Rethinking Load Growth (February 2025)	Whether flexible large loads can be absorbed by existing systems	Argues that flexible load can be absorbed with little new capacity	Notes D and E

Ref	Source	What it is about	What was found	What was taken away from it
[42]	DOE §202(c) order 202-26-02 (January 2026)	The emergency order during Winter Storm Fern	PJM's data centers ordered onto backup generation	The event catalog (§3.5) and the January 2026 step (§3.3)
[43]	DOE §202(c) order 202-25-8 (August 2025)	An earlier emergency order under the same authority	The scope of DOE's emergency powers over grid operators	Note E
[44]	DOE §202(c) order 202-26-23 (June 2026)	A further emergency order for PJM	The same authority used again in summer 2026	The event catalog (§3.5)
[45]	Texas Senate Bill 6 (89th Legislature, 2025)	Texas's large-load law	Loads above 75 MW must join demand response	Note E
[46]	PUCT docket 59220, the Goodnight ruling (2026-07-23)	Curtailed of a co-located campus in Texas	An entire co-located campus made curtailable within 30 minutes	Note E
[47]	NVIDIA, Emerald AI case study	A demonstration of flexible AI load	A 40% reduction in under 60 seconds at five sites	Note E: demonstrated against announced
[48]	EPRI, DCFlex expands to nine demonstration sites	EPRI's flexible-data-center initiative	Nine sites and no quantified results yet	Note E
[49]	Google, A milestone for demand response at our data centers	Google's demand-response contracts	1 GW under contract with no dispatched event reported	Note E
[106]	PJM, zone map (2023-05-11)	PJM's official map of its transmission zones	Where the Dominion zone sits relative to the other 20 zones	Figure 2 (§3.1)

Table 15. Cross-ISO sources (sources/availability/).

Ref	Source	What it is about	What was found	What was taken away from it
[50]	ISO New England, ISO-NE establishes forecast framework for data centers, other large loads (May 2026)	New England's load outlook	"Has not experienced similar growth so far, and only a small amount is expected in the coming decade"	The control-market designation (§3.7); read with ISO-NE's external-interface list (the ATCID) and the NECEC commercial-operation notices, the Canada-tie work (§3.8)
[51]	EIA, Hourly Electric Grid Monitor (EIA-930)	Hourly demand and interchange for every US balancing authority	The ISO-NE to Hydro-Québec interchange series omits NECEC	The NECEC dose check (§3.8)
[52]	MISO, 2026 Long-Term Load Forecast (April 2026)	MISO's own view of data-center growth	Data centers from 9.6 to 266 TWh by 2046, a 28-fold increase	The MISO description in §3.7
[53]	IESO, 2026 Annual Planning Outlook	Ontario's planning outlook	Growth is EV-driven; data-center materialization is called "uncertain"	The IESO description in §3.7
[54]	SPP, High Impact Large Load (HILL) integration	SPP's large-load program	Interconnection agreements promised within 90 days; a near-doubling of peak load expected within ten years	The SPP description in §3.7

Ref	Source	What it is about	What was found	What was taken away from it
[55]	NYISO, Load & Capacity Data Report (the Gold Book)	New York's load and capacity data (Table IV-7)	NYISO's forecast of large-load additions	The NYISO description in §3.7
[56]	University of Texas at Austin, Bureau of Economic Geology, Advancing sustainable data center development in Texas	The "TRAIL Map" white paper, read with ERCOT's large-flexible-load reports	ERCOT's large flexible load was 54 TWh in 2025 with a mostly speculative queue; the TRAIL Map is a policy paper, not a data release	The ERCOT description (§3.6), and the finding that no public facility-level source exists for Texas
[57]	Pecan Street, Waveform release (October 2025)	The Dataport catalog, data dictionary, and the planned 2 kHz release	The free tier stops at 1 second; the waveform release is a request flow	§3.11 and the open 2 kHz question
[58]	PJM Data Miner 2	PJM's public data portal	See Source 1 in §2	Datasets 1 to 6 and 38
[59]	gridstatus.io	A hosted warehouse of ISO data	See Source 2 in §2	Datasets 7 to 9
[60]	ERCOT market information archives	The Texas operator's public archives	See Source 3 in §2	Datasets 10 and 11
[61]	MISO market reports	The Midcontinent operator's report files	See Source 4 in §2	Datasets 12 and 13
[62]	SPP file portal	The Southwest Power Pool's file browser	See Source 5 in §2	Datasets 14 and 15
[63]	NYISO market information system archives	The New York operator's monthly CSV archives	See Source 6 in §2	Datasets 16 to 18
[64]	ISO New England, SMD hourly data workbooks	The New England operator's annual workbooks	See Source 7 in §2	Dataset 19
[65]	IESO public reports	Ontario's public report directory	See Source 8 in §2	Datasets 20 to 22
[66]	CAISO OASIS	The California operator's data API	See Source 9 in §2	Datasets 23 and 24
[67]	CSO PxStat API, table MEC02	The Irish statistics office's API	See Source 12 in §2	Dataset 31
[68]	Pecan Street Dataport	The residential corpus	See Source 14 in §2	Datasets 33 to 36 and 39
[107]	ERCOT, weather zone map	ERCOT's official map of its eight weather zones	Which zone holds which city, and where Far West's Permian load sits	Figure 17 (§3.6)

Table 16. Europe (notes EU-0 to EU-5, K, and L).

Ref	Source	What it is about	What was found	What was taken away from it
[69]	ENTSO-E Transparency Platform	Europe's shared platform for load, price, and capacity data	Parameter names are case-insensitive, size caps are enforced per endpoint, and the A03 curve format needs expanding	The European panel (§2 , source 11)
[70]	ENTSO-E, Introduction guide for new users	The platform's help center	Irish load lives on the CTA identifier; what each document type returns	<code>sources/entsoe-api-constraints.md</code>

Ref	Source	What it is about	What was found	What was taken away from it
[71]	Commission Delegated Regulation (EU) 2024/1364	The EU data-center rating scheme and its database	Article 5(5): facility data is confidential by regulation	Europe cannot supply facility data either (EU-3)
[72]	DG ENER, reply to access-to-documents request 2025/3469 (2025-08-29)	A university's request for the database	Refused in August 2025	EU-3
[73]	ACER, Key developments in European electricity and gas markets, 2026	European wholesale price developments	The European fat tail is negative (German negative-price hours went 301, 457, 573); intra-zone congestion is priced at zero by construction	Why the method does not port to Europe (EU-4)
[74]	CSO, Data centres metered electricity consumption, table MEC02	Measured Irish data-center consumption	The data-center share went from 4.43% to 23.65%	The only measured dose in the project (§3.10)
[75]	CBS, Elektriciteit geleverd aan datacenters, 2017-2024	Measured Dutch data-center consumption	The Dutch share had not stayed flat at 4.6% but tripled, 1.48% to 4.58%	The control dose (§3.10)
[76]	L. Hirth, J. Mühlenpfordt and M. Bulkeley, The ENTSO-E Transparency Platform (Applied Energy, 2018)	How ENTSO-E data deviates from national sources	The deviations are systematic, so mixing sources would add artifacts	The national series were used for validation only
[77]	EirGrid, All-island resource adequacy assessment	The Irish operator's own statistics	Validation values for the Irish series	Validation only
[78]	Terna, Electricity demand of 311.3 TWh in 2025	The Italian operator's own statistics	Validation values for the Italian series	Validation only
[79]	UK Power Networks, open data portal	The data-center demand profiles dataset and its license	CC BY 4.0; 100k calls a day; an export counts as one call	§3.9

Table 17. Additional sources considered (notes I and N).

Ref	Source	What it is about	What was found	What was taken away from it
[80]	PJM, Hourly Electricity Load Forecasting Using Machine Learning Algorithms (FERC technical conference, 2024)	PJM benchmarking four machine-learning models against its production forecast	XGBoost beat the production forecast in every large zone except Dominion; "Dominion data center load is challenging"	The defect PJM sees is unpredictability, not jumpiness (note I)
[81]	EPRI, Powering Intelligence 2026: Annual and Peak Electricity Use	Metered annual and peak use at real facilities	Facilities run at 94% (hyperscale) and 88% (colocation) of their own realized peak; real peaks land at 62 to 80% of nameplate; a footnote locates the volatility at second and sub-second timescales	The duty-cycle numbers that make the hourly null substantive (§3.12)
[82]	EPRI, A Proposed Framework to Assess Headroom for Integrating Data Centers (3002034162)	A four-step method for finding unused grid capacity for large loads: hourly probabilistic screening, nodal	Its only price claim runs the other way (growth lowers the average price through utilization); it relies on data	The mitigation-side view, compared with XFRA

Ref	Source	What it is about	What was found	What was taken away from it
		limits, sub-hourly limits, local equipment	centers ramping down on request	
[83]	EPRI 3002033303, the metered sub-second report	The one metered sub-second source EPRI cites	Paywalled at \$25,000; not read	A lead only
[84]	EPRI, Data Center Load Shape Library (3002033424)	A library of measured data-center load shapes	Paywalled; not read	A lead only
[85]	NERC Large Loads Task Force, April 2025 workshop deck	The task force's 145-page technical workshop	Tesla's and Google's slides are the origin of the "10 to 20 MW several times a second" claim; utilities describe the undervoltage-trip-and-reconnect mechanism	The top rung of the provenance ladder (§3.12)
[86]	NERC, Characteristics and risks of emerging large loads, White Paper 1	The task force's first description of the risk	The reliability framing of large-load behavior, before any limit existed	§3.12
[87]	NERC, Assessment of gaps in existing practices, requirements, and reliability standards, White Paper 2 (March 2026)	Where the standards fall short	Admits that no allowable-oscillation limit exists in any NERC standard and that standard instrumentation cannot characterize the fast band	§3.12
[88]	NERC, Reliability guideline: risk mitigation for emerging large loads (May 2026)	The mitigation framework: voltage ride-through, fast frequency response, and soft-start	Cites arXiv 2508.14318 by name (footnotes 28 and 40) as its evidence base	The regulatory view of the sub-second problem (§3.12)
[89]	ERCOT Large Load Working Group, Large Load Power Variation Requirement Consideration (2026-02-19), with the Electranix framework behind it	The first proposed numerical limit on large-load oscillation	"Load power shall not repetitively exceed 10 MW change in a sliding 5-second window", a flat cap across the whole measured band; oscillation headroom becomes a first-come, first-served locational resource	The limit quoted in §3.12 and §5 , against which a frontier cluster is non-compliant
[90]	E. Choukse et al. (Microsoft, OpenAI, and NVIDIA), Power stabilization for AI training datacenters (arXiv 2508.14318)	Production telemetry from training jobs of tens of thousands of GPUs, and the operators' mitigations	Swings of 10 MW to over 100 MW; spectral energy at 0.2 to 3 Hz; every plot normalized 0 to 1 with no MW axis; no data released; GPUs are more than half of server power	The measured band, and its weight: the best quantification that exists, self-reported and unreplicated
[91]	Y. Li et al. (University of Alberta), The unseen AI disruptions for power grids: LLM-induced transients (arXiv 2409.11416)	Single-GPU power measurements at high resolution	The GPU-level swing that the facility-scale extrapolations start from	A rung of the provenance ladder (§3.12)
[92]	R. Vercellino et al. (National Laboratory of the Rockies), Measurement of generative AI workload power profiles for whole-facility data center	Node-level measurements at 10 Hz	Swing amplitude scales linearly with node count	§3.12

Ref	Source	What it is about	What was found	What was taken away from it
	infrastructure planning (arXiv 2604.07345)			
[93]	C. Chaudhary et al. (Michigan State University), <i>Spatial load correlation in AI data-center-dominated power systems</i> (arXiv 2606.13853)	A simulation of several facilities swinging together on the IEEE 39-bus test system	Assumes a cross-facility correlation of about 0.4; no field measurement exists	The coherence argument (§3.12)
[94]	P. Li et al., <i>Smoothing the ramp, not the peak: scheduling-induced power dynamics of LLM inference and their grid-scale consequences</i> (arXiv 2608.01250)	Load shape under inference rather than training	Ramp-rate risk falls as $1/\sqrt{N}$ because requests are unsynchronized	§3.12
[95]	S. Jadhav and Z. Liu, <i>Machine learning guided cooling system optimization for data center</i> (arXiv 2601.02275)	Machine-learning-guided cooling at Frontier	10-minute data; cannot see the band	Not useful (note I)
[96]	C. Mishra et al., <i>Understanding the inception of 14.7 Hz oscillations emerging from a data center</i> (Sustainable Energy, Grids and Networks, 2025)	A grid oscillation event measured by third parties	A power-electronics control interaction, cured by a firmware upgrade	The only independent rung of the ladder (§3.12)
[97]	J. Sun et al., <i>Data center power system stability, part I: power supply impedance modeling</i> (CSEE Journal of Power and Energy Systems, 2022)	Data-center power-system stability	The measured events are control interactions, not workload synchronization	§3.12
[98]	A. A. Elsayed, A. A. Al-Obaidi, and H. E. Z. Farag, <i>Characterization of high-resolution AI data center training workloads</i> (rs-7943457)	Per-GPU telemetry from 32 training runs on H100 and B200 nodes	The sensor refreshes every ~103 ms, not at 50 Hz; cross-GPU correlation 0.994; the node cycles every 8.5 to 15.9 seconds, and each transition inside the cycle is a step of about a second	The project's own spectrum (§3.12)
[99]	A. Elsayed, <i>High-resolution AI data center training workloads dataset</i> (GitHub)	The dataset behind [98], CC BY 4.0	The sequence-length sweep is not independent runs	Dataset 37 (§2 , source 15)
[100]	SPAN and NVIDIA, <i>XFRA White Paper</i> (2026)	A 12.5 kW always-on compute node plus a 15 kWh battery in new houses	The node runs firm; the battery and the EV charger curtail first (p. 15)	The headroom question note M answers (§3.11)
[101]	Latitude Media, <i>SPAN to launch mini AI data centers for distributed at-home compute</i> (2026)	Coverage of the launch and the pilot	A 100-home PulteGroup pilot; SPAN pays the electricity bill	§3.11
[102]	pv magazine USA, <i>SPAN and NVIDIA to develop AI data centers in your backyard</i> (April 2026)	Coverage of the launch from a second outlet	The "lowering electric bills" pitch, and the same pilot	§3.11

Ref	Source	What it is about	What was found	What was taken away from it
[103]	Google, <i>Data center efficiency</i>	Google's power usage effectiveness	PUE 1.10 (Q2 2026): about 9% overhead	Cooling cannot be the averaging mechanism (§3.12)
[104]	Meta, <i>2024 Responsible Business Practices report</i>	Meta's power usage effectiveness	PUE 1.08 (2023)	§3.12
[105]	A. Gandhi et al., <i>AutoScale: dynamic, robust capacity management for multi-tier data centers</i> (ACM Transactions on Computer Systems, 2012)	Turning servers on and off with demand, as a software-side way of shaping a data center's power draw	Suggested at the 2026-08-24 meeting; not yet read	A future direction (§5)
[108]	A. Grattafiori et al. (Meta), <i>The Llama 3 herd of models</i> (arXiv 2407.21783)	Meta's report on training the Llama 3 models	The 405B model was pre-trained on 16,384 H100 GPUs	A real job size for the scale-up in Table 12 (§3.12)
[109]	Meta Engineering, <i>Building Meta's GenAI infrastructure</i> (March 2024)	Meta's description of its two 2024 training clusters	24,576 H100 GPUs in each cluster	A real cluster size for Table 12 (§3.12)
[110]	NVIDIA, <i>NVIDIA Ethernet networking accelerates world's largest AI supercomputer, built by xAI</i> (October 2024)	The announcement of xAI's Colossus cluster	100,000 H100 GPUs in the first phase	A real cluster size for Table 12 (§3.12)

5. Conclusion

The project originally set out to measure how data-center load behaves and its impact on prices and the grid. Unfortunately, the clearest result is that the lack of data availability makes things very unclear. Facility-level load for a US hyperscale data center does not exist in public data anywhere. Even the government's attempts have failed: EIA's 2021 pilot drew 9 responses from 50 facilities, and its 2024 survey was stopped by a lawsuit and the collected data destroyed. Every US series in this project is a regional aggregate in which data centers are one unlabeled part, and no sub-zonal or per-customer load exists for DOM because PJM dispatches against unpublished bus loads. Furthermore, the attempts to isolate data-center load statistically, by restricting the panel to shoulder months and the 2 to 5 AM window in which little else is running, are not realistic. Unfortunately, they cut out the very events the question is about: inside that window, the price never exceeded \$250 in 2,027 hours.

In Europe, the facility database is confidential by law. Unfortunately, a university's request for it was refused in 2025. Where facility-level data did exist (96 UK sites), it existed as shape only, with no MW, no location, and no price, and the sites are, as far as can be told, conventional colocation and enterprise facilities rather than hyperscale AI training. Therefore, the load signature is different altogether, and at half-hourly resolution the sub-second fluctuations the project was looking for are invisible anyway.

The only series fine-grained enough to see fast fluctuations was the Pecan Street data, which is residential only, and the rs-7943457 GPU telemetry, which is one server node rather than a facility. Trying to isolate data-center load by subtracting residential load from system totals was a dead end as well, because 73 volunteer homes cannot stand in for millions, the sample is tilted to solar and EV owners, whatever is left after the subtraction is all commercial and industrial load, and the data (2018 to 2019) pre-dates modern AI hyperscaler build-outs. The GPU telemetry is fast enough (about 10 Hz) but sees eight GPUs, not the thousands whose synchronized swings the industry claims are about; its cycle repeats well below the claimed band, although each transition inside the cycle is a step of about a second. Moreover, every other dataset held is sampled at one second or slower, while the industry claim lives at 0.2 to 3 Hz, and Microsoft, OpenAI, and NVIDIA, who measured those swings, released no data and no exact, reproducible numbers.

What can and cannot be said. Because of all of the reasons listed above, the conclusions that can be confidently asserted are very limited in scope.

- The volatility threshold the original proposal asked for does not exist in DOM prices, at least not as a single number. The price response to load ramps is a smooth curve.
- Overall, DOM load level has been growing, about 28% from 2023 to 2026. However, its volatility and ramp rate have not, and normalized with respect to overall load level, they fell every year.
- Where load is metered at the facility (UK), it is among the flattest loads on the system, and two-thirds of site-level movement cancels in aggregate. Where it is measured nationally (Ireland), the load-shape change that coincided with it tracks rooftop solar, not data centers. Moreover, both of these are measured at timescales far above sub-second.
- The sub-second swings the industry worries about are real enough to have produced a proposed ERCOT limit (load may not repeatedly change by more than 10 MW within any 5-second window), but the evidence for them is operator-supplied and unreplicated, and the independent analysis on GPU telemetry reproduces the shape of the claim (abrupt steps of about a second, coherent across the node's eight GPUs) but not its numbers or its repetition rate (although it should be noted that a single workstation and hyperscale data centers are on completely different scales). Scaled by GPU count alone, the node's 0.28 to 0.56 kW per-GPU step reaches 10 MW at about 18,000 to 36,000 synchronized GPUs, the size of the training jobs the operators describe, so the claim is at least consistent in magnitude with the telemetry ([§3.12](#)).

- The one thing that can be said with some confidence is narrow: at every timescale observed in load and price data (hourly, 5-minute, half-hourly at the facility, and 1-second at the home), the hypothesized volatility does not show up in either load or prices. The one place it does show up is inside the server: the 10 Hz telemetry of one GPU node records steps of a few kilowatts completed in about a second, repeated every 8 to 16 seconds. That does not prove it is absent from the grid; it means that if it reaches the grid, it lives below the sampling rate of every public dataset, where only reliability, not price, can register it.
- Given that the volatility cannot be seen at any timescale the public data reaches, and that its stated consequences are reliability rather than price, the working explanation is that modern mitigation is absorbing it before any meter sees it. On-site batteries and the uninterruptible power supply (UPS) mitigate the sub-second swings, voltage ride-through and soft-start requirements limit what a facility can do to the grid, behind-the-meter generation takes load off the meter altogether, and the facility-level shapes that do exist (UK) are flat. If that is right, the volatility is real inside the building and invisible outside it, which is consistent with everything measured here, but has not been tested directly.

Where this could go from here.

- Wait for data. Facility-level load becomes public only if someone is made to publish it: the Clean Cloud Act (S. 1475) would give EPA and EIA the authority, and the EU database exists but is closed by regulation. Either would change what this project can do more than any new method would, but both options are also highly unlikely in the near future.
 - Measure load directly. The alternative to waiting is to instrument a training cluster and record its power at 10 Hz or better. Reaching the megawatt swings in question takes hundreds to thousands of GPUs, which is incredibly expensive and most likely outside the scope of what Northwestern can provide, but it is still a possibility that could be considered.
 - Watch the field. The problem is being engineered around in real time: on-site batteries (Tesla's Megapack at xAI's Colossus), soft-start and ride-through requirements (NERC's guideline), ERCOT's oscillation limit, FERC's co-location rules, and behind-the-meter and SMR co-location. These are recent advancements in technology and policy, and they all change what a meter would see. Therefore, the answer to the original question is a moving target, and researching future mitigation strategies from different perspectives may be worthwhile in its own right.
 - Explore software scheduling as a mitigation. Going along with the previous bullet point, the compute-and-communication cycle that produces the sub-second swings is set by the training job's schedule, so a scheduler that staggers or overlaps the phases across GPUs, or fills the communication phase with dummy work, could flatten the power draw at the source before any battery sees it. Whether software can do at the source what batteries do at the meter, and at what cost in training throughput, is an open-ended question.
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Appendix A: The Original Research Proposal

The proposal as submitted, reproduced verbatim from Yu_Tony_SURG_Grant_Proposal.md in the repository. Its bracketed reference numbers point to its own reference list at the end of this appendix, not to §6.

Introduction

Northern Virginia (NOVA), specifically Loudoun and Fairfax counties, handles nearly 70% of the world's internet traffic, earning it the moniker "Data Center Alley" [1]. While this infrastructure is critical for the expanding Artificial Intelligence (AI) economy, it places an unprecedented strain on the local Dominion (DOM) energy grid. The recent (2020-current) rapid expansion of hyperscale data centers, facilities that consume massive amounts of power for cloud computing and AI model training, has shifted the region's energy consumption patterns [1][2]. Current policy discussions often focus on the aggregate capacity needed to meet this demand [1]. However, this macro-level view overlooks the immediate, high-frequency instability caused by the volatile load profiles of modern computing. My project seeks to shift the focus from "how much" energy is consumed to "how volatile" the consumption is and its resulting impacts on grid stability. Specifically, I will model the relationship between high frequency load anomalies and spikes in Locational Marginal Pricing (LMP), a pricing mechanism that sets real-time energy prices at specific grid locations (nodes) by calculating the marginal cost to supply the next unit of electricity [3]. Then, I will determine the critical threshold where future AI infrastructure investment will push NOVA's energy grid into a state of chronic instability, quantifying the "hidden costs" of data center growth on grid stability.

Background/Literature Review

The energy impact of data centers is a subject of intense scrutiny. The 2024 Joint Legislative Audit and Review Commission (JLARC) report on Virginia's data centers provides a comprehensive forecast of aggregate energy demand, predicting that total energy consumption in the state will double by 2040 [1]. While the JLARC report analyzes the need for new generation capacity and the economic implications, it primarily relies on annual and monthly consumption forecasts. This approach treats data center load as a largely static "baseload," an idealized constant hum of flat energy usage. However, recent studies suggest that the shift toward data center use for generative AI training creates "spiky" load profiles that differ significantly from traditional data storage and cloud computing [2], [3]. When these surges occur simultaneously across multiple facilities in a constrained geographic area like Data Center Alley, they risk overloading transmission lines. In the PJM (the organization managing the grid across the northeastern US) energy market, such congestion is reflected in LMP. LMP is calculated as the sum of three components: the system-wide energy price, the marginal cost of transmission congestion, and the cost of marginal losses [4]. Unusually high LMPs indicate that the grid is physically congested and cannot move power efficiently to where it is needed [4], [5]. The increased construction of data centers in NOVA have already had adverse effects on the DOM grid. The 2024 PJM State of the Market report confirms that average congestion costs have already risen significantly, driven specifically by increased energy demand and transmission constraints from the Pleasant View - Ashburn line and the Goose Creek Transformer [6], [7] in the DOM zone. Congestion in these power lines results from the grid's inability to meet hyper-localized demand, causing increased congestion prices in LMP and widespread power outages in the NOVA region. While it is well established that increased demand correlates with higher prices, the specific saturation point of the DOM grid remains unquantified. Literature on self organized criticality suggests that power grids function linearly up to a specific stress limit, beyond which they undergo a phase of large scale cascading failure, transitioning into a chaotic, heavy tailed pricing regime, where extreme price spikes become frequent rather than anomalous [8]. Identifying this tipping point is crucial for policymakers seeking to balance continued AI expansion with the operational stability of the regional grid. Therefore, my research asks: What is the critical

volatility threshold of data center load variance that triggers a non-linear phase transition in Dominion Zone congestion pricing, and at what point in the future will this threshold become the chronic operating state of the NOVA grid?

Methodology

My research will employ a quantitative statistical analysis of historical grid data to isolate data center energy usage from broader consumption trends. I will conduct this project in three distinct phases over the eight week summer period.

Phase 1: Data Acquisition and Preprocessing (Weeks 1-3). I will utilize the PJM Data Miner 2.0 [9] to harvest hourly and, where available, 5-minute interval data for the DOM transmission zone for the years 2020-2025. Specifically, I will extract data from three main datasets: Hourly Metered Load, Real-Time Hourly LMP, and Real-Time Five Minute LMPs. To access the data, PJM provides both a user interface and direct API access [9]. Because API calls are extremely limited to non-members (such as myself), the majority of data will be collected manually through the user interface, with API use when available.

Phase 2: Signal Isolation (Weeks 4-5). A major challenge in this analysis is that aggregate load data includes residential, commercial, and industrial electricity use, in addition to data center use. To address this challenge, I will employ a data filtering strategy to minimize the effect of these other variables on the data. First, I will filter the dataset seasonally to focus on months where air conditioning (AC) and heating use are minimized, specifically targeting the months March-May and September-November, the Spring and Fall months in Virginia. Second, I will isolate the time period to the deep night window, between the hours of 2:00 AM and 5:00 AM. During these hours, variable commercial activity (retail/office) and residential activity (lighting/cooking) are negligible, leaving a load profile dominated by continuous industrial baseloads and data centers. Since the JLARC report identifies data centers as the primary driver of the region's exponential load growth, distinct from historical industrial baseloads, any statistically significant increase in high-frequency volatility during these quiet windows can be reasonably attributed to the operational dynamics of data center expansion [1]. Moreover, to prove the spike is demand-driven (data centers) and not supply-driven (grid maintenance) I will check PJM's publicly available outage logs.

Phase 3: Statistical Correlation and Analysis (Weeks 6-8). I will then analyze the tail distribution of these residual load spikes to identify the system's failure state. Using a Granger Causality test, I will first verify that sudden ramps in residual load reliably predict subsequent spikes in congestion pricing. Moving beyond correlation, I will mathematically isolate the Critical Volatility Threshold. By fitting a Generalized Pareto Distribution (specifically designed to model extreme outliers) to the pricing data, I will determine the specific load variance value (in MW/min) where the grid's pricing response shifts from linear to exponential, a transition often triggered by specific reserve shortage thresholds [10]. This "tipping point" analysis is supported by the non-convex (step-change) nature of electricity pricing, where costs rise disproportionately during scarcity [11]. Once this threshold is defined, I will apply it to the JLARC growth forecasts, forecasting the specific year between 2025 and 2040 when the baseline volatility of the expanded data center fleet will permanently exceed the grid's stability threshold.

Qualifications

I am an undergraduate student majoring in Applied Mathematics. My coursework in Data Structures and Algorithms (CS 214) has equipped me with the necessary algorithmic efficiency to handle large datasets. Previously, at American University, I analyzed the correlation between Medicare Part D spending trends and the COVID-19 pandemic. This past project required an exploratory data analysis workflow similar to what I propose for the PJM data: processing, cleaning, and analyzing millions of distinct data points to identify trend anomalies. Moreover, I'm confident that any gaps in knowledge will be covered by working together

with my research advisor, Professor Ermin Wei, and senior PHD student Lihui, whose work in analyzing the dynamics of energy markets brings the experience of market analysis in energy markets that I lack. Lastly, I am strongly considering graduate study and a career in academia, and early involvement in research will help me assess whether this path aligns with my interests while building the foundational skills needed to continue.

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Appendix B: Figure Index

Table 18. Every figure in this record, the file it is stored in under `assets/final_report/`, and the script that drew it.

Figure	File	Section	Script
1	<code>ordc_two_step.png</code>	§0	<code>scripts/plot_ordc.py</code>
2	<code>pjm_zones_map.png</code>	§3.1	PJM's zone map [106], rasterized from PJM's PDF
3	<code>hourly__dom_load_and_lmp.png</code>	§3.1	<code>scripts/plot_hourly_series.py</code>
4, 5	<code>hourly_filtered__tail_risk_primary.png</code> , <code>hourly_nofilter__tail_risk_primary.png</code>	§3.1	<code>surg-analyze</code> (inside the filter), <code>surg-analyze --no-filter</code> (full panel)
6, 7, 8, 9	<code>F1_premise.png</code> , <code>F2_load_vs_volatility.png</code> , <code>F6_effect_size.png</code> , <code>F8_tail_risk_nofilter.png</code>	§3.2	<code>python -m scripts.plot_subq1_results</code> (inputs from <code>scripts/compute_figure_inputs.py</code>)
10	<code>fivemin_nofilter__tail_risk_cluster.png</code>	§3.2	<code>scripts/run_5min_nofilter.py</code>
11, 12, 13, 14	<code>F4_events_per_month.png</code> , <code>F4b_severity_by_month.png</code> , <code>F11_what_changed.png</code> , <code>F3_prices_over_time.png</code>	§3.3	<code>python -m scripts.plot_subq1_results</code>
15	<code>F7_location.png</code>	§3.4	same
16	<code>F10_nerc_event.png</code>	§3.5	same
17	<code>ercot_weather_zones_map.jpg</code>	§3.6	ERCOT's weather zone map [107]
18, 19	<code>ercot_diagnostic__fig1_volatility_trend_normalized.png</code> , <code>ercot_diagnostic__fig2_level_trend.png</code>	§3.6	<code>scripts/ercot_diagnostic.py</code>
20 to 25	<code><market>_diagnostic__fig1_volatility_trend_normalized.png</code> , <code>...fig2_level_trend.png</code> (MISO, SPP, ISO-NE, NYISO, CAISO, IESO)	§3.7	<code>scripts/<market>_diagnostic.py</code>
26	<code>italy_stage1__fig1_volatility_trend_normalized.png</code> , <code>italy_stage1__fig2_level_trend.png</code>	§3.7	<code>scripts/entsoe_italy_stage1.py</code>
27	<code>isone__necec_me_basis.png</code>	§3.8	<code>scripts/necec_price_test.py</code>
28	<code>ukpn__ukpn_flatness.png</code>	§3.9	<code>scripts/ukpn_flatness.py</code>
29, 30	<code>entsoe__fig_diurnal_profiles.png</code> , <code>entsoe__fig_shape_trends.png</code>	§3.10	<code>scripts/entsoe_ireland.py</code>
31, 32	<code>entsoe__fig_solar_signature.png</code> , <code>entsoe__fig_solar_diurnal.png</code>	§3.10	<code>scripts/entsoe_solar.py</code>
33 to 38	<code>pecanstreet__*.png</code> (8 files)	§3.11	<code>scripts/pecanstreet_headroom.py</code> , <code>scripts/pecanstreet_1sec.py</code>
39	<code>aidc__node_power_timeseries.png</code>	§3.12	<code>scripts/plot_aidc_timeseries.py</code>

Appendix C: Glossary

Table 19. Acronyms and technical terms used in this record, in alphabetical order.

Term	Meaning
A03, A68, A75	ENTSO-E document codes: A03 is the compressed curve format for time series; A68 is installed generation capacity; A75 is actual generation by type
ACER	Agency for the Cooperation of Energy Regulators, the EU's energy-regulator body
API	Application programming interface: a way for a program to request data from a service over the web
Basis	The difference between one zone's price and a reference price; a negative basis means the zone is cheaper
Behind-the-meter (BTM)	Generation or load on the customer's side of the utility meter, invisible to the grid's measurements
Bidding zone	The area within which Europe's day-ahead market treats electricity as freely tradable at one price
Bootstrap	A way of estimating uncertainty by re-running a calculation on many random resamples of the data
BRA	Base residual auction, PJM's annual capacity auction
CAISO	California Independent System Operator
CBS	Centraal Bureau voor de Statistiek, Statistics Netherlands
CC BY 4.0	A Creative Commons license that allows reuse with attribution
Coefficient of variation	The standard deviation divided by the mean: how much a series wanders relative to its size
Congestion price	The part of an LMP caused by transmission limits; it differs from node to node
CRS	Congressional Research Service
CSO	Central Statistics Office of Ireland
CTA	Control area, the ENTSO-E identifier for the Republic of Ireland's load
Decile	One of ten equal groups of observations sorted from lowest to highest
Demand response	Customers reducing load on request, usually for payment, during grid stress
Distribution	The lower-voltage network that delivers power locally (35 kV in this record), as opposed to transmission
DOE	U.S. Department of Energy
DOM	The Dominion transmission zone inside PJM, Dominion Energy's Virginia territory
Dose	The size of a treatment: here, a country's data-center share of electricity consumption
Duty cycle	The fraction of time a device spends at full power rather than idle
EIA	U.S. Energy Information Administration; EIA-930 is its hourly grid monitor
ENTSO-E	European Network of Transmission System Operators for Electricity
EPRI	Electric Power Research Institute
ERCOT	Electric Reliability Council of Texas, the Texas grid operator
EV	Electric vehicle
Fast frequency response	A battery or load reacting within a second to a deviation in grid frequency
FERC	Federal Energy Regulatory Commission
FFT	Fast Fourier transform, the algorithm that splits a signal into its frequencies
GPD	Generalized Pareto distribution, a statistical model of the largest values in a dataset

Term	Meaning
GPU	Graphics processing unit, the chip that does AI training arithmetic
Gradient (of load)	The change in load per unit time, in MW per minute; the volatility measure of this record
Heavy tail	A distribution in which extreme values are far more common than a bell curve would predict
HOEP	Hourly Ontario Energy Price, Ontario's single province-wide price until 2025
Hub	A trading location whose price is the average over a group of nodes
HVDC	High-voltage direct current, the technology used for long transmission links
IESO	Independent Electricity System Operator, Ontario's grid operator
Interface (external tie)	A transmission connection between one grid and a neighboring one
ISO	Independent system operator: the organization that runs a regional grid and its wholesale market
ISO-NE	ISO New England
JLARC	Joint Legislative Audit and Review Commission, Virginia's legislative audit agency
LBMP	Locational-based marginal price, NYISO's name for LMP
LBNL	Lawrence Berkeley National Laboratory
LLM	Large language model
LLTF	NERC's Large Loads Task Force
LMP	Locational marginal price, the wholesale electricity price at a specific point on the grid
Load factor	Average load divided by peak load: how fully a system or facility is used
Load zone	A sub-region of an ISO for which load and prices are published
LRZ	Local resource zone, MISO's planning regions
MAD	Mid-Atlantic/Dominion, a PJM reserve sub-zone
Marginal loss price	The part of an LMP that pays for the energy lost as heat in the wires
MISO	Midcontinent Independent System Operator
MW, MWh, GW, TWh	Megawatt (power), megawatt-hour (energy), gigawatt (1,000 MW), terawatt-hour (1,000,000 MWh)
NECEC	New England Clean Energy Connect, a 1,200 MW HVDC line from Québec to Maine
NERC	North American Electric Reliability Corporation, the grid reliability regulator
Node (pricing node)	A specific point on the grid at which the ISO computes a price
Nyquist frequency	The highest frequency a sampled series can resolve, half the sampling rate
NYISO	New York Independent System Operator
OASIS	Open Access Same-time Information System, CAISO's data API
ORDC	Operating reserve demand curve, PJM's schedule of penalties for reserve shortages
Percentile / quantile	The value below which a given share of observations fall; τ is the share as a fraction
PJM	PJM Interconnection, the grid operator for the Mid-Atlantic and parts of the Midwest
Placebo	A comparison case that received no treatment, used to judge whether a change could be chance
Pre-registration	Writing down a test and its decision rule before looking at the data

Term	Meaning
PSD	Power spectral density: how much of a signal's variation sits at each frequency
PUE	Power usage effectiveness, total facility power divided by computing-equipment power
PV	Photovoltaic (solar panels)
Quantile regression	A regression that estimates how a chosen percentile of the outcome, rather than its mean, responds
R^2	The share of the variation in an outcome that a regression explains (0 none, 1 all)
Ramp	A change in load from one interval to the next; used interchangeably with gradient
RTEP	Regional Transmission Expansion Plan, PJM's transmission planning process
RTO	Regional transmission organization; used for PJM's whole footprint
SCC	Virginia State Corporation Commission, the state utility regulator
SEM	Single Electricity Market, the all-island Irish market
Settlement point	A location at which ERCOT settles energy transactions (a hub or a load zone)
Shadow price	The cost of moving one more MW across a transmission element that is at its limit
Shoulder months	Spring and fall, when heating and cooling loads are lowest
SMD	Standard Market Design, the 2003 ISO-NE market redesign whose name its data files carry
SMR	Small modular reactor
Soft-start	Ramping a facility's load up gradually rather than switching it on at once
SPP	Southwest Power Pool
Stage 1	The project's level-versus-volatility diagnostic, run in every market
System energy price	The part of an LMP that is the same everywhere in PJM; carries the reserve-shortage penalty
TAR	Threshold autoregression, Hansen's threshold regression method
Transmission	The high-voltage backbone (500 kV in this record) that moves bulk power across a region
UKPN	UK Power Networks, the distribution network operator for London and the South East
UPS	Uninterruptible power supply
Voltage ride-through	Staying connected through a voltage dip instead of tripping to backup power
XFRA	SPAN and NVIDIA's program for a compute node plus battery in new houses
Z	This record's name for the load-volatility variable, the absolute load gradient in MW per minute
ξ (xi)	The shape parameter of a generalized Pareto distribution; larger means a heavier tail